

pilot's guide

EFIS-85C(14)/86C(14) Electronic Flight Instrument System

Includes TWR-850 Turbulence Weather Radar System operation

pilot's guide

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COLLINS EFIS-85C(14)/86C(14)

ELECTRONIC FLIGHT INSTRUMENT SYSTEM

PILOT'S GUIDE (5-TUBE VERSION)

The following lists specify the Collins part numbers for the equipment used in the EFIS-85C(14)/86C(14) 5-tube system. Verify that these part numbers are the same as specified in the list of equipment for your airplane.

EFIS-85C(14) SYSTEM

EQUIPMENT TYPE	COLLINS PART NUMBER
EFD-85 (4 req'd) or EFD-85A (4req'd)	622-6020-011, -012, -021, -022, -121, or -122 622-6464-011, -012, -021, or -022
MFD-85B (1 req'd) or ¹ MFD-85C (1 req'd)	622-7876-011, -012, -211, or -212 622-9849-001, -002, -011, -012, -201, or -202
DPU-85R (2 req'd)	622-9007-001, -002, -003, or -004
MPU-85R (1 req'd)	622-9008-001, -002, -003, or -004
DCP-85F (2 req'd) or DCP-85H (2 req'd)	622-8148-001, -002, -201, or -202 622-9165-001, -002, -201, or -202
CHP-85C (1 req'd) or CHP-86C (2 req'd)	622-6404-001, or -002 622-7875-001, -002, -201, or -202
RDP-300 (1 req'd) or CEU-85 (1 req'd) or CEU-85A (1 req'd)	622-5110-001 622-7877-001 622-9890-001
^{2, 3} WXP-850A or ^{2, 3} WXP-850B	⁴ 622-8393-001, -002, -003, -004, -011, -012, -013, or -014 ⁴ 622-8394-001, -002, -003, -004, -011, -012, -013, or -014
Notes: ¹ Used in installations that show TCAS traffic on the MFD. ² Although not part of the EFIS system, the WXP-850() Weather Radar Panel controls the display of weather radar information on EFIS displays. This pilot's guide includes a description of the WXP-850() operation. ³ One or two may be installed. Installations with slaved radar operation require two WXPs. ⁴ -001, -002, -003, and -004 versions of WXP-850A and -850B are used in installations with -001 and -002 versions of DPU-85R and MPU-85R. -011, -012, -013, and -014 versions of WXP-850A and -850B are used in installations with -003 and later versions of DPU-85R and MPU-85R.	

(Equipment list for the EFIS-86C(14) is on the following page.)

EFIS-86C(14) SYSTEM

EQUIPMENT TYPE	COLLINS PART NUMBER
EFD-86 (4 req'd) or EFD-86A (4 req'd)	622-6342-011, -012, -013, -014, -021, -022, -023, -024, -121, or -122
MFD-85B (1 req'd) or ¹ MFD-85C (1 req'd)	622-6535-011, -012, -021, or -022
DPU-86R (2 req'd)	622-7876-011, -012, -211, or -212
MPU-86R (1 req'd)	622-9849-001, -002, -011, -012, -201, or -202
DCP-85F (2 req'd) or DCP-85H (2 req'd)	622-9009-001, -002, -003, or -004
CHP-85C (1 req'd) or CHP-86C (2 req'd)	622-9010-001, -002, -003, or -004
RDP-300 (1 req'd) or CEU-85 (1 req'd) or CEU-85A (1 req'd)	622-8148-001, -002, -201, or -202
^{2,3} WXP-850A or ^{2,3} WXP-850B	622-9165-001, -002, -201, or -202
	622-6404-001, or -002
	622-7875-001, -002, -201, or -202
	622-5110-001
	622-7877-001
	622-9890-001
	⁴ 622-8393-001, -002, -003, -004, -011, -012, -013, or -014
	⁴ 622-8394-001, -002, -003, -004, -011, -012, -013, or -014
Notes: ¹Used in installations that show TCAS traffic on the MFD. ²Although not part of the EFIS-86B(14) system, the WXP-850() Weather Radar Panel controls the display of weather radar information on EFIS displays. This pilot's guide includes a description of the WXP-850() operation. ³One or two may be installed. Installations with slaved radar operation require two WXP's. ⁴-.001, -.002, -.003, and -.004 versions of WXP-850A and -850B are used in installations with -.001 and -.002 versions of DPU-86R and MPU-86R. -.011, -.012, -.013, and -.014 versions of WXP-850A and -850B are used in installations with -.003 and later versions of DPU-86R and MPU-86R.	

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TABLE OF CONTENTS

INTRODUCTION	1
SYSTEM DESCRIPTION.....	3
EFD-85()/86() Electronic Flight Displays	4
EADI (Electronic Attitude Director Indicator)	4
EHSI (Electronic Horizontal Situation Indicator)	5
MFD (Multifunction Display).....	6
DCP (Display Control Panel)	7
CHP (Course Heading Panel).....	7
DPU (Display Processor Unit)	8
MPU (Multifunction Processor Unit).....	8
WXP (Weather Radar Panel)	8
OPERATING CONTROLS.....	9
DCP-85F/85H Display Control Panel Operation.....	9
CHP-85C/86C Course Heading Panel Operation	13
WXP-850() Weather Radar Panel	17
MFD-85() Multifunction Display Operation.....	20
MFD-85B Controls	20
MFD-85C Controls	23
EFIS-86C(14) Display Formats.....	25
EADI Display Formats	25
EADI Displays and Annunciators	26
Attitude Display	26
Aircraft Symbol	26
Command Bar Display.....	26
Inclinometer	26
Attitude Sensor Annunciation.....	27
Flight Control Mode Annunciation	27
Autopilot/Yaw Damper and Trim/Sync Annunciation	27
Marker Beacon Display	28
Vertical Deviation Display.....	28
Radio Altitude Display.....	29
Radio Altitude Test	29
Decision Height Display.....	29
Decision Height Annunciator	29
Lateral Deviation Display	29
Acceleration (When On The Ground) or Mach Display	30
Speed Displays	31
EADI Excessive Attitude Display	38
EADI Warning Flags	39
Attitude Flag	39
Drive Unit (MPU or DPU) Flag.....	39
Vertical Deviation Flags.....	39
Radio Altimeter Flag	40
Display Control Panel Flag	40
Lateral Deviation Flag	40
Accelerometer/Mach Flag	40
Cross-Side Data Flag	40
Speed Deviation Flags	40

Flight Director Flag	41
Indicated Airspeed Flag	41
EADI Comparators and Master Warn	41
EADI Comparator Warnings	41
Comparator Reset	41
Master Warn and Master Warn Reset	41
Pitch and Roll Attitude Comparators	42
IAS Comparator	43
ILS Localizer Comparator	43
ILS Glideslope Comparator	43
Radio Altitude Comparator	44
EADI Category II Excessive Deviation Warnings	44
Excessive ILS Localizer Deviation	45
Excessive ILS Glideslope Deviation	45
Excessive Indicated Airspeed Deviation	46
CAT II Outputs	46
EHSI Display Formats	49
Rose, Sector, or Sector With Map Formats	49
Sector Format	49
Sector With Map Format	50
EHSI Displays and Annunciators in the Rose Format	50
Bearing Pointer Display	50
LNAV Annunciation	50
Compass Display and Heading Sensor Type	51
Selected Heading Display	51
Active Selected Course Display	52
To/From Arrow	52
TWR-850 Weather Annunciation	53
Data Display	53
Wind Display	54
Second Course Display	54
Vertical Deviation Display	55
Lateral Deviation Bar	56
LIN/ANG/XTK and B/C (Back Course) Annunciations	56
Waypoint or DME Identifier and Waypoint Alert	56
Distance Display	56
EHSI Displays and Annunciators in the Sector Format	59
Compass Sector Display	59
Selected Heading Display	59
Active Selected Course Display	59
Bearing Pointer Display	59
Vertical Deviation Display	59
Range Display	59
To/From Display	60
Weather Radar Display	60
Second Course Display	60
Peripheral Displays	60
EHSI Displays and Annunciators in the Sector with Map Format	63
Compass Sector Display	63
Active Selected Course Display	63
Second Course Display	64
Other Map Format Displays	65

EHSI Warning Flags	65
Distance Flag	66
Heading Flag	66
Drive Unit (MPU or DPU) Flag	66
Data Flags	66
Wind Flag	66
Vertical Deviation Flags	66
Display Control Panel Flag	66
Cross-Side Data Flag	66
Bearing Pointer Flag	66
Active Selected Course Flag	66
Second Course Flag (Rose Format)	67
Second Course Flag (Sector and Sector with Map Formats)	67
EHSI Comparators and Master Warn	67
EHSI Comparator Warnings	67
Comparator Reset	67
Master Warn and Master Warn Reset	67
Heading Comparator	67
Comparison Limits	68
Annunciation	68
EHSI Category II Excessive Deviation Warnings	68
MFD-85B DISPLAY FORMATS	69
Overview of Standard Functions	70
Weather Radar	70
Navigation Information	70
Page Data	71
Overview of Additional Functions	71
Waypoint Definition	71
Remote Data	71
Weather Radar (RDR) Display Mode	72
Navigation (NAV) Display Mode	76
Display Features and Elements	76
NAV MENU Display	80
Heading Up Map Display	82
North Up With Aircraft Centered Map Display	84
North Up Maximum View Map Display	86
Plan View Map Display	88
Remote Data (RMT) Display Mode	90
RMT #	90
Extended Data	91
Page (PGE) and Emergency (EMG) Display Modes	101
▼ (Line Advance)	104
▲ (Line Reverse)	104
SKP (Skip)	104
RCL (Recall)	104
CLR (Clear)	105
Joystick	105
Warning Flags and Comparators	106
NAV Flags	106
Heading Flag	106
Heading Comparator	106
MPU FAIL Flag	106

MFD-85C DISPLAY FORMATS	109
Overview	109
TCAS Traffic Display	109
RDR and NAV Displays	109
RMT, PGE, and EMG Displays	109
Traffic (TFC) Display Mode	110
Traffic Display	110
TFC Line Key Menu	112
Annunciators and Flags	114
Composite Display	120
TCAS System Diagnostic Display	121
Controls	122
Radar (RDR) and Navigation (NAV) Display Modes	124
RDR Display	124
NAV Display	126
NAV MENU Display	126
RMT, PGE, and EMG Display Modes	128
REVERSIONARY MODES OF OPERATION	129
Electronic Flight Display (EFD) Failure	129
EADI Failure (top EFD)	129
EHSI Failure (bottom EFD)	129
Display Processor Unit (DPU) Failure	131
Display Control Panel (DCP) Failure	134
Multifunction Processor Unit (MPU) Failure	136
Multifunction Display (MFD) Failure	136
SELF-TEST	137
EFIS Self-Test	137
Sensor Self-Test	138
Attitude Heading Sensors	138
Navigation Sensors	138
Radio Altitude Sensors	138
Air Data Sensors	138
PROGRAMMING MFD PGE AND EMG DATA PAGES	139
Programming with CEU-85 or CEU-85A Checklist Entry Units	139
Programming with the RDP-300 Remote Data Programmer	140
Programming Preparations	140
Operating the RDP-300 Remote Data Programmer	144
Keyboard Description	144
Operation	146
Programming a New MFD Multifunction Display	148
Changing an Already Programmed MFD Multifunction Display	151
EFIS SYSTEM BLOCK DIAGRAM	152

INTRODUCTION

This pilot's guide presents general information about the EFIS-85C(14)/86C(14) Electronic Flight Instrument System. It is not intended to be a training manual. It is a guide to understanding the general operation of the system as designed by Collins General Aviation Division.

This pilot's guide describes all variations of the EFIS-85C(14)/86C(14) 5-tube systems that consist of the equipment listed on the title page of this book. This includes descriptions of system components, typical operating procedures, annunciators, and warning flags. The specific information shown on the displays in any given aircraft installation is determined by the equipment installed in the aircraft and the aircraft's wiring. Occasionally references to a specific version of equipment are made in order to show and describe system or display variations. When this occurs, the status number for a specific unit may be given so that you can refer to your aircraft manuals to determine which version of the unit is installed in your aircraft. Not all of the elements and/or annunciators shown in the figures and described in this pilot's guide show on any given system installation. Some display elements and/or annunciators may not show on the displays at the same time as other elements and/or annunciators as shown in the figures. Also, some display information can be selected or deselected depending on the regime of flight. You must refer to the appropriate aircraft flight manuals for specific information for your aircraft's EFIS installation.

Supplemental information describing the EFIS-85C(14)/86C(14) in considerable technical depth is available in the form of manuals and brochures, as well as formal training courses. Contact your local Collins sales office for cost and availability information.

The EFIS-85C(14)/86C(14) 5-tube systems use color CRTs to display both the pilot's and copilot's ADI and HSI and a center mounted multifunction display. The CRT displays are called EADI (Electronic ADI), EHSI (Electronic HSI), and MFD (Multifunction Display) respectively. The 85C(14) system uses 5-inch by 5-inch CRTs for the EADI and EHSI and the 86C(14) system uses 5-inch by 6-inch CRTs for the EADI and EHSI. Both systems use a 4-inch by 5-inch CRT for the MFD. The figures and descriptions throughout this pilot's guide show and refer to the 86C(14) system. The 85C(14) system is the same as the 86C(14) except for the 5-inch by 5-inch CRTs.

EFIS-85C(14)/86C(14) Electronic Flight Instrument Systems use input data from the following sources:

- VOR, localizer, glideslope, TACAN, and microwave landing system (MLS) receivers (MLS consists of microwave azimuth (MAZ) and microwave glidepath (MGP))
- pitch, roll, heading, rate, and acceleration data from an AHS (attitude heading system) or conventional vertical gyro, compass system, and longitudinal accelerometers
- radio altimeters
- air data systems and angle of attack systems
- distance measuring systems
- flight control system
- long range navigation systems (LNAV's)(RNV, INS, VLF, LRN, LNV, FMS, GPS, etc.)
- vertical navigation systems
- weather radar systems
- and automatic direction finding systems.

Input data from the above sources is used by EFIS-85C(14)/86C(14) systems to show V-bar or crosspointer steering commands and other navigational data for flight control purposes as well as information of an advisory nature. Also, extensive monitoring and comparator circuits in the system give warning flags and other types of data flagging techniques to indicate possible equipment malfunctions.

WARNING

THE EFIS-85C(14)/86C(14) ELECTRONIC FLIGHT INSTRUMENT SYSTEMS AND RELATED COMPONENTS EXHIBIT A VERY HIGH DEGREE OF FUNCTIONAL INTEGRITY. NEVERTHELESS, USERS MUST RECOGNIZE THAT IT IS NOT PRACTICAL TO PROVIDE MONITORING FOR ALL CONCEIVABLE SYSTEM FAILURES AND THAT, HOWEVER UNLIKELY, IT IS POSSIBLE THAT ERRONEOUS OPERATION COULD OCCUR WITHOUT A FAULT INDICATION. THE PILOT HAS THE RESPONSIBILITY TO DETECT SUCH AN OCCURRENCE BY MEANS OF CROSS-CHECKS WITH REDUNDANT OR CORRELATED INFORMATION AVAILABLE IN THE COCKPIT.

WARNING

SUNGLASSES WITH POLARIZED LENSES OR LENSES THAT ARE DESIGNED TO FILTER SPECIFIC COLORS/FREQUENCIES OF LIGHT MAY ADVERSELY AFFECT A PILOT'S ABILITY TO SEE SOME COLORS SHOWN ON EFIS DISPLAYS. SOME ELEMENTS ON THE DISPLAY CAN BE COMPLETELY INVISIBLE WHILE WEARING THESE TYPES OF SUNGLASSES. ALSO, THE COLOR OF SOME ELEMENTS MAY BE CHANGED. FOR EXAMPLE, SOME BLUE LIGHT FILTERING LENSES CAN CHANGE MAGENTA TO RED.

SYSTEM DESCRIPTION

Collins EFIS-85C(14)/86C(14) Electronic Flight Instrument Systems can be installed in a variety of configurations. 5-tube systems have an EADI and EHSI on both the pilot's side and copilot's side, and a center mounted MFD. The system may be installed as a stand-alone flight instrument system, although it is normally installed with a flight control system such as the Collins FCS-80, APS-65, or FCS-85.

EFIS-85C(14) and EFIS-86C(14) 5-tube systems consist of the equipment listed in the following table. To assist in a clearer understanding of the relationships of the equipment, a simplified functional block diagram, in the back of this pilot's guide, shows how a typical system is interconnected.

EFIS-85C(14)	QTY	EFIS-86C(14)	QTY
EFD-85() Electronic Flight Display	4	EFD-86() Electronic Flight Display	4
MFD-85B Multifunction Display	1	MFD-85B Multifunction Display	1
or		or	
MFD-85C Multifunction Display	1	MFD-85C Multifunction Display	1
DCP-85F Display Control Panel	2	DCP-85F Display Control Panel	2
or		or	
DCP-85H Display Control Panel	2	DCP-85H Display Control Panel	2
CHP-85C Course Heading Panel	1	CHP-85C Course Heading Panel	1
or		or	
CHP-86C Course Heading Panel	2	CHP-86C Course Heading Panel	2
DPU-85R Display Processor Unit	2	DPU-86R Display Processor Unit	2
MPU-85R Multifunction Processor Unit	1	MPU-86R Multifunction Processor Unit	1

Although not part of the EFIS system, the WXP-850() Weather Radar Panel controls the operation of the TWR-850 Turbulence Weather Radar System which shows weather radar information on the EHSI and MFD displays. This pilot's guide includes a description of the operation of the TWR-850().

An EFIS-85C(14)/86C(14) system receives data from the following sources:

- VOR/localizer/glideslope/TACAN/microwave landing system receivers
- pitch, roll, heading, rate, and acceleration data from an AHS (attitude heading system)
or
- conventional vertical gyro, compass system, and longitudinal accelerometers
- radio altimeters
- air data systems
- angle of attack systems
- distance measuring systems
- flight control systems
- long range navigation systems (RNV, INS, VLF, LRN, LNV, FMS, GPS, etc.)
- vertical navigation systems
- weather radar systems
- automatic direction finding systems.

EFIS uses the data to show:

- attitude and heading information
- V-bar or cross-pointer steering commands
- navigational data for flight control purposes
- weather radar
- NAVAID/waypoint locations
- airspeed and mach speed
- wind vectors
- up to four formats of moving map displays
- FCS mode annunciation
- autopilot/yaw damper engage status
- attitude, heading, and CAT II comparator warnings
- CAT II excessive deviation
- decision height
- TCAS traffic displays
- checklists
- and diagnostic messages.

In addition, extensive monitoring and comparator circuits provide warning flags and other types of data flagging techniques to indicate possible equipment malfunctions.

An EFIS-85C(14)/86C(14) system interfaces with various sensors, flight control systems and other devices. Aircraft wiring straps, connected when the EFIS system is installed in the aircraft, identify the devices connected to the system for proper display of the data provided by each device. Refer to the appropriate aircraft manuals for your aircraft to determine the sensors and other systems installed.

NOTE

EFIS-85C(14)/86C(14) Electronic Flight Instrument Systems operate normally until the +27.5-V dc power bus voltage drops to less than +18 V dc. Below +18 V dc, the EADIs, EHSIs, and MFD displays turn off. They automatically turn back on when the voltage rises above +18 V dc.

The following paragraphs describe the equipment that is part of the EFIS-85C(14)/86C(14) system.

EFD-85()/86() Electronic Flight Displays

EFD-85()/86() Electronic flight displays are used as ADI (EADI) and HSI (EHSI) displays. In 5-tube systems four EFD are required, two on each side of the cockpit. EFIS-85C(14) systems use the EFD-85() which has 5-inch by 5-inch color CRTs and EFIS-86C(14) systems use the EFD-86() which has 5-inch by 6-inch color CRTs. Following is a description of the information shown on the EFD in both the EADI and EHSI positions:

EADI (Electronic Attitude Director Indicator)

EADI displays show:

- aircraft attitude information

- flight control system mode annunciation
- lateral deviation (VOR, localizer, TACAN, MAZ, or LNAV)
- vertical deviation (glideslope, MGP, or VNAV)
- autopilot engage annunciation
- attitude comparator warnings
- decision height set and annunciation.

In addition, an EADI shows pilot selected information for specific functions or operating modes such as:

- flight control system steering commands
- attitude sensor source annunciation
- marker beacon annunciation
- radio altitude
- fast/slow or angle-of-attack deviation
- indicated airspeed with airspeed bugs and an airspeed trend vector
- mach speed
- altitude alert
- approach mode CAT II comparator warnings
- Cat II excessive deviation for ILS or MLS and airspeed.

Wiring straps, connected when the system is installed in the aircraft, select display functions such as:

- cross-pointer or V-bar command bars
- enable or inhibit of fast/slow or angle of attack display
- enable or inhibit a rising runway lateral deviation symbol
- enable or inhibit a composite mode display format for reversion display modes
- enable or inhibit the display of digital air data information.

EHSI (Electronic Horizontal Situation Indicator)

EHSI displays show the aircraft's horizontal navigation situation. In the EFIS-85C(14)/86C(14) system, an EHSI has three display modes: ROSE, SECTOR, and SECTOR with MAP (or MAP). In the ROSE display format the EHSI shows:

- compass heading and sensor type (true, magnetic, or DG mode)
- selected heading bug
- active selected course arrow, sensor annunciation and digital read-out
- active course lateral deviation bar and scale
- to/from indication
- distance to station/waypoint
- second course arrow and sensor annunciation for a second navigation sensor source
- second course deviation bar (with no deviation scale)
- approach mode and VNAV vertical deviation
- NAV data for ground speed, time-to-go, elapsed time and true airspeed
- weather radar target alert
- waypoint or DME identifier and waypoint alert
- "DR" and message ("MSG") alert from an LNAV or RNAV set as the active navigation sensor
- heading comparator warn flag
- approach mode comparator warn flags (when Cat II approach mode is selected)
- linear, angular, and cross-track deviation and back course localizer annunciation

- DME hold annunciation
- bearing pointer and sensor annunciator
- wind speed and direction.
- reversionary source annunciation

The SECTOR format shows all the same information as the ROSE format with the following differences.

- An expanded compass segment shows in place of the compass rose (shows approximately 80 degrees, 40 on each side of the lubber line).
- The aircraft symbol is located at the bottom center of the display.
- Second course NAV source shows as a symbol on the display in place of or with a second course arrow (depends on the version of DPU/MPU installed).
- Weather radar information shows on the display.

MAP format shows the same information as the SECTOR format with the following differences.

- Nav aids and waypoint symbols show on the display for the active course and the second course.
- Course lines show to and from VOR symbols.
- Idents or waypoint numbers show next to the DME, VOR, VORTAC and waypoint symbols.
- Waypoint strings connected with desired track lines to show a flight plan.

EFD-85() and EFD-86() displays also have an output for a cockpit mounted, installer supplied high temperature warning annunciator. This annunciator turns on when the internal temperature of the EFD reaches +95° C (+203° F) in still air.

MFD (Multifunction Display)

One MFD-85B or MFD-85C is mounted in the center of the panel. The information shown on an MFD depends on the associated equipment (LNAV, sensors, etc.) installed in the aircraft, the version of MPU installed and the version of MFD installed.

Typically the MFD-85B display shows:

- weather radar
- three selections of pictorial navigation maps: heading up, north up aircraft centered, and north up max view (a fourth map, plan view, is available from a compatible LNAV in some installations)
- remote data (up to three remote inputs for data from a compatible LNAV source)
- extended data pages (CAT II checklist and information available from compatible LNAV sources)
- and page and emergency data checklists.

MFD-85C shows all the same information of an MFD-85B with the added ability to show a TCAS traffic display and the associated TCAS traffic display options. TCAS traffic can show on the MFD with or without both radar and navigation information. Installations with an MFD-85C use the -003 or -004 versions of MPU and DPU to drive the EFIS displays and -011, -012, -013, or -014 versions of WXP-850A/85B to control the TWR-850.

Both the MFD-85B and MFD-85C have a high-temperature warning output that can be connected to a cockpit mounted, installer supplied annunciator. This annunciator turns on if the internal temperature of the MFD reaches +78°C (+172°F) in still air.

DCP (Display Control Panel)

DCP-85F/85H Display Control Panels control the information that shows on the EADI and EHSI displays. The DCP-85H control is the same as the DCP-85F except for the added selection for ADF sensors. In a 5-tube EFIS system, two DCPs are required, one on each side of the cockpit. On EHSI displays the DCP controls:

- display format selection (ROSE, SECTOR, or SECTOR with MAP)
- display range selection (when radar is not shown on the display)
- active course and navigation sensor selection
- bearing pointer selection
- weather radar
- second course information from a second navigation sensor
- and display brightness.

On EADI displays the DCP controls:

- decision height set
- radio altimeter test
- and display brightness.

CAUTION

Operating EFIS system displays at maximum brightness (DCP DIM controls fully clockwise) for extended periods of time may eventually result in a condition known as 'imprinting' on the displays. Imprinting is evidenced by the ability to "see" an image on the display when the display is turned off, or by "seeing" an image other than the one desired. This last condition occurs when, for example, a display that was used as an EADI is removed from service and reinstalled as an EHSI, or vice versa.

Proper management of the DCP-85F/85H DIM controls minimizes imprinting and helps to maximize the service life of EFD and MFD displays. We recommend that the intensity controls not be set to the maximum bright (full clockwise) position unless cockpit conditions (such as direct sunlight, etc.) require maximum brightness. Also, when maximum brightness is no longer required, adjust the DIM controls to lower the brightness of the displays to a comfortable viewing level.

There are other steps pilots can take to contribute greatly to extending the life of the displays and minimize the chances of imprinting. For example, adjust the DIM controls for minimum brightness (fully counterclockwise) while at the terminal or waiting for passengers or during other periods of activity on the ground when the EFIS is not in use.

CHP (Course Heading Panel)

CHP-85C/86C controls also select/set the information that shows on EADI and EHSI displays. CHP-85C controls have two CRS knobs, one for each side of the cockpit, and a common HDG select knob. CHP-86C controls have a single CRS knob and a single HDG knob for control of one side of the cockpit. In 5-tube system installations either one CHP-85C or two CHP-86Cs are installed for control of:

- course arrows and heading cursors.
- course direct to and heading sync.
- NAV data (ground speed, time-to-go, elapsed time and true airspeed) and timer functions.

DPU (Display Processor Unit)

A DPU processes sensor and control inputs and gives the necessary deflection, video signals, and power to the EFDs. A single DPU is capable of driving two EFDs with different deflection and video signals. Typically this includes one EADI and one EHSI on one side of the cockpit. DPUs also have a high-temperature warning output that can be connected to a cockpit mounted, installer supplied annunciator. This annunciator turns on if the internal temperature of the DPU reaches +100°C (+212°F) in still air.

There are a number of versions of the DPU that are used in the EFIS-85C(14)/86C(14) 5-tube system. Each later version includes a general update of the previously existing software in the unit. Many of the changes are very subtle and may not be seen visually while some vary the information that shows on the EADI, EHSI, or MFD. A very brief description of a visually identifiable feature for each version is listed below:

- -001 version DPU is the first version and the basic configuration.
- -002 (baseline) version DPU, the visually identifiable features shown on the displays are the addition, to the EHSI display only, of unique pseudo localizer and glideslope pointers for second course lateral and vertical deviation, and an LNAV heading bug.
- -003 version DPU added the ability to show TCAS traffic on an MFD-85C display (derived from the -002 baseline version).
- -004 version DPU added the ability to display GPS data for lateral and vertical deviation when received from a compatible FMS (derived from the -003 version).

As always, refer to the appropriate aircraft flight manuals for a complete list of the equipment installed in your aircraft.

MPU (Multifunction Processor Unit)

The MPU processes sensor and control inputs and gives the necessary deflection, and video signals to the MFD. The MPU can also give deflection and video signals to EADI and EHSI displays in cross-side data reversion modes in the event of failures to either or both DPUs. The MPU is similar to the DPU except the MPU has expanded input/output capabilities for a second set of aircraft sensors. The MPU also has a high-temperature warning output that can be connected to a cockpit mounted, installer supplied, annunciator. This annunciator turns on if the internal temperature of the MPU reaches +135°C (+275°F) in still air.

As with the DPU there are a number of versions of the MPU that are used in the EFIS-85C(14)/86C(14) 5-tube system. Each version includes a general update of the previously existing software in the unit which may include many changes. The version number for the MPU and DPU are always the same for a given installation. The features that identify the version of DPU installed in a system also identify the version of MPU. Refer to the description of the DPU above and to the appropriate aircraft flight manuals to determine the version of MPU installed in your aircraft.

WXP (Weather Radar Panel)

The WXP-850() is not part of the EFIS system but does provide the mode control, range selection, and other system operating controls to display weather radar information on the MFD and the EHSI displays. (See EFIS equipment list notes, front section, for detailed compatibility status information.)

OPERATING CONTROLS

DCP-85F/85H Display Control Panel Operation

The controls on the DCP-85() Display Control Panel select the information that shows on the EADI and EHSI displays. For the EHSI the DCP selects the display format and various operating functions. For the EADI, the DCP controls set the decision height and operate the radio altimeter self-test (RA TST). The DCP also controls display dimming for both the EADI and EHSI. Each of the controls on the DCP-85() are described in the shaded boxes below and on the following pages. An example of the associated information that shows on the display is also shown. The DCP-85F and -85H are the same except that the -85H has additional sensor selections for ADF.

To initialize the EFIS system in the same configuration with the same sensors selected as when power was turned off to the system, the DCP stores mode status information in its memory. The information stored includes:

- EHSI display format
- active, second and bearing navigation sensor selection
- display range
- weather radar display mode on or off
- EADI decision height setting
- and selected course, heading and NAV data from the CHP.

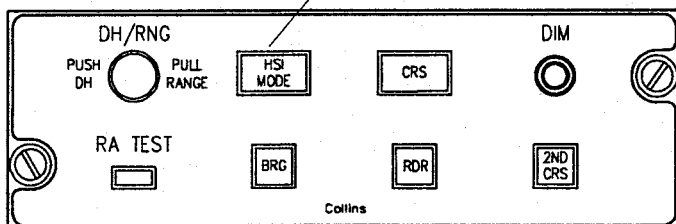
HSI MODE Button

Sequentially, push the HSI MODE button to step through the EHSI display format selections.

ROSE — The EHSI full compass rose format is displayed in a manner similar to a conventional HSI.

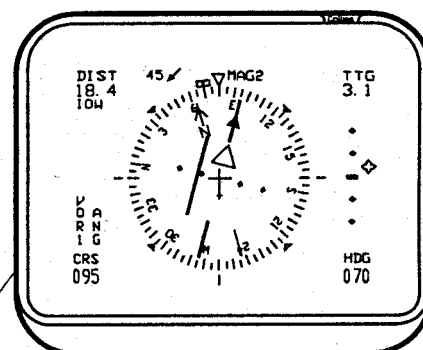
SECTOR — This format shows an expanded (approximately 80 degrees) compass segment across the top of the display with the airplane symbol centered at the bottom. Information similar to that presented in the full compass rose mode is shown in an enlarged, easily readable format. Weather radar information may also be displayed in this format.

SECTOR WITH MAP — This format also uses the expanded compass format but places VOR and/or waypoint symbols for both active selected course and preset second course in proper rho-theta position with respect to the airplane symbol and selected range. Weather radar information may also be displayed in this format.

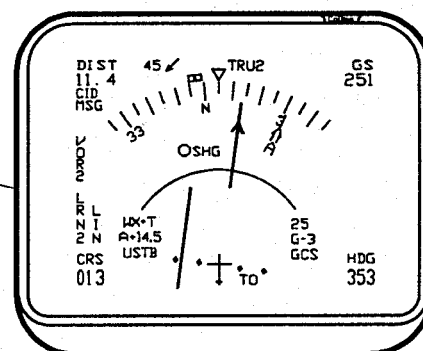


CGO-4753-11-AC-1

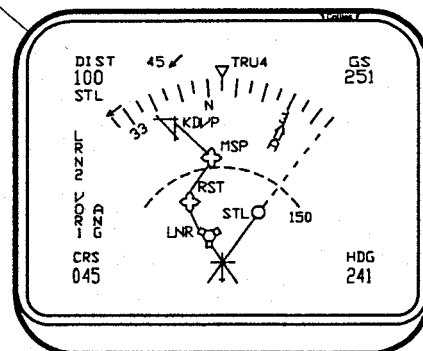
DCP-85F/85H Display Control Panel



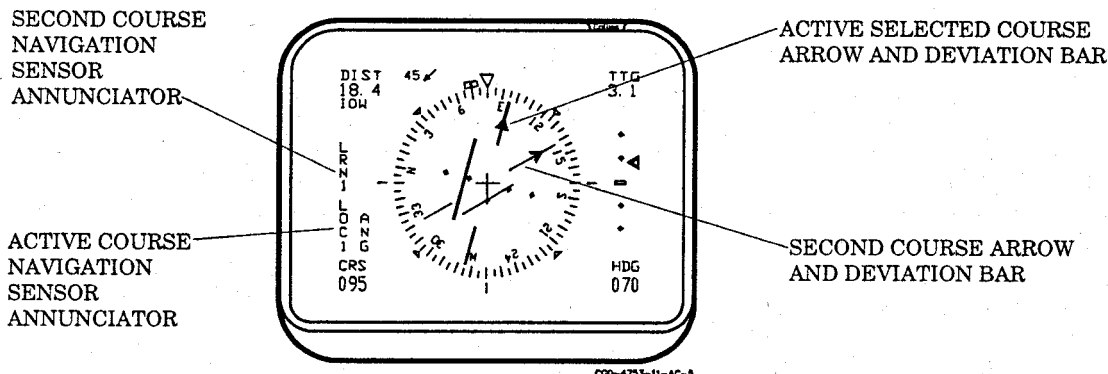
CGO-4753-11-AC-2



CGO-4753-11-AC-3



CGO-4753-11-AC-4



CRS (Course) Button

Sequentially push the CRS button to step through the selections of navigation sensors to be connected to the EHSI active selected course arrow and lateral deviation bar (solid double bar course arrow).

The possible selections depend upon interconnect wiring (both strapping and installer supplied switching). The possible selections are VOR1/LOC1, LRN1 (which may be an INS, VLF, LRN, LNV, GPS or FMS), MLS1, *ADF1, VOR2, LRN2, MLS2 and *ADF2 (*not available in TCAS versions). Holding the CRS button for longer than 1/2 second causes the course deletion to cycle directly to the primary VOR/ LOC system (VOR1/LOC1 for the pilot, and VOR2/LOC2 for the copilot).

The active selected course navigation sensor is annunciated in the lower left center of the EHSI.

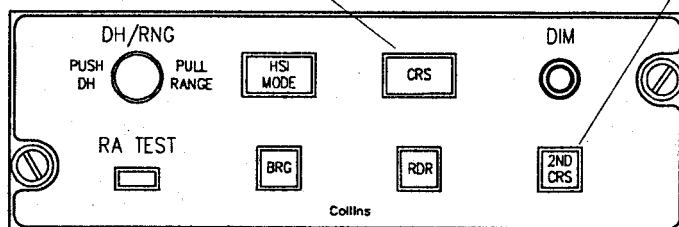
Active selected course data is stored if power is lost or if the system is turned off.

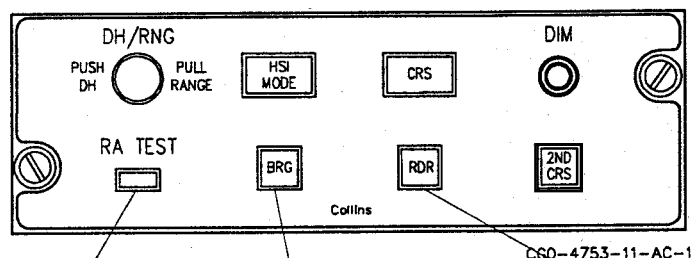
2ND CRS (Second Course) Button

Sequentially push the 2ND CRS button to allow information from a second navigation sensor to be displayed on the EHSI.

The second course arrow is displayed in cyan as a solid single bar course arrow. The possible selections depend upon interconnect wiring (both strapping and installer supplied switching). The applicable sensor is annunciated in cyan above the active selected course arrow sensor annunciator.

At power on, no second course arrow is displayed. Sequentially pushing the 2ND CRS button selects VOR1/LOC1, VOR2/LOC2, LRN1 (which may be an INS, VLF, LRN, LNV, GPS or FMS), LRN2 and off for the pilot's DCP; and VOR2/LOC2, VOR1/LOC1, LRN2, LRN1 and off for the copilot's DCP. MLS and *ADF can also be on the second course arrow. Holding the 2ND CRS button for longer than approximately 1/2 second causes the second course selection to cycle directly to the off state.





RA TEST (Radio Altimeter Test) Button

Push the RA TEST button to start the radio altimeter test function.

When the RA TEST button is pushed, the EADI displays the radio altitude that is output by the radio altimeter when it is in self-test (50 ft for ALT-50/55).

The DH annunciator shows on the EADI display during the radio altimeter test if the DH setting is greater than the test altitude.

BRG (Bearing) Button

Sequentially push the BRG button to step through the selections for number 1 (single arrow) and number 2 (double arrow) bearing pointers.

The possible selections depend upon strapping in the interconnect wiring. The possible selections and their sequence of selection are VOR1, VOR2, ADF1, ADF2, LRN1 (which may be an INS, VLF, LRN, LNV, GPS or FMS), LRN2 and off for the pilot's DCP; and VOR2, VOR1, ADF2, ADF1, LRN2, LRN1 and off for the copilot's DCP. VOR selections are removed from the sequence if the corresponding localizer is tuned. Holding the BRG button for longer than approximately 1/2 second causes the bearing selection to cycle directly to the "off" state.

Bearing pointer data is stored if power is lost or if the system is turned off.

RDR (Weather Radar) Button

Push the RDR button to add weather radar information to the EHSI when either of the sector formats are being displayed and WX, WX+T or TURB has been selected on the WXP-850 ().

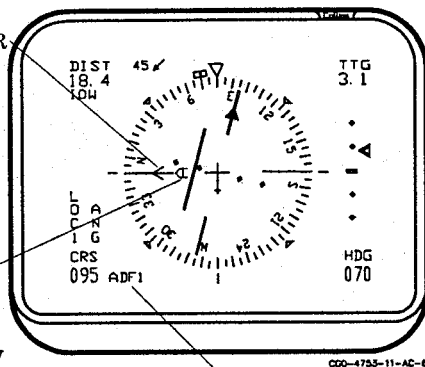
At power on radar data is not displayed. The first push of the RDR button adds weather radar information to the display. The second push removes the radar information.

When weather radar information is being displayed, range is controlled by the RANGE control on the WXP. One-half of the selected range appears at the right end of the solid cyan range arc and the weather radar mode appears at the left end of the range arc.

BEARING POINTER

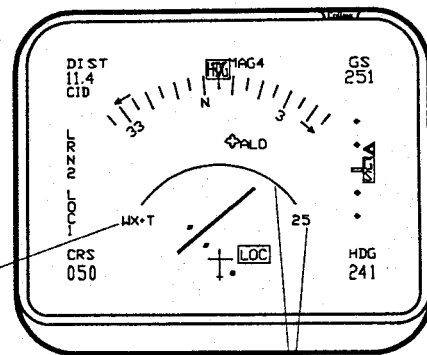
- #1 SENSOR IS A SINGLE SHAFT
- #2 SENSOR IS A DUAL SHAFT
- #3 SENSOR IS A TRIPLE SHAFT

- 'V' = VOR
- 'A' = ADF
- 'T' = TACAN
- 'N' = LNAV OR RNAV



BEARING POINTER ANNUNCIATOR
(REMOVED FROM VIEW AFTER
5 SECONDS)

WEATHER RADAR MODE



SOLID RANGE ARC AND
HALF-SCALE RANGE
ANNUNCIATION

DH/RNG (Decision Height/ Range) Knob

The dual function DH/RNG knob sets the decision height shown on the EADI and selects the range on the EHSI when in the sector formats and when weather radar (RDR button) is not selected.

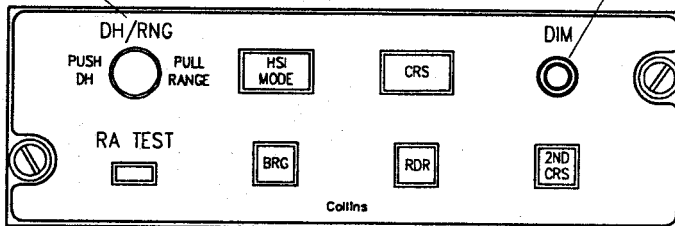
PUSH DH Position — When pushed in, the decision height altitude appears on the EADI, and rotating the knob changes the DH setting in ± 10 foot steps (clockwise rotation increases DH with maximum DH being 999 feet). A rapid turn of the knob gives a minimum 300 foot change in the DH. The DH display is removed when the DH/RNG knob is pulled out. It remains out of view until the knob is pushed in again, or until the aircraft descends through 1000 feet agl (radio altitude) at which time the DH display appears flashing for 10 seconds and is then removed from view.

PULL RANGE Position — When pulled out, rotating the knob selects range on the EHSI when either of the sector formats is being displayed and when RDR is not selected. A dashed cyan range arc is shown at midrange and one-half of the full scale range is displayed at the right end of the range arc. Available full scale ranges are 5, 10, 25, 50, 100, 200, 300 and 600 nmi.

DIM Knobs

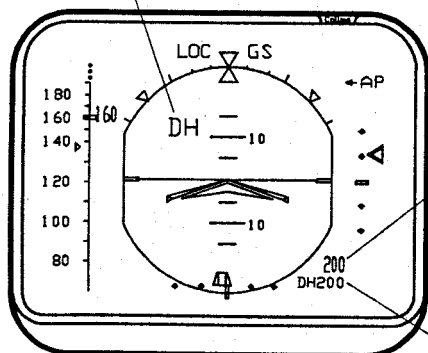
Rotate the DIM knobs to change the intensity of the EADI and/or EHSI.

The DIM control consists of a pair of concentric knobs to control each CRT's intensity. One knob controls the EADI; the other controls the EHSI. The knobs are sized such that they normally turn together, but may be turned independently to balance the displays. (See CAUTION on page 7 relating to operating the displays at maximum brightness.)



CGO-4753-11-AC-1

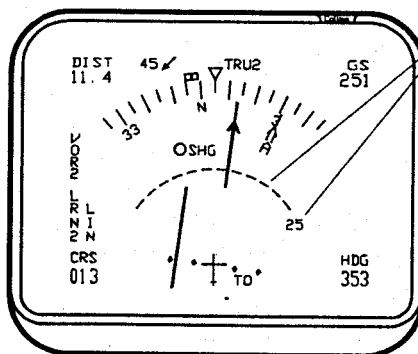
DECISION HEIGHT
ANNUNCIATOR



CGO-4753-11-AC-8

RADIO
ALTITUDE

DECISION
HEIGHT SET



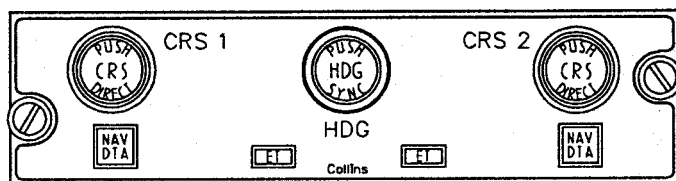
CGO-4753-11-AC-8

DASHED RANGE
ARC AND HALF-
SCALE RANGE
ANNUNCIATION

CHP-85C/86C Course Heading Panel Operation

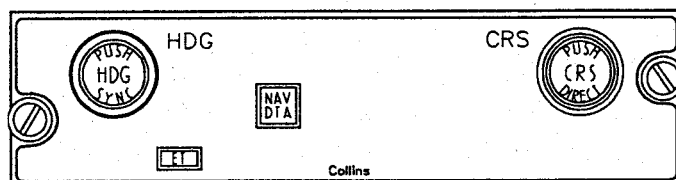
EFIS-85C(14)/86C(14) 5-tube system installations have one CHP-85C or two CHP-86C Course Heading Panel controls. Both CHPs control the same functions but in slightly different ways. The CHP-85C has independent course and NAV data controls for both sides (pilot and copilot) of the EFIS system and selected heading controls (HDG knob and PUSH SYNC button) that control both sides of the system at the same time. The CHP-86C independently controls all course, heading and NAV data selections on only one side of the EFIS system and two are required in 5-tube systems. Both types of CHP store the settings for the heading cursors, course arrows and NAV data selections in the DCP control for recall when power is turned on to the EFIS system.

Each of the controls on the CHP are described in the shaded boxes below and on the following pages. An example of the associated information that shows on the display is also shown. The CHP-85C is used to describe the functions and controls of both types of CHP. The differences for the CHP-86C are minor as described above and are noted as necessary in the text.



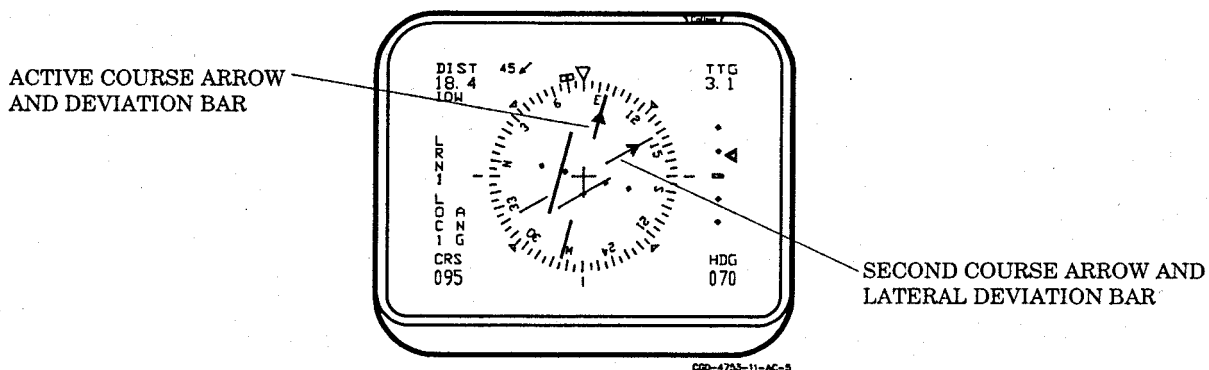
CGO-4753-11-AC-10

CHP-85C Course Heading Panel



CGO-4753-11-AC-11

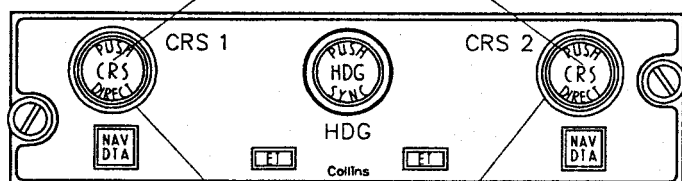
CHP-86C Course Heading Panel



PUSH CRS DIRECT (Course Direct To) Button

Push and release the PUSH CRS DIRECT button to automatically center the deviation bar with the course arrow pointing directly to the station.

On the EHSI display associated with the PUSH CRS DIRECT button, the selected active or second course arrow (selected by the appropriate DCP button) rotates to point to the station and the deviation bar centers.



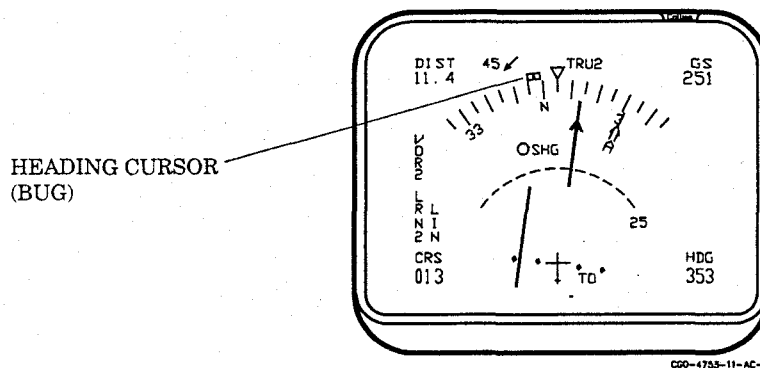
CGO-4753-11-AC-10

CRS (Course Select) Knob

Turn the CRS 1 knob (pilot's side) or the CRS 2 knob (copilot's side) on the CHP-85C to set active and second course arrows shown on the respective EHSI and the course lines shown on the MFD. For CHP-86C installations turn the CRS knob to set active and second course arrows on the CHPs' associated EHSI.

Course control knobs rotate continuously in either direction to set the position of active and second course arrows (and MFD course lines) for the NAV sensors as selected on the DCPs. The DCP's CRS button selects the active course arrow NAV sensor as does the 2ND CRS button for the second course arrow. The selected course value is stored if power is lost or if the system is turned off.

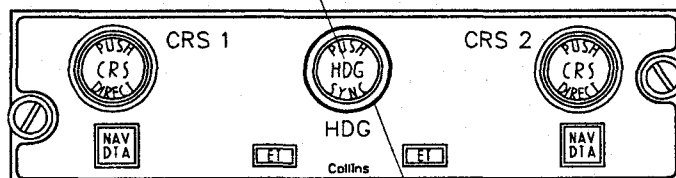
In installations with CHP-86C controls, the CRS knob on the left side (pilot's) CHP also sets the selected course line shown on the MFD display. (The right side CHP can control the MFD course line when control is transferred through cross-side switching.)



PUSH HDG SYNC (Heading Sync) Button

Push and release the PUSH HDG SYNC button to set the heading cursor to the aircraft's current heading shown at the lubber line on EHSI and MFD displays.

For CHP-86C installations the PUSH HDG SYNC buttons control the heading cursor for only the associated EHSI. Also, the PUSH HDG SYNC button on the pilot's side CHP controls the heading cursor on the MFD. (The right side CHP can control the MFD heading cursor when control is transferred through cross-side switching.)



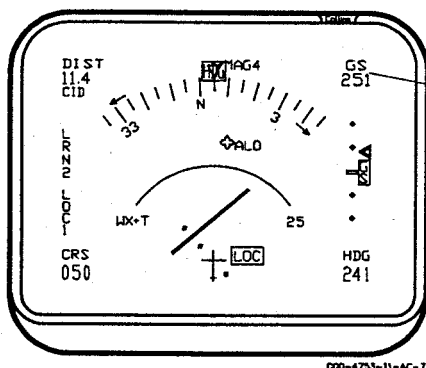
HDG (Heading Select) Knob

In installations with CHP-85C controls, turn the HDG knob to set the position of the magenta selected heading cursors on both the pilot and copilot EHSI and the MFD at the same time and to the same setting. In installations with CHP-86C controls, turn the HDG knob to set the selected heading cursor on only the associated EHSI display.

The HDG knob rotates continuously in either direction to set the position of the selected heading cursor(s).

In installations with CHP-86Cs, the HDG knob on the left side (pilot's) CHP also controls the magenta heading cursor on the MFD. (The right side CHP can control the MFD heading cursor when control is transferred through cross-side switching.)

In installations with later versions (-002 and later) of DPU/MPU and a compatible LNAV set to the heading mode, the HDG knob can also control an LNAV heading cursor. The LNAV heading cursor (a white box open on one side) and the magenta heading cursor move together when the HDG knob is turned.



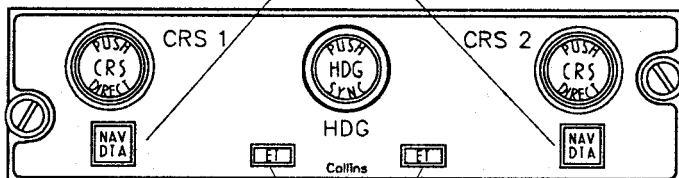
000-4753-11-AC-7

NAV DTA (Navigation Data) Button

Push the NAV DTA button to select ET (elapsed time), GS (ground speed), TTG (time-to-go), or TAS (true airspeed) to show in the upper right corner of the EHSI. Each push of the NAV DTA button selects the next data type in the sequential order as just presented.

TTG and GSP are taken from the navigation sensor driving the active selected course arrow. The TTG and GSP are dashed and the TAS display is blanked if valid data is not being received. FTTG and FGSP options come from an LNAV; but are available only if the active sensor is FLOC and the DME cannot compute the distance data. Installations with -001 version DPU and MPU do not show FTTG and FGSP data. ET is from an elapsed timer in the CHP.

In installations with the CHP-85C, the NAV DTA button on the left side of the unit controls the pilot's EHSI and the right side NAV DTA button controls the copilot's EHSI. For installations with the CHP-86C, the NAV DTA button controls the EHSI associated with the control.



CGO-4753-11-AC-10

ET (Elapsed Timer) Button

Push the ET button to operate the elapsed timer functions of the NAV data display.

In installations with the CHP-85C, the ET button on the left side of the unit controls the timer on the pilot's EHSI and the right side ET button controls the timer on the copilot's EHSI. For installations with the CHP-86C, the ET button controls the timer for the EHSI associated with the control.

The timer has three modes: start, stop and reset. Each time the ET button is pushed, the timer advances to the next mode in the sequence. After starting the timer, TTG, GSP or TAS may be selected with the NAV DATA button and the timer will continue to count.

ET is set to 00:00 when power is applied to the EFIS system.

WXP-850() Weather Radar Panel

The WXP-850() Weather Radar Panel controls the weather radar display information shown on the MFD display when it is set to the RDR display mode. The same information also shows on EHSI displays that are in the Sector or Sector with Map formats; and the RDR mode is selected on the DCP. Radar information shown on the displays includes: any precipitation detected by the radar (in four colors green, yellow, red, and magenta), a half-range arc, radar operating mode annunciators, and PAC alerts. The WXP also controls the range selection for the displays when they are set to a radar display mode.

Each of the controls for the WXP-850A/850B are described in the shaded boxes below and on the following pages. Also shown is an example of the radar information as it displays on an MFD-85B. The MFD-85C shows the same radar information as the MFD-85B, but includes a TCAS traffic display mode.

The WXP-850A shown has an SLV button. Units with an SLV button are used in installations with dual WXP controls. Installations with only one WXP use a version that does not have an SLV button. The WXP-850B (not shown) has the same controls as the WXP-850A, but some of them are positioned in different locations on the panel.

MODE Control

Turn the MODE knob to select the operating mode of the TWR-850 radar.

OFF—Turns off the TWR-850 system.

STBY—Turns on the TWR-850 system but does not allow it to transmit.

TEST—Turns on the radar self-test display.

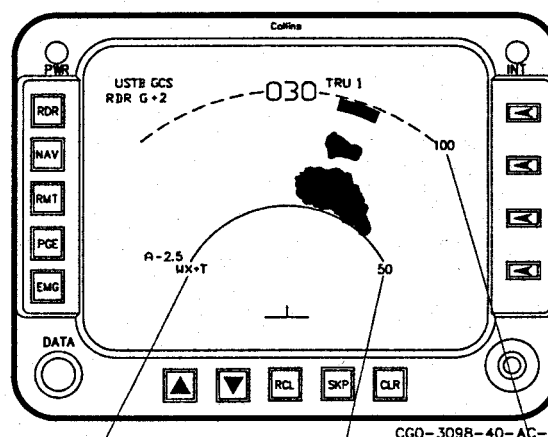
TGT—Turns on the radar receiver-transmitter to detect precipitation, but does not show any detected weather on the displays.

MAP—Turns on receiver-transmitter in the ground mapping mode.

WX—Turns on the receiver-transmitter in the basic weather detection mode. Detected precipitation may show as green, yellow, red, or magenta targets.

WX+T—Turns on the receiver-transmitter in the basic weather detection mode and also turns on the precipitation-related turbulence detection mode. Precipitation-related turbulence is detectable on only the 5, 10, 25, and 50 mile ranges.

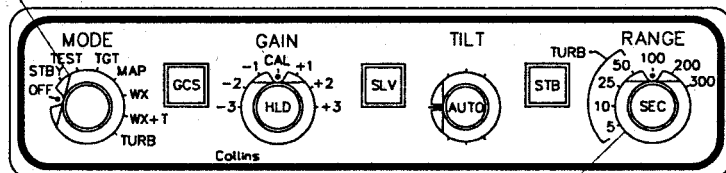
TURB—Turns off all targets except detected precipitation-related turbulence.



RADAR MODE
ANNUNCIATOR

HALF-RANGE
ANNUNCIATOR

FULL-RANGE
ANNUNCIATOR



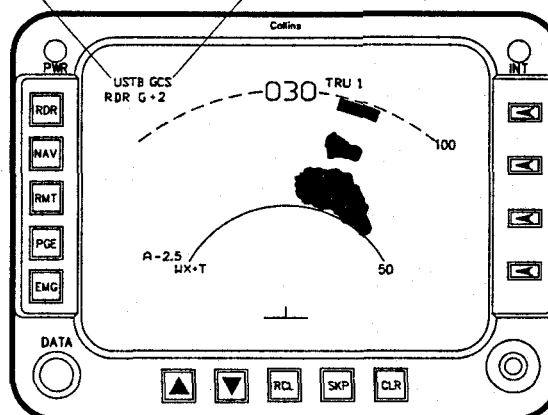
RANGE Control

Turn the RANGE knob to set the desired maximum display range.

Turbulence is detectable in only the 5, 10, 25, or 50 mile ranges. Display range is controlled by the WXP any time radar information shows on the display. The selected range is annunciated on the EFIS displays at the half-range and full-range scale marks. (Full-scale range is not annunciated on some EFIS displays).

STABILIZATION OFF
ANNUNCIATOR

GROUND CLUTTER
SUPPRESSION ANNUNCIATOR



CGO-3098-40-AC-2

SLV (Slave) Button

Push the SLV button on one WXP to select or deselect slaved operation.

The SLV button is a latching push on/push off button. Slaved mode is turned on when the button is in the in position and turned off when it is in the out position. Slaved operation and the SLV button exist only on versions of the WXP that are used in installations with dual WXP controls.

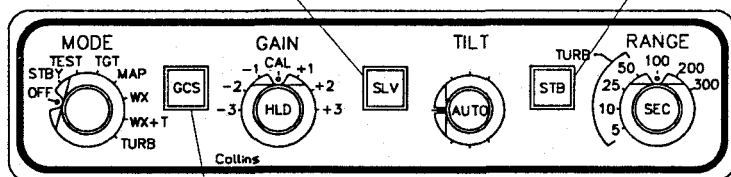
Selecting slaved operation on one WXP causes the EFIS display on that side to show the mode, range, tilt, and gain selected by the other WXP. All radar annunciators, except the half-range arc, change to yellow to indicate cross-side control. If the WXP's on both sides of the cockpit are set to the slaved mode, the radar system is automatically set to the standby mode, and the STBY mode annunciator shows on the EFIS displays.

STB (Stabilization) Button

Push the STB button to turn on or off the radar antenna stabilization circuits.

The STB button is a latching push on/push off button. Stabilization is turned on when the button is in the in position and turned off when it is in the out position.

Normal operation is with STB turned on (STB button pushed in). Antenna stabilization is turned off in the event of an attitude input signal failure to the radar system. When STB is turned off, the annunciator USTB shows on the EFIS displays.



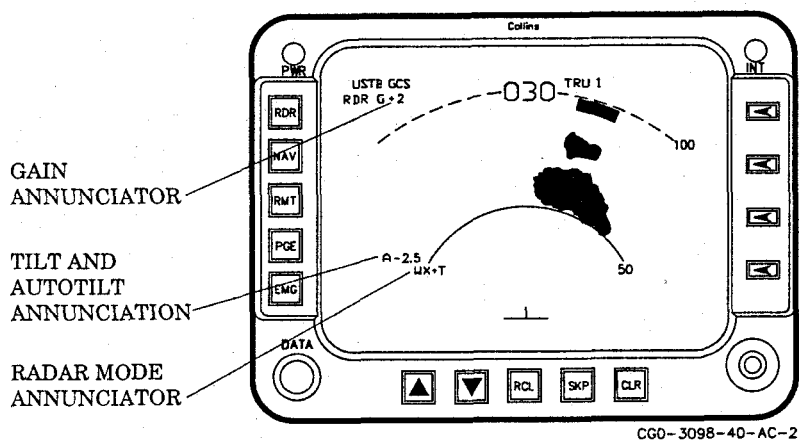
CGO-3018-11-AC-10

GCS (Ground Clutter Suppression) Button

Push the GCS button to instantly reduce the intensity of ground returns.

The GCS button is a push on/push off momentary button. Each push of the button toggles the GCS mode on or off.

When turned on, GCS separates precipitation returns from ground returns making the precipitation easier to see. GCS may also reduce the intensity of, or completely eliminate, weaker precipitation returns. For this reason, **do not leave GCS turned on continuously**. (Some systems automatically turn off GCS after approximately 10 seconds.) GCS operates in only the WX, WX+T or TURB modes. When turned on, "GCS" is annunciated on the EFIS displays.

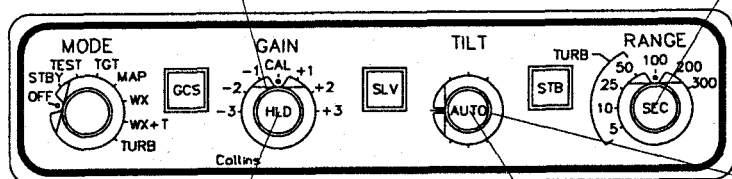


CGO-3098-40-AC-2

GAIN Control

Turn the GAIN knob to increase or decrease the gain from the CAL (calibrated) setting.

Normal system operation is with the knob in the CAL position. Settings other than CAL are annunciated on EFIS (i.e. G+2, G-3, etc.).



CGO-3018-11-AC-10

HLD (Hold) Button

Push the HLD button to "freeze" the display for detailed analysis of targets.

The HLD button is a push on/push off momentary button. Each push of the button toggles the display from normal operation to a display hold mode.

The display hold mode stays on until the HLD button is pushed to turn it off, or until the operating mode, display range, gain setting, GCS on or off, SLV on or off, or SEC on or off is changed, or until the mode switch is turned OFF. Targets shown on the EFIS displays are not updated in the display hold mode. The display hold mode is annunciated on EFIS as "HOLD" alternating with the selected mode (i.e., HOLD-WX-HOLD-WX etc.).

AUTO (Autotilt) Switch

Pull out on the AUTO knob to select the autotilt feature.

Autotilt is designed to reduce pilot workload. First the pilot uses the TILT control to adjust the antenna to the desired tilt angle for the present range and flight altitude. After this is done, select autotilt by pulling out on the AUTO knob. With autotilt selected, changing ranges or altitudes causes the antenna to automatically tilt up or down to maintain the same ratio of tilt angle to range that were present prior to selecting autotilt. Annunciation is on EFIS with an "A" placed before the tilt angle (i.e., A+5.7, etc.).

SEC (Sector Scan) Button

Push the SEC button to reduce the normal 120 degree scan to a 60 degree scan (30 degrees either side of center).

The SEC button is a push on/push off momentary button. Each push of the button toggles the sector scan mode on or off.

Reducing the sector scan increases the radar display update rate. When dual WXP's are installed, either WXP can turn on or off the SEC scan mode. When sector scan is turned on, all EFIS displays in the radar display mode show the reduced sector scan display. The SEC display mode stays on until turned off by a push of the SEC button. The SEC display mode is identified by the reduced size of the range arcs.

TILT Control

Turn the TILT knob to manually set the tilt angle of the radar antenna.

Tilt can be set at each quarter of a degree to a maximum of ± 15 degrees. However, tilt settings are annunciated on EFIS only to one decimal point (i.e., +10.5, -5.2, etc.). The second digit after the decimal point (either a 5 or a 0) is not shown.

MFD-85() Multifunction Display Operation

The MFD is a multicolor CRT display unit that mounts in the center of the instrument panel. The MFD normally receives its display information from the MPU, except when reversionary display modes are selected by optional, installer supplied, cockpit mounted reversion switches. Standard information shown on the MFD includes: weather radar, three or four selections of pictorial navigation maps (heading up, north up with aircraft centered, north up max view, and map plan if a compatible LNAV is installed), and page data. Up to three remote data inputs also show information on the MFD, when a compatible LNAV is installed.

Some of the information available on the MFD is dependent on the equipment installed in the aircraft and the aircraft wiring. Not all of the features described may be in any given installation. Refer to the appropriate aircraft flight manuals for specific information for your aircraft.

This pilot's guide describes the operation of two versions of the MFD Multifunction Display, the MFD-85B and MFD-85C. The MFD-85C has all the capabilities of the MFD-85B plus the added capability to show TCAS traffic displays. Controls for the MFD-85C are similar to the MFD-85B with the exception that the NAV and RDR display modes are selected from the same button and the TFC button is added for selection of the TCAS traffic display mode. Also, when the TFC display mode is selected, additional TCAS display functions are selectable via the line select keys along the right side of the MFD. The following paragraphs describe the controls first for the MFD-85B and then the MFD-85C.

MFD-85B Controls

Each of the controls on the MFD-85B are described in the shaded boxes below and on the following pages. All of the controls, except the PWR button, INT knob, and the Joystick, are momentary type switches for the selection of display modes and options.

PWR (Power) Button

Push this button to turn the MFD on or off.

The PWR button is a latching push on/push off type switch. The MFD is turned on when the button is pushed in and off when it is in the out position.

The MFD stores the selected display mode(s) and options in memory when power is turned off. When power is turned on, the MFD shows the stored display mode(s) and options.

WARNING

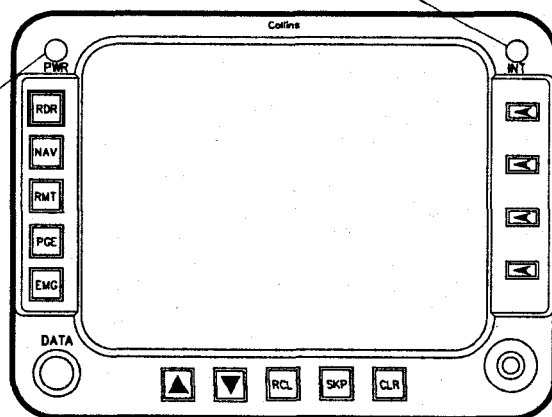
Turning off power on the MFD does not turn off power to weather radar systems and does not put radar systems in a non-transmitting mode. The WXP controls the radar receiver-transmitter operating modes. For installations with a WXP-850() the radar transmits when the MODE switch is set to the TGT, MAP, WX, WX+T or TURB modes.

INT (Intensity) Control

Turn the INT fingertip knob to adjust the MFD display brightness.

Turn the knob counterclockwise to dim the display and clockwise to brighten the display.

(See CAUTION on page 7 about operating displays at maximum brightness.)



MFD-85B

CGO-3098-40-AC-1

RDR (Radar) Button

Push and release this button to turn on or off the radar (RDR) display mode.

The RDR and NAV display modes can be turned on at the same time. When any other display mode is selected (RMT, PGE or EMG) the display goes directly to that mode. Also, from the RMT, PGE or EMG display modes, pushing the RDR button changes the display directly to the RDR mode.

NAV (Navigation) Button

Push and release this button to turn on or off the navigation (NAV) display mode.

The NAV and RDR display modes can be turned on at the same time. When any other display mode is selected (RMT, PGE, or EMG) the display goes directly to that mode. Also, from the RMT, PGE or EMG display modes, pushing the NAV button changes the display directly to the NAV mode.

RMT (Remote) Button

Push and release this button to show the remote data source line-key menu for the line select keys.

When the RMT button is pushed, the MFD goes directly to the remote line key menu display mode. To exit the RMT display mode, push any other display mode button. The MFD changes directly to the selected mode.

PGE (Page) and EMG (Emergency) Buttons

Push and release these buttons as necessary to show page or emergency data information.

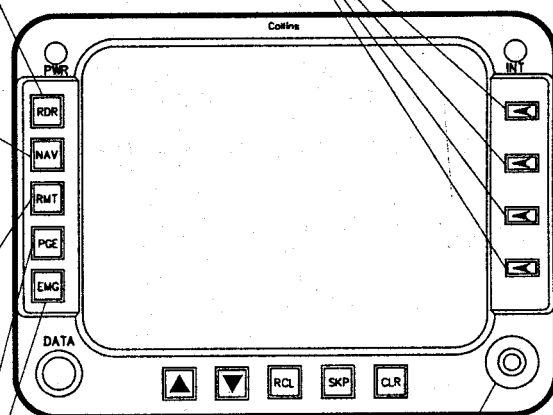
When a PGE or EMG button is pushed the MFD goes directly to the line-key menus or display pages for the data that is stored in the MFD. To exit PGE and RMT display modes, push any other display mode button. The MFD changes directly to the selected mode.

Up to 100 pages of data are available for checklists and emergency use. These buttons are also used to load page and emergency data through the data jack on the MFD.

Display Line Keys

Push the necessary line key to select or deselect display options or modes from the display menus that show to the left of the keys.

Line-Key menus show for the NAV, RMT, PGE and EMG display modes.



Joystick

The "joystick" is a multiple position switch that is used in NAV, RMT, PGE and EMG mode operations.

In the NAV menu mode, move the joystick as indicated to select the number of NAVAIDS and airports to show on the display in the NAV mode.

In NAV mode, move the joystick as necessary to position a waypoint on the MFD display that can be transferred to compatible LNAV systems.

In RMT mode, move the joystick as necessary to slew through pages or chapters of data of a compatible remote system.

NOTE

5- and 600-nmi ranges are also available with full-range arc annunciations of 5- and 600-nmi, and half-range arc annunciations of 2.5- and 300-nmi respectively, when the RDR mode is not selected. When RDR is selected and the MODE switch on the WXP is in STBY, TEST, TGT, MAP, WX, WX+T or TURB; the radar RANGE control has priority over the navigation map range selected with the line advance and reverse buttons.

Line Reverse (Delta) Button

Push this button to move the cursor back (up the list) in the PGE or EMG modes or to increase NAV map display range when RDR is not selected.

Any time that radar information shows on the MFD the display range is controlled by the WXP.

Line Advance (Inverted Delta) Button

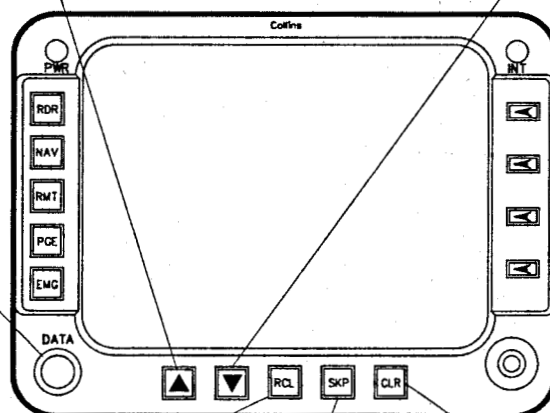
Push this button to move the cursor ahead (down the list) in the PGE or EMG modes or to decrease NAV map display range when RDR is not selected.

Any time that radar information shows on the MFD the display range is controlled by the WXP.

DATA Jack

Input port for remote programming, indexing, and revision of page and emergency data.

The Collins CEU-85 or CEU-85A Checklist Entry Unit or the RDP-300 Remote Data Programmer plug into the DATA jack to load or revise page and emergency data.



RCL (Recall) Button

Push the RCL button to view previously skipped lines when in the PGE or EMG mode.

If in PGE or EMG mode and NAV or RMT is selected, then pushing RCL immediately returns the display to the line in the PGE or EMG chapter that was being displayed prior to selecting NAV or RMT.

SKP (Skip) Button

Push the SKP button to move the cursor past a line in the PGE or EMG mode without changing its color.

Skipped lines are recallable at any time prior to chapter exit.

CLR (Clear) Button

Push the CLR button to reset all lines of a selected chapter to yellow in the PGE or EMG mode.

MFD-85C Controls

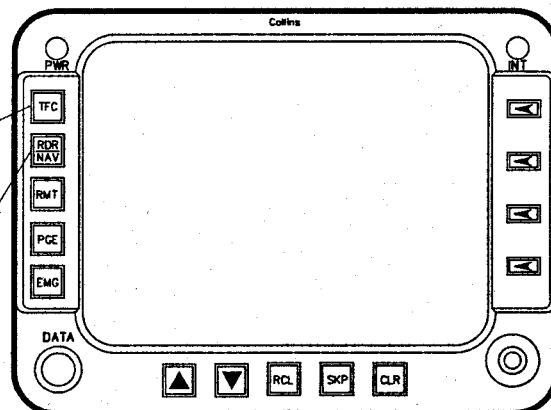
The MFD-85C controls the NAV and RDR display modes from one mode button ($\frac{\text{RDR}}{\text{NAV}}$ button) and has an additional mode button for TCAS (TFC button). All other controls are the same as the MFD-85B. Operation of the controls on the MFD-85C is essentially the same as that of the MFD-85B. Only the controls and operations that are different from the MFD-85B are described in this section. Refer to the MFD-85B control descriptions, for information about the controls that are not described in this section.

TFC (Traffic) Button

Push the TFC button to turn on the TCAS traffic display mode and/or the TCAS line-key menu along the right side of the display.

When the MFD is in any display mode other than the TFC display mode, pushing the TFC button changes the MFD directly to the TFC display mode. Pushing the TFC button or any line key when the MFD is in the TFC display mode, turns on the TFC display line-key menu. The TFC line-key menu turns off after approximately 5 seconds of no selection activity.

To exit the TFC display mode, push any other display mode button to change to that mode.



CGO-3098-40-AC-22

$\frac{\text{RDR}}{\text{NAV}}$ (Radar and Navigation) Button

Push the $\frac{\text{RDR}}{\text{NAV}}$ button to turn on and off the radar (RDR) and navigation (NAV) display modes.

Each push of the $\frac{\text{RDR}}{\text{NAV}}$ button cycles the MFD through a sequence of radar and navigation display modes. The first push of the $\frac{\text{RDR}}{\text{NAV}}$ button turns on the RDR display mode; the second push turns on the NAV mode with the RDR mode; the third push turns off the RDR mode and leaves the NAV mode on; and a fourth push of the button turns off the NAV mode and turns on the RDR mode. Push any other display mode button, at any time, to exit the currently selected RDR and/or NAV display mode(s) and change the MFD directly to that mode. Selection of any other mode causes the MFD to store, in memory, the currently selected NAV and/or RDR display mode(s) and display options. From any other display mode, push the $\frac{\text{RDR}}{\text{NAV}}$ button at any time to return to the last selected RDR and/or NAV display mode(s).

WARNING

Changing the MFD display from the RDR display mode or turning off the RDR display mode does not turn off or change the operating mode of the radar receiver/transmitter. The WXP controls the weather radar receiver/transmitter operating modes. For installations with a WXP-850() the radar transmits when the MODE switch is set to the TGT, MAP, WX, WX+T or TURB modes.

NOTE

Any time that the RDR mode shows on the display, with or without the NAV mode, the WXP controls the display range.

EFIS-86C(14) Display Formats

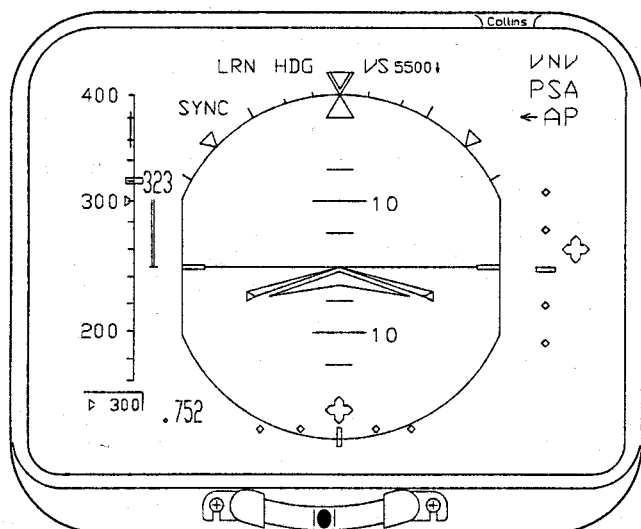
EADI Display Formats

This section describes the functions of the EADI displays that are used in the EFIS-86C Electronic Flight Instrument System. The EADI has one basic format and shows the familiar attitude display with lateral and vertical computed steering signals needed to give the commands to intercept and maintain a desired flight path. Also presented are FCS mode annunciation, autopilot engage status, attitude comparator warnings and decision height set. Other parameters are displayed as a function of pilot selection or operational mode. The EADI provides the following display information:

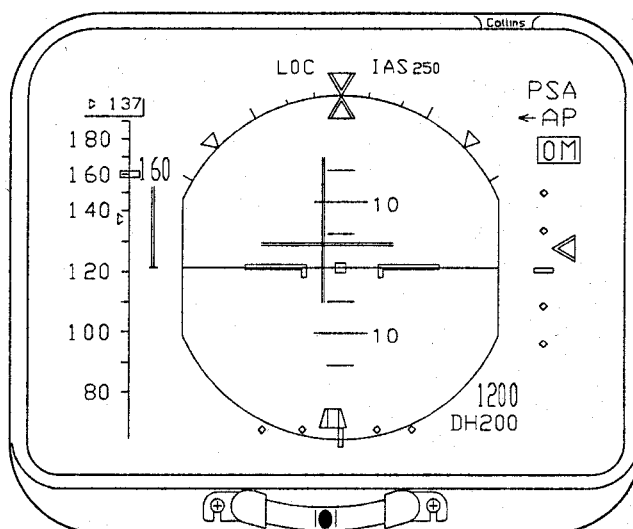
- Attitude display and sensor annunciation (ATT1 or ATT2, ATT3 or ATT4)
- Attitude comparator warnings
- Flight director commands and FCS mode annunciation
- Autopilot/yaw damper engage status
- Primary vertical deviation from GS, MGP, PGS (LNAV pseudo glideslope), FGS (LNAV controlled glideslope) or VNAV sensors
- Second course related vertical deviation from PGS or FGS
- Lateral deviation from VOR, LOC, PLOC (LNAV pseudo localizer), FLOC (LNAV controlled localizer), MAZ, TCN, or LNAV (FMS, RNV, LRN, etc) sensors
- LOC/GS or MAZ/MGP deviation displayed only when a LOC/MLS frequency is tuned and not in back LOC/MLS. LOC/MAZ deviation is displayed in back LOC/MAZ mode, GS/MGP is not displayed
- Second course related lateral deviation from PLOC or FLOC
- LNAV (GPS) "SCALING" annunciation for GPS vertical and lateral deviation data
- LNAV APPR mode that allows a compatible LNAV to control pitch commands (if available) and bank commands during approach
- Pseudo APPR mode that allows a compatible LNAV to control pseudo LOC and GS deviations
- CAT II excessive deviation and comparator warning for ILS, indicated airspeed, and radio altitude
- Radio altitude is displayed only when the radio altimeter system is within range or in self-test mode
- Decision height set and DH annunciation
- Air data information (IAS, airspeed trend, reference airspeed bugs, MACH)
- Speed deviation from a fast/slow or an angle-of-attack system
- Marker beacon
- Flags

In addition, interconnect wiring straps allow several EADI format changes to be made. These are:

- V-bar commands or cross-pointer commands (and style of airplane symbol)
- FCS lateral modes may be shown on the left and vertical on the right, or they may be swapped (lateral on the right, vertical on the left)
- Inhibit speed deviation display
- Rising runway symbol or no rising runway symbol
- Autopilot mistrim annunciator display
- Composite format or no composite format in reversionary mode
- Inhibit display of digital air data parameters



CGO-4755-10-AC-1



CGO-4755-10-AC-2

EADI WITH V-BARS

EADI WITH CROSS-POINTERS

EADI Displays and Annunciators

Attitude Display—Pitch attitude is indicated by the position of the blue sky and brown ground area containing the pitch scale with reference to the airplane symbol. Pitch attitude marks are in 5-degree increments up to ± 30 degrees. Expanded markings are shown from there up to ± 90 degrees.

Roll attitude is indicated by the position of the moving roll attitude pointer with reference to the fixed roll attitude scale. Roll attitude reference marks are located at 0, 10, 20, 30, 45, and 60 degrees on the roll attitude scale.

Aircraft Symbol—This symbol is a stationary representation of the aircraft. Aircraft pitch and roll attitudes are shown by the relationship between the aircraft symbol and the movable pitch and roll attitude display. The aircraft is maneuvered so that the aircraft symbol is 'flown into' the command bars until the two are aligned to satisfy the commands of the selected flight director mode.

Command Bar Display—The magenta command bars are in view when the flight director is being used. The command bars are out of view when the flight director is turned off or flagged, or when the attitude is extreme.

The command bars display computed steering commands to capture and maintain a desired flight path. The command bars move up or down to command a climb or descent, and turn right or left to command a right or left bank. V-bar or cross-pointer command bars are provided as an EFIS system strapping option.

Inclinometer—The inclinometer or slip indicator consists of a weighted ball in a liquid-filled curved tube. It is attached to the lower front of the EADI, can be either lighted or unlighted, and is used as an aid to coordinated maneuvers.

SELECTED ATTITUDE SENSOR	ANNUNCIATION SHOWN ON PILOT'S EADI	ANNUNCIATION SHOWN ON COPILOT'S EADI
P=#1, C=#2	None	None
P=#1, C=#1	ATT1 (yellow)	ATT1 (yellow)
P=#2, C=#2	ATT2 (yellow)	ATT2 (yellow)
P=#3, C=#2	ATT3 (white)	None
P=#3, C=#3	ATT3 (yellow)	ATT3 (yellow)
P=#1, C=#3	None	ATT3 (white)
P=#1, C=#4	None	ATT4 (white)
P=#4, C=#4	ATT4 (yellow)	ATT4 (yellow)
P=#3, C=#4	ATT3 (white)	ATT4 (white)

NOTE: The first five lines apply to both three and four sensor installations.

Attitude Sensor Annunciation—Attitude sensor annunciation is displayed in the right center portion (to the left of the vertical deviation scale) of the EADI when any attitude sensor except the primary sensor is selected. A maximum of four attitude sensors, either gyro, ARINC 429 or AHRS (attitude heading reference system) may be installed with the EFIS-86C system.

The pilot's primary sensor is #1 and his secondary sensor is #3. The copilot's primary sensor is #2 and his secondary sensor is #3 with three attitude sensors installed, or #4 with four attitude sensors installed. The above table shows combinations of sensors and annunciations. In the first column, 'P' stands for the pilot's attitude sensor, and 'C' stands for the copilot's attitude sensor.

Flight Control Mode Annunciation—The FCS (flight control system) vertical and lateral modes are annunciated along the top of the EADI. Vertical modes are shown to the right of the lubber line and lateral modes are shown to the left of the lubber line. The active vertical and lateral modes are shown in green and the armed modes and submodes in white. The vertical mode commanded reference data are shown in green and the advisory reference data are shown in white when available.

When a new active mode is displayed, either by being selected or by automatic capture, the mode annunciation flashes for 5 seconds before becoming steady.

Autopilot/Yaw Damper and Trim/Sync Annunciation

autopilot/yaw damper annunciation—The autopilot and yaw damper annunciation is displayed at the upper right of the EADI. The autopilot and yaw damper engaged annunciation is 'AP' and 'YD' for a single flight control system, and 'AP →' or '←AP' and 'YD' for a dual flight control system. The engage annunciators are green.

If a disengagement occurs (for both the autopilot and yaw damper), the 'AP' turns yellow (or red if strapped on the DCP) and flashes. Upon acknowledgment by the crew (through the FCS), the engage messages disappear for a single FCS, or the 'AP' and 'YD' disappear and the arrow turns white and becomes steady for a dual FCS.

trim and vertical sync annunciation—The trim annunciation is displayed at the upper left of the EADI. The trim annunciation is a boxed message in yellow that indicates the axis that is mistrimmed (A = aileron, E = elevator and R = rudder). The -003 and later versions of the DPU and MPU have a strapping option for R = roll, P = pitch and Y = yaw annunciation. The annunciation flashes for 5 seconds each time a mistrim occurs and is removed from view immediately when the input is removed.

The annunciation may also be an unboxed 'SYNC', or no message at all (blank). The 'SYNC' message is a flight director function that overrides the trim messages on the side (pilot's or copilot's) that the SYNC button is actuated.

Marker Beacon Display—Marker beacon passage is indicated by a box with 'OM', 'MM', or 'IM' inscribed appearing at the upper right side of the EADI. Outer marker is cyan, middle marker is yellow, and airway or inner marker is white. All marker beacon annunciations flash when they are displayed.

Vertical Deviation Display

A scale and pointer on the right side of the EADI indicates deviation from a selected vertical flight path. The scale consists of two white dots above and below a yellow center index. The aircraft is below the selected vertical flight path if the pointer is deflected upward. The EFIS-86C system can simultaneously display vertical deviation from primary sensors and second course related sensors depending on DPU/MPU status configurations, selected nav sensor and strapping options.

primary vertical deviation sensors—One of five primary vertical deviation sensors may be displayed. These are: GS, MGP, PGS (LNAV pseudo glideslope), FGS (LNAV controlled glideslope), and VNAV. The deviation pointers for the primary vertical deviation sensors are always displayed to the right of the vertical deviation scale. The primary deviation pointers for GS, MGP, PGS, and FGS are dual green triangles for on-side sensors, and dual yellow triangles for cross-side sensors. The VNAV pointer is a green star-shaped symbol for an on-side sensor and yellow if cross-side.

When GS is the sensor and back localizer mode is selected, the glideslope scale is removed from view and 'B/C' is annunciated vertically to the right of the active course sensor annunciation. The glideslope scale and pointer are removed from view when operating in back course conditions during an LNAV approach.

When LNAV pseudo glideslope is the sensor, the letters 'PGS' are displayed vertically to the right of the top two dots on the vertical deviation scale. When LNAV controlled glideslope is the sensor, the letters 'FGS' are displayed.

Vertical deviation scaling varies with the data source/sensor, LNAV mode and autopilot mode selections.

- For GS, MGP, FGS, and PGS, each of the enroute inner deviation scale dots represent 1/4-degree displacement and each of the outer dots represent 1/2-degree displacement from the vertical flight path.
- For VNAV deviation in -001 version DPU/MPU, each of the inner deviation scale dots represent 200 feet of displacement and each of the outer dots represent 400 feet of displacement from the vertical flight path.
- For VNAV deviation in -002 and -003 version DPU/MPU, each of the inner deviation scale dots represent 500 feet of displacement and each of the outer dots represent 1000 feet of displacement from the vertical flight path.
- For VNAV deviation from an LNAV set to FMS approach mode, -002 and -003 versions DPU/MPU each of the inner deviation scale dots represent 100 feet of displacement and each of the outer dots represent 200 feet of displacement from the vertical flight path.
- For the -004 DPU/MPU version, if the LNAV is using GPS vertical navigation data, the FMS will send the EFIS a scaling word which can range from 1024 feet per dot enroute to 1.00 foot per dot in approach.

second course related vertical deviation sensors—One of two second course related vertical deviation sensors may be displayed. These are: PGS (LNAV pseudo glideslope) and FGS (LNAV controlled glideslope). The deviation pointers for these sensors are cyan triangles and are always displayed to the left of the vertical deviation scale with the apex of the triangles pointing to the right.

The -001 versions of DPU and MPU can show 2ND CRS vertical deviation on the EADI and EHSI at the same time. The -002 and later versions DPU/MPU show only the active primary vertical deviation on the EADI and the 2ND CRS vertical deviation shows on the EHSI.

Radio Altitude Display—Radio altitude is displayed in the lower right corner in green digits. The radio altitude display appears automatically when the radio altitude system is within range (2500 feet with the ALT-55 or standard ARINC radio altimeter, or 2000 feet with the ALT-50) and changes in 50-foot increments above 1000 feet, and 10-foot increments below 1000 feet.

Radio Altitude Test—When the RA TEST button on the DCP is pushed, the radio altimeter is placed in self-test and outputs a fixed value. If the decision height happens to be set higher than this value, the DH annunciator turns on during the test.

Decision Height Display—Decision height up to a maximum of 999 feet is displayed in the lower right corner of the display with the letters 'DH' to the left of the digits. This indication is cyan in color if set from the on-side DCP or yellow if from the cross-side DCP.

If the DH/RNG knob on the DCP is pushed in, the decision height display is in view continuously and may be changed by rotating the DH/RNG knob. A rapid turn of the knob gives a minimum 300 -foot change in the DH. If the DH/RNG knob is pulled out, the decision height display is removed from view. It remains out of view until the knob is pushed in again, or until the aircraft descends through 1000 feet agl (radio altitude) at which time the DH display appears flashing for 10 seconds and is then removed from view.

Decision Height Annunciator—When decision height is reached during descent, the yellow letters 'DH' appear in the upper left center portion of the attitude display, flash for 5 seconds, and then become steady.

The DH annunciator is off when the strut switch is compressed (weight-on-wheels), and for 20 seconds after the strut switch is extended. Also, the DH annunciator remains off if DH is set to zero or turns off if the radio altimeter fails.

Lateral Deviation Display

A scale and pointer in the bottom center of the EADI indicates deviation from a selected lateral navigation path. The scale is the same for all sensor types and consists of two white dots on either side of a white center index. Various deviation pointer symbols move laterally along the deviation scale. All on-side sensor pointers show in green and cross-side sensor pointers show in yellow. The EFIS-86C system can simultaneously display lateral deviation from primary sensors and second course related sensors.

primary lateral deviation sensors—The primary lateral deviation pointer may be driven by VOR, LOC, MAZ, PLOC (LNAV pseudo localizer), FLOC (LNAV controlled localizer), TCN, or LNAV (FMS, LNV, RNV, LRN, VLF, INS, GPS, etc) sensors. The scaling for lateral deviation changes, as follows, depending on the selected sensor, autopilot mode, and LNAV mode:

- For all angular sensors (VOR, TCN, RNV, etc.) the scale is 5° per dot.
- For VOR sensors in linear mode the enroute scale is 5 nmi per dot.
- For VOR sensors in linear mode with an autopilot in approach mode the scale is 1 nmi per dot.
- For an LNAV sensor the enroute scale is 3.75 nmi per dot.
- For an LNAV in approach mode the scale is 1 nmi per dot.
- For GPS (-004 DPU/MPU), the FMS will send the EFIS a scaling word which can range from 8 nmi per dot enroute to 0.015 nmi per dot in approach. The EFIS limits the scale factor at the upper and lower bounds.

In the LNAV approach mode, the annunciator 'APPR' shows just above the left side of the lateral deviation to indicate the approach mode deviation scaling is in use. In the case of GPS, if the scaling word is changing from one level to the next, a 'SCALING' annunciation is shown at a position located above the 'APPR' annunciator location. When conditions exist such that a final approach can be made, the 'APPR' annunciation is displayed if the final approach scaling is linear. The 'LIN' annunciation is removed if the final approach segment simulates a pseudo localizer approach. In GPS, if the GPS data is invalid the scale reverts to previous operations.

When an approach is being flown and the appropriate sensor is selected, the pointer changes to a rising runway symbol after descending to a radio altitude of 200 feet. At 200 feet of radio altitude, the runway symbol begins expanding both vertically and laterally until at zero-foot altitude, the top edge of the runway symbol just touches the bottom edge of the aircraft symbol. The bottom edge of the runway symbol remains adjacent to the deviation scale. In -002 and later versions DPU and MPU, the pointer symbol for PLOC (LNAV pseudo localizer) is a diamond-shaped rectangle. In these systems, if PLOC is the sensor during an approach, the diamond-shaped pointer does *not* change to the rising runway symbol at 200 feet radio altitude.

second course related lateral deviation sensors—Second course lateral deviation is displayed on the EADI only in configurations using the -001 version DPU and MPU. DPU and MPU -002 and later versions show second course lateral deviation only on the EHSI (see EHSI Display Formats section).

In the case of the -001 DPU/MPU, one of two second course related lateral deviation sensors may be displayed. These are: PLOC (LNAV pseudo localizer) and FLOC (LNAV controlled localizer).

If pseudo ILS approach data is received from the LNAV selected as the active sensor, the EFIS-86C uses this data in place of real ILS data. This is annunciated by the letters 'PLOC' located above the right two lateral deviation dots. A second deviation pointer, cyan in color, is also displayed showing the deviation from the second course. This should coincide with the actual runway heading. When the pseudo localizer is within capture range, the second course PLOC replaces the active sensor, and the only remaining deviation is PLOC.

The LNAV controlled localizer is selected on the LNAV's CDU and is displayed on the EADI when the LNAV is selected as the active sensor. The operation is the same as the pseudo ILS described above except that the annunciation is 'FLOC'.

Acceleration (When On The Ground) or Mach Display—The same display location in the lower left corner is used for acceleration when on the ground, or airborne Mach readout. When on the ground, a three-digit magenta readout preceded by a decimal point represents longitudinal acceleration in g's. A leading minus sign indicates deceleration. In flight, the three-digit green readout displays Mach number in .002 Mach increments. The readout comes into view above .300 Mach and falls out of view below .290 Mach.

Speed Displays—The EFIS-86C may display indicated airspeed, fast/slow speed deviation from a digital air data system, angle of attack system, or an analog air data system, or no speed display at all. These are installation strapping options.

IAS (Indicated Airspeed) Display—IAS is shown by one of two scales on the left side of the EADI. The two scales are referred to as the 'low-IAS' and the 'high-IAS' scales. The determination of which scale is to be displayed is based on actual airspeed. The scales and scale markings are white.

The low-IAS scale covers the range from 60 to 190 knots and is divided into 10-knot increments. The high-IAS scale covers the range from 160 to 400 knots and is divided into 20-knot increments. Scale markings are on the left of the vertical scales. The low-IAS scale is displayed at power on. (See page 35.)

There are three continuation dots at the upper end of the low-IAS scale and three continuation dots at the bottom end of the high-IAS scale. These dots serve to remind the pilot that the currently displayed scale is continued in the direction of the dots. If the low-IAS scale is shown, as "IAS increases toward the dots at the top, the pilot knows that the scale will change over and continue on the high-IAS scale. Conversely, if the high-IAS scale is shown, as IAS decreases toward the dots at the bottom, the pilot knows that the scale will change over and continue on the low-IAS scale.

The scale changeover point from low-IAS to high-IAS occurs as follows:

- IAS 190 knots, *or*
- IAS 160 knots *and* flaps up *and* trend limited at upper edge *and* acceleration has been 1.5 kts per second for more than 30 knots.

The scale changeover point from high-IAS to low-IAS occurs as follows:

- IAS 160 knots, *or*
- IAS 170 knots *and* trend limited at lower edge, *or*
- IAS 190 knots *and* flaps extended *and* scale has not switched from low to high with flaps extended within the last 30 seconds

If the difference between the pilot's and copilot's air data is less than 10 knots, then the copilot's scale switching follows the pilot's side. If the difference between the two sides is greater than 10 knots or if the pilot's side is flagged, then the copilot's scale switching is independent of the pilot's side.

Actual IAS is shown by the position of a small green rectangular marker extending across the low- or high-IAS scale, and is repeated in green digits to the right of the marker. The marker and digital readout move below the end of the low-IAS scale for values below 60 knots until they have moved out of the viewing area on the display. For actual IAS values above 400 knots, the marker and digital readout park at the upper end of the high-IAS scale with the digital readout continuing to display actual IAS.

On the high-IAS scale, V_{mo} is indicated by two parallel red lines that extend upward from V_{mo} a maximum of 1/2 inch. The right hand red line covers the scale. Above V_{mo}, the actual IAS marker, digital readout, and Mach display turn red.

Selection of cross-side air data is indicated by the green actual IAS marker, digital readout, and Mach display all turning yellow. In addition, a yellow 'XADC' (cross-side air data) annunciation appears on the left side of the EADI just above the horizon line. If actual IAS becomes greater than V_{mo}, the yellow values turn red until actual IAS drops below V_{mo}.

Reference Airspeed—Reference indicated airspeed is displayed by a cyan colored triangular shaped bug to the left of the scale. Under certain special equipment and interconnect conditions, reference airspeed is repeated digitally in a half-box at the upper left of the low-IAS scale or at the lower left of the high-IAS scale. When in view, the digital readout and half-box are also cyan.

The reference IAS bug setting has to be within the range of the IAS scale presently in view before the bug is displayed. The bug never appears outside the limits of either scale, and never appears next to the three continuation dots at the upper end of the low-IAS scale or the bottom end of the high-IAS scale.

The reference airspeed digital readout and half-box appear only under the following conditions:

- In view full time whenever the reference IAS bug is *not* on the IAS scale presently being displayed.
- If the reference IAS bug is on scale, the digits and half-box appear whenever the reference IAS is being changed and for 10 seconds thereafter.
- Whenever a scale changeover occurs, the digits and half-box will be in view for 10 seconds.
- If not already in view, the digits and half-box appear for 20 seconds when descending through 1000 feet radio altitude in approach.

The reference airspeed display is removed from view whenever the reference airspeed is set below 70 knots. If a failure is detected in the reference airspeed, the bug and digital readout are removed from view.

In the -003 and later DPU/MPU and also depending upon installation configuration, up to eight reference IAS bugs can be displayed along the IAS scale. All bugs have a programmed initial value at power up (these values come from the air data computer and are determined at certification). These bugs are set/controlled by the ARP-86() Air Data Reference Panel. The symbols used to represent the bugs come from the air data computer and may vary depending upon user preference. Bug colors (determined in the air data module during certification) may be cyan, green, amber, red, magenta, blue, or white. Once set, the bugs move up or down the IAS scales. The -001 and -002 DPU and MPU can display only the reference IAS bug as presented in the paragraphs above.

IAS Acceleration Trend—Extending upward or downward from the center of the actual IAS digital readout is an acceleration trend vector consisting of two parallel magenta bars. It is in view only when the strut is extended, but is removed if radio altitude is less than 30 feet or if speed is less than 50 knots. When the trend vector reaches the display edge limit or the maximum length limit, a horizontal line shows across the end of the trend vector.

A digital readout of acceleration (in g's) shows in the lower left corner of the display when the airplane is on the ground and the source of longitudinal acceleration is valid. The acceleration readout consists of a decimal point, three magenta numbers, and minus sign (the minus sign shows only during ground deceleration).

The acceleration trend information is derived from actual IAS, longitudinal acceleration, and pitch. The trend vector shows on the display even if an acceleration fault is detected. Detection of a fault shows a red box with the letters 'ACC' inscribed in place of the acceleration numbers when the airplane is on the ground and for 20 seconds after strut extension. This flag flashes for 10 seconds and stays on. Push the master warn reset to stop the flashing prior to the 10 second timeout. Also, if longitudinal acceleration is invalid, the 'ACC' flag shows during the first 4 seconds of the self-test.

The loss of pitch angle data from the attitude source degrades the trend vector, but continues to show on the display for -001 and -002 versions of the DPU and MPU. In -003 and later versions

DPU/MPU, detection of a loss of pitch angle data from the attitude source turns off the display of the trend vector.

Speed Deviation Display—If the IAS scale and display are not used in a particular installation, then one of three sensors may be selected as the input for the fast/slow speed deviation display. These are:

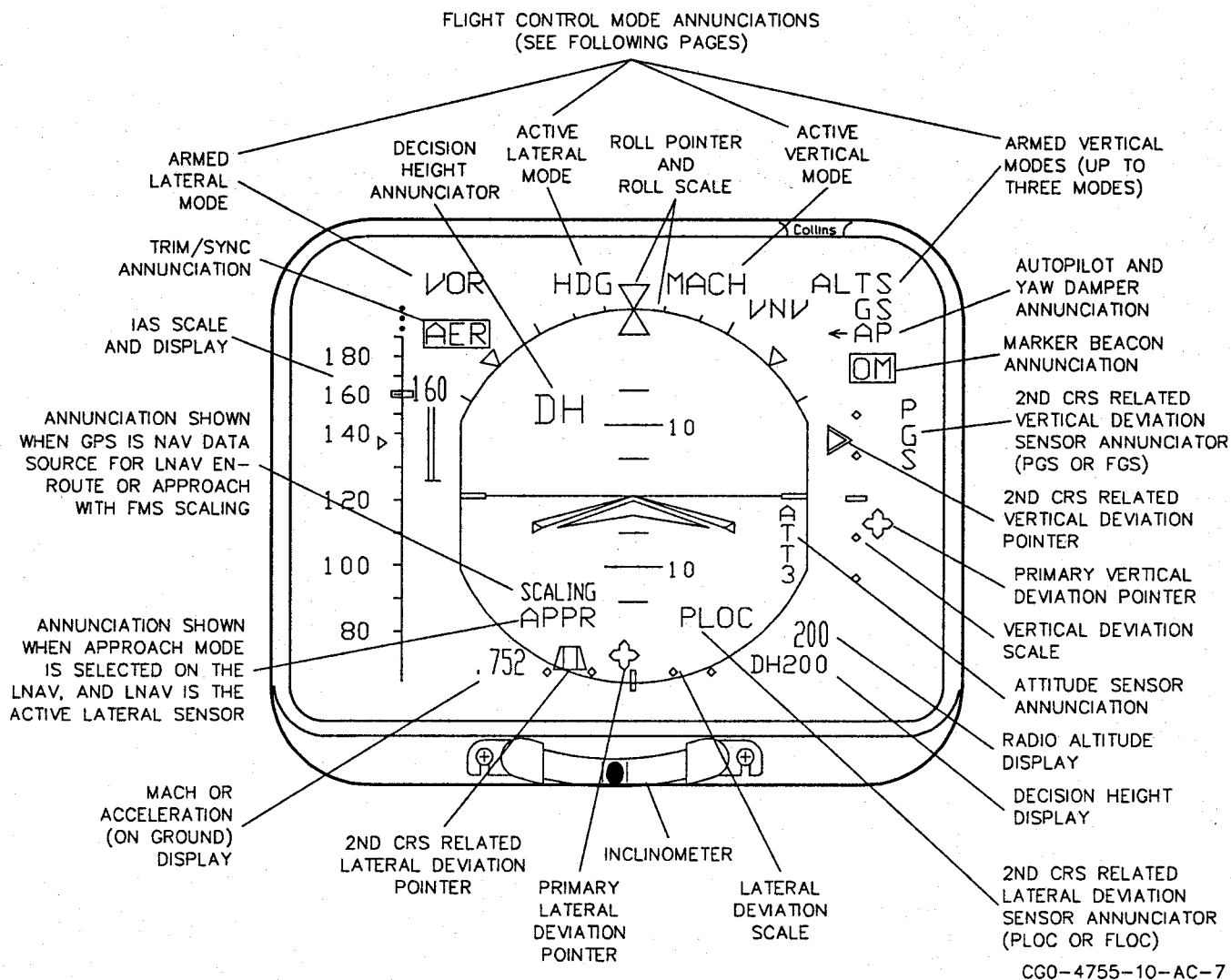
- Digital air data system
- AOA (angle of attack) system
- Analog air data system

The speed deviation display is located on the left side of the EADI (in the same location that the IAS display would normally occupy). The speed deviation scale consists of four dots with the letters 'F' and 'S' to the left of the top and bottom dots respectively. The speed deviation pointer is green and consists of two concentric squares which move in relation to the scale such that if the pointer is next to the upper dot, then actual airspeed is 10 knots higher than desired.

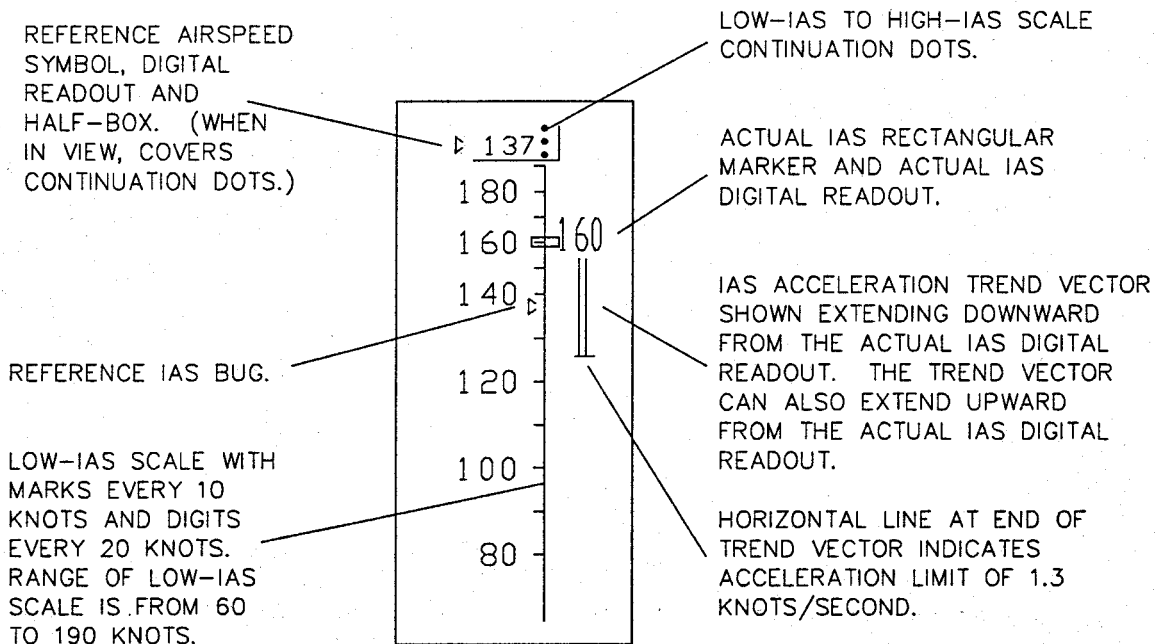
With a digital air data system, the EFIS-86C compares actual IAS with the reference IAS being received from the air data system, and positions the pointer accordingly. With either the AOA or analog air data input, the EFIS-86C positions the pointer with reference to the analog voltage received from the respective system.

When speed deviation is from a digital air data system, the digital reference airspeed is displayed below the letter 'S' on the speed scale. When speed deviation is AOA based, the letters 'AOA' appear below the speed scale, and if speed deviation is analog air data based, the letters 'SPD' appear below the speed scale. These displays are always cyan in color.

The display shown below is for illustrative purposes only. All of the pointers, symbols, annunciators, etc. may not appear on an actual display.

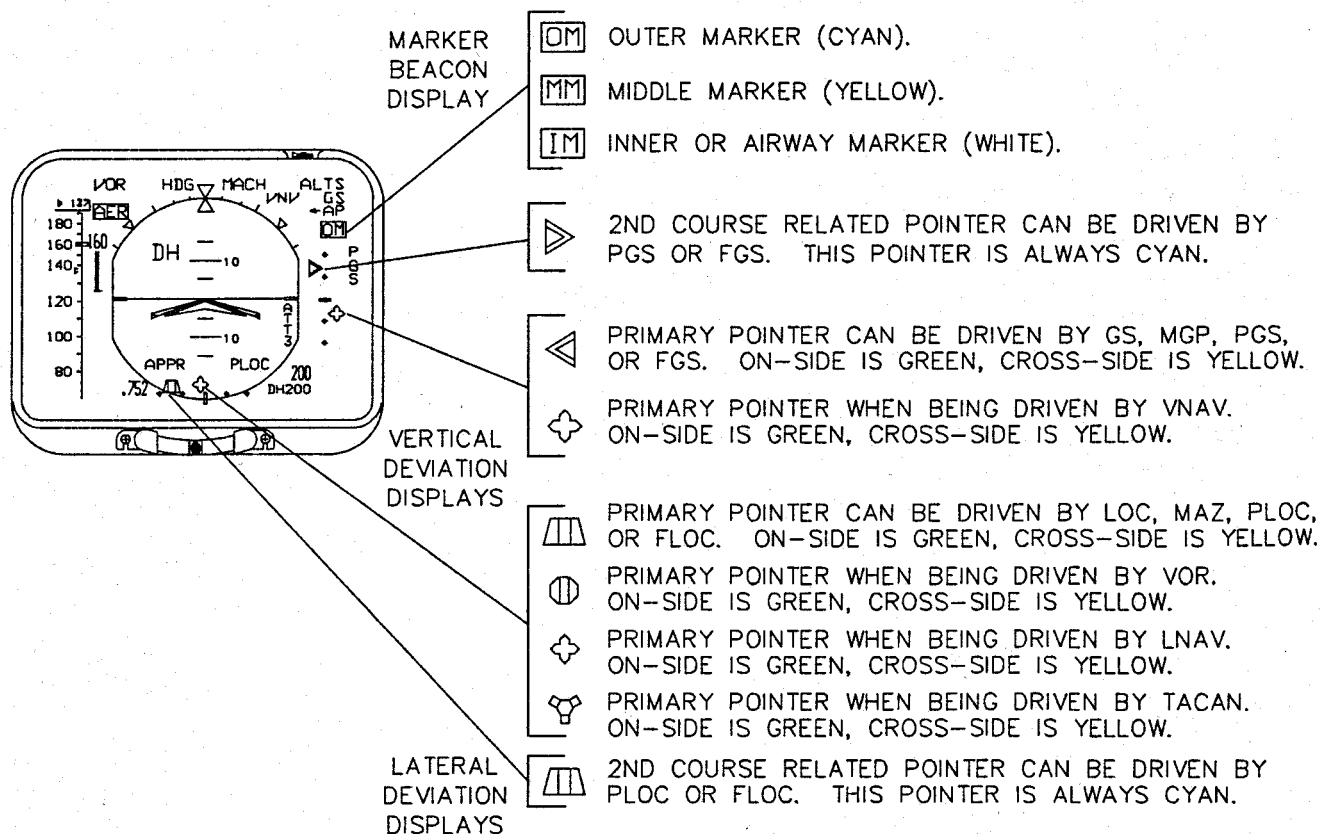


CGO-4755-10-AC-7

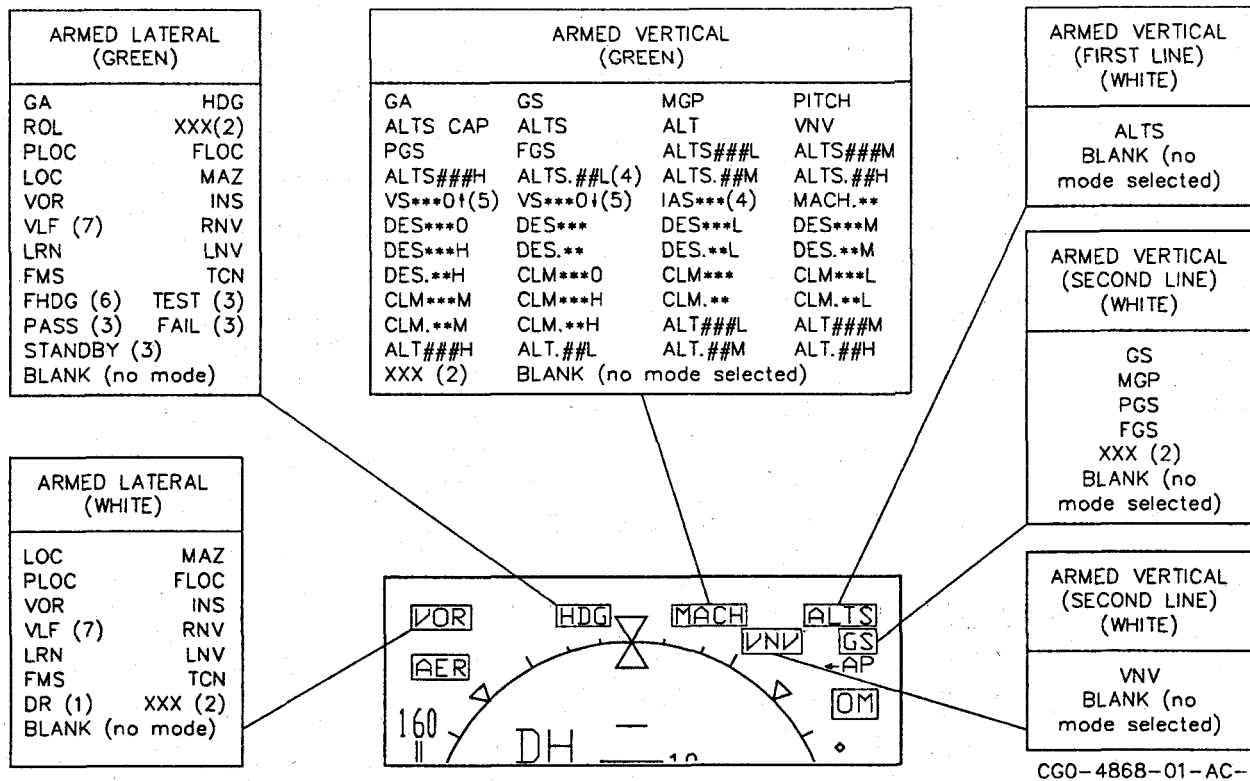


IAS SCALE AND DISPLAY (LOW-IAS SCALE SHOWN)

CGO-4755-10-AC-5



CGO-4755-10-AC-6

**LATERAL AND VERTICAL MODE NOTES:**

- (1) 'DR' IS NOT DISPLAYED WITH LATERAL CAPTURE MODES LOC OR MAZ. THIS 'DR' IS FROM THE FLIGHT CONTROL SYSTEM ONLY.
- (2) WHITE Xs ARE PLACED IN BOTH ARM AND ACTIVE POSITIONS WHEN NAV SENSOR IS ARMED OR CAPTURED WITH AN UNIDENTIFIED NAV SENSOR CONFIGURATION STRAP COMBINATION.
- (3) THESE ANNUNCIATIONS ARE FROM THE FLIGHT CONTROL SYSTEM AND ARE YELLOW.
- (4) * = GREEN DIGITS AND # = WHITE DIGITS. THESE DIGITS ARE ADVISORY VALUES FROM THE FLIGHT CONTROL SYSTEM.
- (5) VS REFERENCE NUMBERS TURN TO FLASHING YELLOW '9999' IF VS EXCEEDS 9999 FEET/MINUTE.
- (6) 'FHDG' INDICATES FMS HEADING MODE WHICH ALLOWS THE FMS TO CONTROL THE HEADING REFERENCE CURSOR DISPLAY.
- (7) THIS ANNUNCIATOR REPLACED BY "GPS" IN GPS CONFIGURATION (-004).

LATERAL AND VERTICAL MODE DESCRIPTIONS**DESCRIPTION OF LATERAL MODES**

GA Go around mode.

LOC Nav mode, localizer arm or capture.

MAZ Nav mode, microwave azimuth arm or capture.

VOR Nav mode, VOR arm or capture.

INS Nav mode, inertial navigation system arm or capture.

VLF Nav mode, very low frequency navigation system arm or capture.

GPS Nav mode, global positioning system arm or capture.

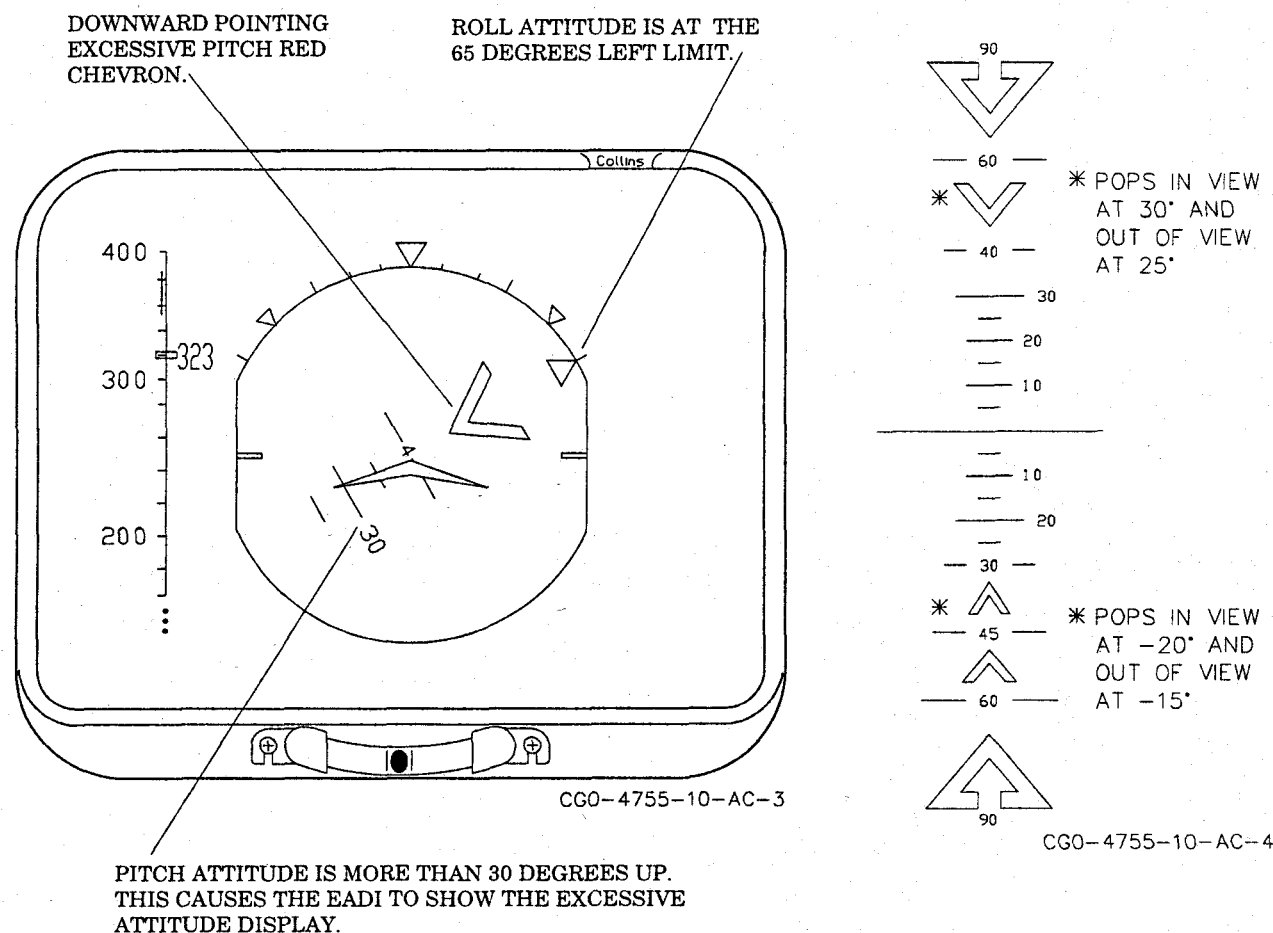
RNV Nav mode, area navigation system arm or capture.

LRN Nav mode, long range navigation system arm or capture.

LNV Nav mode, lateral navigation system arm or capture.
 TCN Nav mode, TACAN arm or capture.
 FMS Nav mode, flight management system arm or capture.
 FLOC Nav mode, LNAV controlled localizer capture.
 PLOC Nav mode, LNAV pseudo localizer capture.
 FHDG Nav mode, LNAV heading mode.
 DR Dead reckoning mode.
 ROL Roll hold mode.
 HDG Heading mode.
 TEST Flight control system self-test.
 PASS Flight control system mode tests passed.
 FAIL Flight control system mode tests failed.
 STANDBY Flight control system in standby mode.
 (BLANK) No lateral mode selected.

DESCRIPTION OF VERTICAL MODES

GA Go around mode.
 GS Glideslope arm or capture.
 PGS LNAV pseudo glideslope.
 FGS LNAV controlled glideslope.
 MGP Microwave glidepath arm or capture.
 PITCH Pitch hold mode.
 ALTS Altitude preselect track.
 ALTS CAP. Altitude preselect capture ('CAP' turns on at capture and turns off at track).
 ALT Altitude hold mode.
 VNV Vertical navigation mode.
 ALTS##L Altitude preselect with low speed climb profile and pre-programmed IAS.
 ALTS##M Altitude preselect with mid speed climb profile and pre-programmed IAS.
 ALTS##H Altitude preselect with high speed climb profile and pre-programmed IAS.
 ALTS.#L Altitude preselect with low speed climb profile and pre-programmed Mach.
 ALTS.#M Altitude preselect with mid speed climb profile and pre-programmed Mach.
 ALTS.#H Altitude preselect with high speed climb profile and pre-programmed Mach.
 VS**O↑ Vertical speed mode showing feet per minute up.
 VS**O↓ Vertical speed mode showing feet per minute down.
 IAS *** Indicated airspeed hold showing reference airspeed.
 MACH.** Mach hold showing Mach number.
 DES**O Descend mode showing rate of descent in feet per minute.
 DES*** Descend mode with pre-programmed IAS.
 DES***L Descend mode with low speed descend profile and pre-programmed IAS.
 DES***M Descend mode with mid speed descend profile and pre-programmed IAS.
 DES***H Descend mode with high speed descend profile and pre-programmed IAS.
 DES.** Descend mode with pre-programmed Mach.
 DES.**L Descend mode with low speed descend profile and pre-programmed Mach.
 DES.**M Descend mode with mid speed descend profile and pre-programmed Mach.
 DES.**H Descend mode with high speed descend profile and pre-programmed Mach.
 CLM**O Climb mode showing rate of climb in feet per minute.
 CLM*** Climb mode with pre-programmed IAS.
 CLM***L Climb mode with low speed climb profile and pre-programmed IAS.
 CLM***M Climb mode with mid speed climb profile and pre-programmed IAS.
 CLM***H Climb mode with high speed climb profile and pre-programmed IAS.
 CLM.** Climb mode with pre-programmed MACH.
 CLM.**L Climb mode with low speed climb profile and pre-programmed Mach.
 CLM.**M Climb mode with mid speed climb profile and pre-programmed Mach.
 CLM.**H Climb mode with high speed climb profile and pre-programmed Mach.
 ALT##L Altitude hold mode with low speed climb profile and pre-programmed IAS.
 ALT##M Altitude hold mode with mid speed climb profile and pre-programmed IAS.
 ALT##H Altitude hold mode with high speed climb profile and pre-programmed IAS.
 ALT.#L Altitude hold mode with low speed climb profile and pre-programmed Mach.
 ALT.#M Altitude hold mode with mid speed climb profile and pre-programmed Mach.
 ALT.#H Altitude hold mode with high speed climb profile and pre-programmed Mach.
 (BLANK) No vertical modes selected.



EADI Excessive Attitude Display

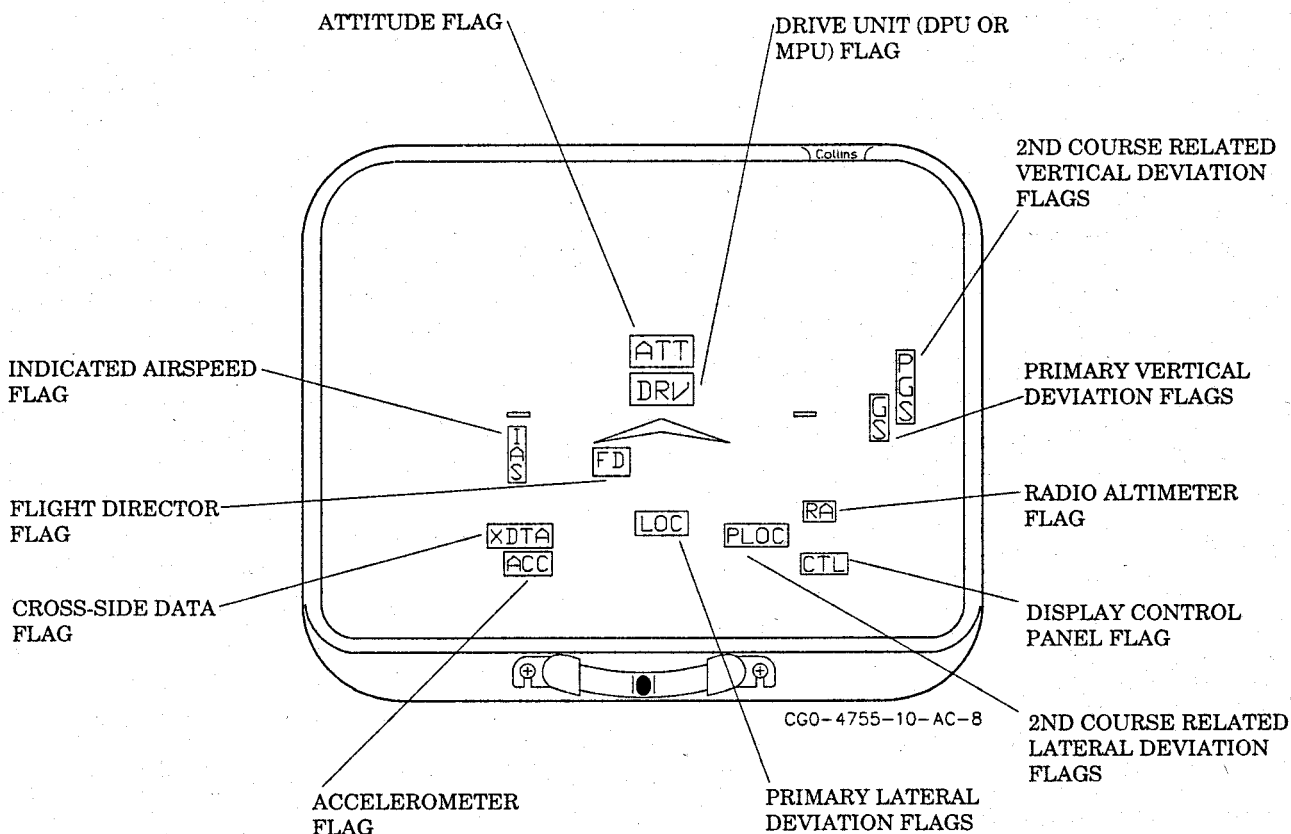
Excessive attitude is when the pitch attitude exceeds 30 degrees up or 20 degrees down, or when the roll attitude exceeds 65 degrees right or left. In the extreme pitch attitudes, a portion of the blue (sky) or brown (ground) area is retained in view to aid in interpretation of pitch attitude. The pitch markings continue to move to provide accurate pitch information. When in an excessive attitude condition, all information except the airplane symbol, attitude display, and the IAS display is removed from the EADI. All information is restored to the EADI after the pitch attitude decreases to +25 degrees (pitch up) or increases to -15 degrees (pitch down), or after the roll attitude decreases to ± 60 degrees.

If the excessive attitude condition is due to excessive pitch, then upward or downward pointing (as appropriate) excessive pitch red chevrons pop into view as part of the pitch display. These chevrons are out of view until the extreme pitch attitude is entered at pitch angle greater than 30 degrees pitch up, or 20 degrees pitch down. The red upward or downward pointing chevrons indicate which direction (up or down) is required to correct for the excessive pitch attitude. In the EFIS-86C system, there are two downward pointing chevrons for excessive pitch up attitudes, and three upward pointing chevrons for excessive pitch down attitudes. If the aircraft passes through the vertical, the attitude display switches 180 degrees in the roll axis and continues to provide the correct ground-sky reference during the inverted portion of the maneuver.

EADI Warning Flags

NOTE

Unless otherwise noted, when an EADI warning flag appears, it flashes for 10 seconds and then becomes steady. The flashing can be eliminated by pushing the master warn reset.



Attitude Flag—If a failure of the attitude sensor is detected, the pitch scale, roll scale, roll pointer, sky/ground display, and command bars disappear and a red box with the letters 'ATT' inscribed appears above the airplane symbol. The ATT flag remains in view until an alternate sensor is selected or until the fault is cleared.

Drive Unit (MPU or DPU) Flag—If an MPU or DPU fails, the non-flashing letters 'DRV' appear in a red box centered on the display. Nothing is removed from the EADI when a DRV flag appears.

Vertical Deviation Flags

primary vertical deviation sensor flags—A detected failure of any of the primary vertical deviation sensors removes the vertical deviation scale and pointer from view and replaces them with the following flags. If a failure of the glideslope or MLS glidepath receiver is detected, a red box with the letters 'GS' or 'MGP' inscribed appears. If an LNAV failure is detected while it is being used for LNAV pseudo or controlled glideslope, a red box with the letters 'PGS' or 'FGS' appears. If a VNAV failure is detected, a red box with the letters 'VNV' appears.

second course related vertical deviation flags—In -001 version DPU and MPU, if an LNAV failure is detected while it is being used for second course related LNAV pseudo or controlled glideslope, the cyan 'PGS' or 'FGS' annunciations are replaced by a red box with the letters 'PGS' or 'FGS' inscribed. For -002 and later versions DPU/MPU, the EADI does not show vertical second course sensors or the associated flags.

Radio Altimeter Flag—If a failure of the radio altimeter system is detected, the radio altitude display is replaced by a red box with the letters 'RA' inscribed. The DH set display and the DH annunciator are also removed from view.

Display Control Panel Flag—If a failure of the display control panel occurs, a red box with the letters 'CTL' inscribed appears in the lower right corner replacing the DH set display. Flight control system mode annunciators that are derived from the DCP are also removed from view. The EFIS-86C continues to operate in the mode that was active prior to the DCP failure.

Lateral Deviation Flag

primary lateral deviation sensor flags—A detected failure of any of the primary vertical deviation sensors removes the vertical deviation scale and pointer from view and replaces them with the following flags. If a failure of the localizer or MLS azimuth receiver is detected, a red box with the letters 'LOC' or 'MAZ' inscribed appears. If an LNAV failure is detected while it is being used for LNAV pseudo or controlled localizer, a red box with the letters 'PLOC' or 'FLOC' appears. For the remaining sensors, a red box with the letters 'VOR', 'TCN', 'RNV', or 'LRN' appears.

second course related lateral deviation flag—For the -001 version DPU and MPU, if an LNAV failure is detected while it is being used for second course related LNAV pseudo or controlled localizer, the cyan 'PLOC' (LNAV pseudo localizer) and FLOC (LNAV controlled localizer) are replaced by a red box with the letters 'PLOC' and 'FLOC' inscribed. The -002 and later DPU/MPU versions second course related lateral deviation sensors do *not* show on the EADI. The second course sensors show only on the EHSD.

Accelerometer/Mach Flag—If an accelerometer failure is detected, the digital readout is replaced by a red box with the letters 'ACC' inscribed. This flag can only be displayed when the aircraft is on the ground. For a Mach flag, the Mach digits are blanked.

Cross-Side Data Flag—If a failure of the cross-side data bus occurs, a red box with the letters 'XDTA' inscribed appears in the lower left of the EADI. Data from the cross-side sensors is no longer available, and any display driven by cross-side data is flagged.

Speed Deviation Flags—If the speed deviation is digital air data based and a failure is detected on the digital air data or referenced airspeed bus, the fast/slow scale, pointer, and digital readout are removed from view.

If an AOA failure is detected when displaying AOA based speed deviation, the speed deviation scale and pointer disappear and the letters 'AOA' that are beneath the scale turn red and are boxed.

If a detected failure of an analog air data based speed deviation input occurs, the speed deviation scale and pointer disappear and the letters 'SPD' appear at the left center of the display in red and are boxed.

Flight Director Flag—If the flight director system fails, the command bars disappear and a red box with the letters 'FD' inscribed appears at the lower left of the aircraft symbol.

Indicated Airspeed Flag—If a failure of the digital air data system is detected, the IAS display in the left center of the EADI is replaced by a red box with the letters 'IAS' inscribed. The IAS scale and pointers, reference airspeed pointer and display, acceleration trend vector, and Vmo display are removed from view.

EADI Comparators and Master Warn

Comparator monitoring is performed in each DPU and MPU in the EFIS-86C system. On-side data is read and stored by each DPU and the MPU. Each DPU sends its information to the cross-side DPU so that each of the processors has the information required to perform the comparison function. When the MPU is driving the EADI, it provides the comparator function in place of the failed DPU.

EADI Comparator Warnings

NOTE

When an EADI comparator warning occurs, it appears nonflashing. The messages disappear when the external comparator reset switch is actuated, if the comparator error is no longer present.

In addition to the comparator warnings displayed on the EADI, the EFIS-86C provides individual attitude and ILS comparison outputs from each DPU and MPU when in drive transfer. These comparator warning outputs are used to turn on remote mounted attitude and ILS annunciators (not part of the EFIS-86C system).

Comparator Reset—Momentarily actuating an installer supplied comparator reset switch turns off the EADI comparator warnings and the external comparator warning flag annunciator(s). If the condition causing the compare error is no longer present, the EADI comparator warning display and the external comparator warning annunciator(s) remain off when the comparator reset switch is released. If the warn condition is still present when the comparator reset switch is released, the EADI comparator warning display and the external comparator warning annunciator(s) turn on again, but the master warn annunciator does not.

Master Warn and Master Warn Reset—A master warn which serves as an overall compare function output is provided from each DPU and from the MPU when in drive transfer. These outputs are used to turn on installer supplied master warning annunciators (not part of the EFIS-86C system).

Momentarily actuating an installer supplied master warn reset switch stops the flashing of the EADI comparators and of the remote mounted attitude and ILS comparators, and turns off the master warning annunciator.



- Either side has an ATT flag
- A cross-side data failure is detected

GLIDESLOPE CAPTURED	SENSOR SOURCE TYPE DIFFERENCE	PITCH OR ROLL DIFFERENCE MORE THAN
Yes	Yes	5°
Yes	No	3°
No	Yes	6°
No	No	4°

3rd Edition
29 Apr 97

IAS Comparator

Enable Logic—A comparison of the two indicated airspeeds is made when:

- Cross-side data is valid
- Both air data computers are valid
- IAS scale is displayed
- Either IAS 90 knots

Comparison Limits—A comparator warning occurs when the enable conditions are met and the two indicated airspeeds differ by more than 10 knots.

Annunciation—When an indicated airspeed comparator warning is detected, the letters 'IAS', boxed and in yellow, appear to the left of the aircraft symbol. The external master warn comparator (if installed) also turns on.

ILS Localizer Comparator

Enable Logic—A comparison of the two localizer deviations is made when:

- On-side DCP is valid and its active course is VOR/LOC
- Cross-side data is valid
- Both localizer signals tuned and valid for at least 15 seconds (localizer tuned and valid on both VORs)
- Alternate DCP is not selected
- Glideslope is captured on the side that the autopilot is coupled to

Comparison Limits—A comparator warning occurs when the enable conditions are met and the two localizer deviations differ by more than 30 microamperes + 1/8 of the absolute value of deviation 1 + deviation 2.

Annunciation—When a localizer comparator warning is detected, a yellow box with the letters 'LOC' inscribed appear above the lateral deviation scale. The external 'ILS' comparator annunciator (if installed) and the master warn comparator also turn on.

ILS Glideslope Comparator

Enable Logic—A comparison of the two glideslope deviations is made when:

- Both DCPs are valid and their active courses are VOR/LOC
- Cross-side data is valid
- Both glideslope valid and both localizer signals tuned for at least 15 seconds
- Alternate DCP is not selected
- Both sides not in back course
- Glideslope is captured on the side that the autopilot is coupled to

Comparison Limits—A comparator warning occurs when the enable conditions are met and the two glideslope deviations differ by more than 40 microamperes + 1/8 of the absolute value of deviation 1 + deviation 2.

Annunciation—When a glideslope comparator warning is detected, a yellow box with the letters 'GS' inscribed appear near the center of the vertical deviation scale. The external 'ILS' comparator annunciator (if installed) and the master warn comparator also turn on.

Radio Altitude Comparator

Enable Logic—A comparison of the two radio altitudes is made when:

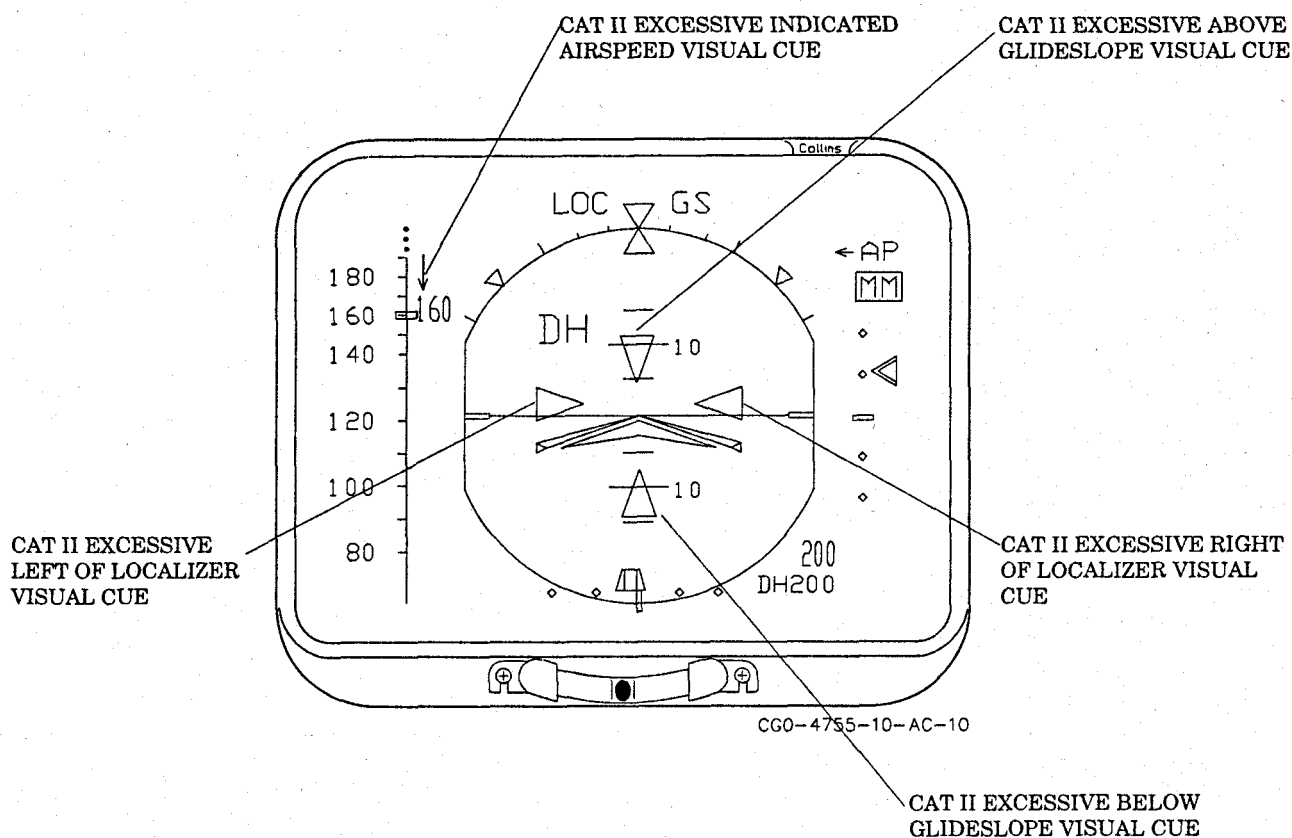
- Cross-side data is valid
- Both radio altimeters are valid

Comparison Limits—A comparator warning occurs when the enable conditions are met, either radio altimeter is on scale, and the difference between the two radio altitudes is 40 ft plus desensitizing values (depending upon radio altitude).

Annunciation—When a radio altitude comparator warning is detected, a yellow box with the letters 'RA' inscribed appears next to the radio altitude readout.

EADI Category II Excessive Deviation Warnings

The EFIS-86C system provides special visual cues to indicate excessive deviation of the ILS localizer and ILS glideslope, and excessive indicated airspeed deviation during a Category II approach.



Excessive ILS Localizer Deviation

Enable Logic—The on-side localizer deviation is compared to the CAT II limit when all of the following conditions are met:

- On-side CAT II A or B request received
- VOR/LOC is the active course from the on-side DCP
- On-side radio altitude is less than 600 feet
- On-side localizer is tuned and valid
- On-side back course is not detected
- Go around is not selected on either side
- On-side DCP is valid and cross-side DCP is not selected

Monitor Limits—An excessive deviation warning occurs when the enable conditions are met and the localizer deviation exceeds 25 microamperes for a CAT IIA request, or 20 microamperes for a CAT IIB request.

The on-side excessive deviation warning is also displayed when the cross-side system detects an excessive deviation condition, cross-side data is valid, and a localizer scale is in view.

Annunciation—When an excessive deviation warning occurs, a red triangular-shaped pointer appears to the right or left of the aircraft symbol. The apex of the triangle points in the direction of appropriate correction (fly right if the triangle is to the left of the aircraft symbol, and fly left if the triangle is to the right of the aircraft symbol).

Neither the external master warn or ILS comparator warn annunciators are turned on for the excessive deviation condition.

Excessive ILS Glideslope Deviation

Enable Logic—The on-side glideslope deviation is compared to the CAT II limit when all of the following conditions are met:

- On-side CAT II A or B request received
- VOR/LOC is the active course from the on-side DCP
- On-side radio altitude is between 90 and 600 feet
- On-side glideslope is valid
- On-side localizer is tuned and valid for 15 seconds
- On-side back course is not detected
- Go around is not selected on either side
- On-side DCP is valid and cross-side DCP is not selected

Monitor Limits—An excessive deviation warning occurs when the enable conditions are met and the glideslope deviation exceeds 75 microamperes for a CAT IIA request, or 65 microamperes for a CAT IIB request. The on-side excessive deviation warning is also displayed when the cross-side system detects an excessive deviation condition, cross-side data is valid, and a glideslope scale is in view.

Annunciation—When an excessive deviation warning occurs, a red triangular-shaped pointer appears above or below the aircraft symbol. The apex of the triangle points in the direction of appropriate correction (fly down if the triangle is above the aircraft symbol, and fly up if the triangle is below the aircraft symbol).

Neither the external master warn or ILS comparator warn annunciators are turned on for the excessive deviation condition.

Excessive Indicated Airspeed Deviation

Enable Logic—The on-side IAS deviation is compared to the CAT II limit when all of the following conditions are met:

- On-side CAT II request received
- VOR/LOC is the active course from the on-side DCP
- On-side radio altitude is between 15 and 1000 feet
- On-side glideslope is valid
- On-side localizer is tuned and valid for 15 seconds
- On-side back course is not detected
- Go around is not selected on either side
- On-side DCP is valid and cross-side DCP is not selected
- Excessive IAS deviation is enabled

Monitor Limits—An excessive deviation warning occurs when the enable conditions are met and the IAS deviates from the IAS reference by more than +10 knots or -5 knots.

The on-side excessive deviation warning is also displayed when the cross-side system detects an excessive deviation condition, cross-side data is valid, and a glideslope scale is in view.

Annunciation—When an excessive deviation warning occurs, a yellow arrow appears above or below the green rectangular IAS marker pointing in the direction of appropriate correction (lower IAS if the arrow is above the IAS marker, and increase IAS if the arrow is below the IAS marker).

The external master warn annunciator is not turned on for the excessive deviation condition.

Category II Outputs

The EFIS-86C provides two Category II outputs that can aid the pilot in determining if Category II approach monitoring requirements are met. These outputs are the 'green' output and the 'amber' output, with each output driving appropriately colored annunciators.

NOTE

The pilot's and copilot's CAT II select switches, switching relays, and annunciator lights (CAT II green and amber light on both sides) are installer supplied items and are not furnished as part of the EFIS-86C system.

The CAT II 'amber' annunciator flashes for 2 seconds following a CAT II request to verify that the annunciator is working. Then, if all of the conditions listed below are met, the CAT II 'green' annunciator turns on. If all of the conditions listed below are not met, the CAT II 'amber' annunciator either turns on steady or flashes on and off. The on steady or flashing state of the 'amber' annunciator is determined by whether the 'amber' annunciation is caused by a condition 1 function, or by a condition 2 function (both of these conditions are listed in following paragraphs).

NOTE

It is *always* the pilot's responsibility to decide if the approach is to be made.

Green CAT II Annunciator—The green Category II annunciator turns on when all of the following conditions are met:

- At least 2 seconds have elapsed since a CAT II request was made on either side.
- Cross-side data valid for both sides.
- LOC tuned and valid for 15 seconds on both sides.
- GS valid for 15 seconds on both sides.
- Back course not detected for either side.
- Radio altimeter valid and on scale for both sides.
- Attitude valid for both sides.
- Heading valid for both sides.
- Alternate attitude sensor not selected on either side.
- Alternate heading sensor not selected on either side.
- On-side localizer selected as active course on both sides.
- DCP valid and not alternate DCP selected for either side.
- Sector with MAP format not selected on either DCPs.
- Flight director valid and command bars in view on the side that the autopilot is engaged and coupled.
- True heading not selected on either side.
- On-side dual drive transfer not selected.
- Air data valid on both sides.
- Composite format not selected on either side.
- Heading comparator warning not present on either side.
- Attitude comparator warning not present on either side.
- Localizer comparator warning not present on either side.
- Glideslope comparator warning not present on either side.
- Cross-side air data not present on both side.
- Autopilot is engaged on either side.
- If radio altitude is 200 feet, excessive localizer deviation is not present on either side.
- If radio altitude is 200 feet, excessive glideslope deviation is not present on either side.

If decision height is reached and the green annunciator has not yet turned on, it will not do so at a lower altitude. This indicates that the approach should not be continued to Category II limits.

Amber CAT II Annunciator—If *any* of the following condition 1 functions occur *below* 200 feet of radio altitude, or *any* of the condition 2 functions occur, the amber Category II annunciator flashes on and off.

If *none* of the condition 2 functions listed below is detected, and *any* of the condition 1 functions occur *above* 200 feet of radio altitude, the amber Category II annunciator turns on steady.

Condition 1 Functions

- Active course is not the on-side localizer.
- Either side radio altitude flag is in view.
- Either side air data flag is in view.
- Either side localizer flag is in view.
- Either side glideslope flag is in view.
- Either side heading flag is in view.
- Either side attitude flag is in view.
- Either side DCP flag is in view.
- Cross-side flag is in view.
- Either side is in back course.
- Either side has true heading.
- Either EHSI is displaying a sector with MAP format.
- Both attitudes are from the same sector.
- Both headings are from the same sector.
- Composite format is being displayed on either side.
- Heading comparator warning is present.
- Localizer comparator warning is present.
- Glideslope comparator warning is present.
- Dual drive transfer selected.
- Cross-side air data on either side.

Condition 2 Functions

- Any of the following occurs on the side that the autopilot is coupled:
 - On-side flight director flag in view.
 - On-side attitude flag in view.
 - On-side heading flag in view.
 - Glideslope capture and the on-side localizer flag in view.
 - Glideslope capture and the on-side glide slope flag in view.
- Attitude comparator warning is present.
- Autopilot is disconnected.
- Either of the following occurs below 200 feet of radio altitude:
 - Excessive localizer deviation is present.
 - Excessive glideslope deviation is present.

EHSI Display Formats

The EHSI brings together numerous displays to show a map-like presentation of the aircraft's horizontal navigation situation. The EHSI shows aircraft heading and aircraft displacement relative to VOR radials, localizer, and glideslope beams. Deviation from other lateral and vertical sensors may also be shown.

The EHSI can display three formats that are selected with the HSI MODE button on the DCP-85F/85H Display Control Panel. These are the full compass rose, sector, and sector with map formats. The EHSI provides the following display information:

Rose, Sector, or Sector With Map Formats

- Aircraft symbol
- Heading display and sensor type (MAG#, TRU#, or DG#) annunciation
- Selected heading cursor
- Active selected arrow pointer and lateral deviation bar
- Active selected course digital display
- Annunciation of deviation type (ANG, LIN, XTK, or B/C)
- Active selected course navigation sensor annunciation
- Active selected course to/from arrow (rose format)
- Active selected course 'TO'/'FR' annunciation (sector and sector with map formats)
- Second course arrow and course deviation (no deviation scale or to/from information is provided)
- Second course navigation sensor annunciation
- Vertical deviation display
- Bearing pointer display and sensor annunciation
- Distance to VOR/DME or waypoint
- Waypoint or DME identifier and waypoint alert
- DME hold annunciation
- Navigation data display (GS/TTG/ET, or TAS)
- Wind direction/speed from a compatible LNAV
- DR and MSG annunciation from a compatible LNAV
- Separate (second) heading cursor controlled by a compatible LNAV
- Weather radar target alert
- Reversionary source annunciation
- Flags and comparator warnings

Sector Format

- Expanded compass segment across the top of the display
- Aircraft symbol relocated at bottom center of display
- Second course VOR/DME/VOR-DME/waypoint symbol with identifier or number placed in rho-theta position with respect to the aircraft symbol and selected range
- Second course arrow automatically appears when a compatible LNAV is in a NAV to NAV capture mode
- Weather radar range and mode annunciation
- Weather radar display
- Range is selected from the WXP when a weather radar mode is selected or with the DH/RNG knob on the DCP when weather radar is not selected

Sector With Map Format

- Also uses the expanded compass segment
- Places VOR and/or waypoint symbols for both active selected and second course in proper rho-theta position with respect to the aircraft symbol and selected range
- DME ident is shown next to the VOR/DME symbols
- Selected course lines are drawn through the symbols
- Waypoint strings (up to five waypoints) are shown with their numbers or identifiers next to the symbol and with the symbols connected by desired track lines
- Weather radar display and range selection same as in sector format

EHSI Displays and Annunciators in the Rose Format

NOTE

For all EHSI formats, when the navigation sensor is selected from the on-side DCP, the course arrow, deviation bar, to-from arrow, digital course readout, navigation sensor annunciator, distance display, bearing pointer sensor annunciator, data display, and LIN/ANG annunciation are green. If control has been transferred to the cross-side, these displays are yellow for all navigation sensors.

Bearing Pointer Display—The bearing pointer indicates relative bearing to the selected NAVAID. In the EFIS-86C system, the bearing pointer is magenta and may be driven by a #1, #2, or #3 sensor (available sensors are determined during installation).

Bearing pointer selection is made using the BRG button on the DCP. Sequentially pushing the BRG button steps through the available sensors. The selected sensor is annunciated in magenta in the lower left of the EHSI for 5 seconds after a selection is made. This annunciation then blanks until another sensor is selected. If the BRG button is pushed and held for approximately 1/2 second, the bearing pointer turns off.

The letters 'A' (ADF), 'V' (VOR), 'T' (TCN), or 'N' (LNAV) appear beneath the head of the pointer to indicate the sensor type. If the bearing is from a #1 sensor, the pointer has a single bar on both the head and tail of the pointer. If the bearing is from a #2 or #3 sensor, the pointer has a double or triple bar head and tail respectively.

When VOR is the sensor and a localizer frequency is tuned, the bearing pointer is removed from view. The pointer is also removed from view if a failure is detected in the sensor currently driving the pointer.

If the heading sensor is not compatible with the navigation sensor, the letter 'A', 'V', 'T', or 'N' flashes as follows. For ADF, the 'A' flashes for 10 seconds then becomes steady, except GPS -004 version DPU/MPU do not flash. For VOR, TCN, or LNAV, the 'V', 'T', or 'N' flashes continuously until a compatible selection has been made (for LNAV, the flashing 'N' will be corrected and not flash if magnetic variation is available).

LNAV Annunciation—When the active selected course is from an LNAV, 'MSG' (flashing yellow) and 'DR' (yellow) annunciations may appear as received from the LNAV. 'MSG' appears below the waypoint identifier at the upper left of the EHSI, and 'DR' appears to the left of the distance display, also at the upper left of the EHSI.

The letters 'APPR', in white, are annunciated above the left dots of the lateral deviation scale if a compatible LNAV is controlling pitch and bank commands during an approach. For GPS -004 version DPU/MPU, a white 'SCALING' annunciation shows above 'APPR'.

Compass Display and Heading Sensor Type—The compass display is a full 360-degree compass rose with letters at the cardinal points and numbers at the 30-degree marks. Aircraft heading is read against the lubber line which has a 'MAG#', 'TRU#', 'DG#', or 'HDG#' annunciation to the right of the lubber line to indicate whether the source of heading information is magnetic or true referenced, or if DG (directional gyro) mode is selected or heading failure exists. Markings are provided every 45 degrees around the perimeter of the card to aid in procedure turns.

The '#' symbol behind the 'MAG', 'TRU', 'DG', or 'HDG' annunciation may be a '1', '2', '3', or '4', depending upon the number of sensors in the installation. The on-side 'MAG' annunciation is not displayed unless both sides have that sensor selected. Any time the left and right heading displays are from the same sensor, annunciations are displayed in yellow on both EHSIs. Secondary sensor annunciation is in white if not the same on both sides.

The following table describes the heading sensor annunciation. In all columns, the number symbol (#) shown before the digit represents either 'MAG', 'TRU', 'DG', or 'HDG'. In the first column, 'P' stands for the pilot's heading sensor, and 'C' stands for the copilot's heading sensor.

SELECTED HEADING SENSOR	ANNUNCIATION SHOWN ON PILOT'S EHSI	ANNUNCIATION SHOWN ON COPILOT'S EHSI
P= #1, C= #2	Note (1)	Note (2)
P= #1, C= #1	#1 (yellow)	#1 (yellow)
P= #2, C= #2	#2 (yellow)	#2 (yellow)
P= #3, C= #2	#3 (white)	Note (2)
P= #3, C= #3	#3 (yellow)	#3 (yellow)
P= #1, C= #3	Note (1)	#3 (white)
P= #1, C= #4	Note (1)	#4 (white)
P= #4, C= #4	#4 (yellow)	#4 (yellow)
P= #3, C= #4	#3 (white)	#4 (white)
NOTE (1) : Blank if MAG heading is selected, 'TRU1', 'DG1', or 'HDG1' for the other selections. NOTE (2) : Blank if MAG heading is selected, 'TRU2', 'DG2', or 'HDG2' for the other selections. NOTE (3) : The first five lines apply to both three and four sensor installations. NOTE (4) : The legend MAG was used as an example; it can be replaced by TRU, DG, or HDG.		

Selected Heading Display—Selected heading is shown by the location of two adjacent magenta rectangles (heading cursor) with respect to the compass card. The selected heading is repeated digitally in magenta in the lower right corner of the EHSI below the magenta letters 'HDG'.

The HDG knob on the CHP may be used to set the heading cursor or it can be made to rotate to the lubber line by momentarily pushing the PUSH HDG SYNC button on the HDG knob (the heading sync function may also be remotely actuated by mode logic from some digital autopilots). If a single CHP-85C is used and the left side DCP and cross-side data are valid, the right side selected heading cursor follows the left side. If two CHP-86Cs are used, or if cross-side data is invalid, each side's heading cursor acts independently.

In -002 and later version DPU and MPU, if a compatible LNAV is selected, it may be used to control the heading cursor instead of using the HDG knob on the CHP. A separate LNAV heading cursor shows around the outside edge of the compass card to show the selected LNAV heading. The cursor

shows in green as a single line arc parallel to the arc of the compass card with three perpendicular lines, one at each end and one in the center, from the arc to the outer edge of the compass card. When centered over the magenta heading cursor, it surrounds it on three sides. An annunciator 'FDHG#' (# is the navigator number) and a numeric readout show in green above the magenta HDG annunciator in the lower right corner of the display. A heading change entered from the LNAV will cause both heading cursors to slew to the entered heading. The present position of the selected heading cursor is also transmitted on the MPU bus so the LNAV can stay updated with any change in the selected heading. This feature does not inhibit the CHP from controlling the heading cursor. If a secondary LNAV is selected on either side, it will then control the heading cursor in place of that side's primary LNAV.

Active Selected Course Display—Active selected course is shown by the relationship of the solid double bar course arrow with respect to the compass card. The aircraft symbol pictorially shows actual aircraft position in relation to this selected course. The active selected course is repeated digitally in the lower left corner of the EHSI below the letters 'CRS'. The active selected course navigation sensor is annunciated at the lower left center of the EHSI.

The course is selected with the CRS knob on the CHP when in VOR, LOC, RNV, TCN, ADF, or MLS mode, or is automatically set by a lateral navigation system (INS, VLF, GPS, LRN, LNV, FMS, etc) when an LNAV is selected. (ADF course selection has been removed from -004 GPS version DPU and MPU). Once set, the course arrow rotates with the compass card to provide a continuous display of the selected course and course error with respect to aircraft heading. This data is supplied to the flight control system. If a localizer frequency has been tuned, VOR/LOC has been selected, and the selected course is more than 105 degrees from the lubber line, the ANG legend to the right of the selected course annunciation changes to B/C (back course).

If a VOR approach is being flown, the letters 'APPR', in green (on-side sensor) or yellow (cross-side sensor), are annunciated in the lower left quadrant of the EHSI as well as in the lower portion of the EADI. This annunciation indicates lateral deviation rescaling has occurred in support of VOR approach condition.

NOTE

During a VOR approach, select APPR mode only when within normal approach distances and not until after VOR capture. The autopilot automatically changes to dead reckoning (DR) mode when a VOR station is crossed. DR is *not* a recommended approach mode. **If the VOR approach is one that crosses the VOR on the way to the airport, select HDG mode at station passage instead of continuing with NAV or APPR mode (ensure the heading cursor is under the lubber line before selecting HDG mode).** The reason for this is if the distance between the VOR and the airport is short, the system may not have time to come out of DR mode after crossing the station.

If the selected navigation sensor is incompatible with the heading type (TRU or MAG), the sensor annunciation flashes until a compatible selection is made. If the sensor is an LNAV and magnetic variation is available, the LNAV data is corrected and no flashing occurs.

To/From Arrow—The to-from arrow indicates that the selected course is either 'to' or 'from' the selected NAVAID. The to-from arrow can also indicate the course to the next waypoint or the course from an overflowed waypoint when an LNAV is the navigation sensor. If a compatible LNAV is installed and LNAV is selected, the 'to' arrow flashes to indicate waypoint alert. The to-from arrow is removed from view whenever a localizer frequency is selected.

TWR-850 Weather Annunciation—When TGT (target alert) mode is selected on the WXP-850, the letters 'TGT' appear in cyan. Target alert notifies the pilot of the following two weather phenomena:

- When weather targets that are producing rainfall rates greater than 0.5 inch (12.7 millimetre) per hour are detected.
- When turbulence consisting of *precipitation* areas where *horizontal* wind velocity shifts in excess of 16.40 feet-per-second (5 metres-per-second) are detected.

NOTE

Doppler turbulence detection techniques used in the TWR-850 Turbulence Weather Radar System rely on the presence of at least light precipitation. The pilot should note that the TWR-850 system is **not** capable of detecting clear air turbulence.

The TGT mode provides a target alert 'window' from 7- to 200-nmi range (50 nmi maximum range for turbulence detection) and ± 15 degrees of dead ahead, regardless of the range selected on the WXP.

A flashing yellow box with the letters 'TGT' inscribed appears when a *precipitation* target is detected within the target alert 'window'. A flashing yellow box with the letters 'TRB' inscribed appears when an area of *precipitation related turbulence* is detected within the target alert 'window'. If areas of precipitation and turbulence are both detected within the target alert 'window', the flashing annunciation alternates between 'TGT' and 'TRB' but at a faster rate of flashing.

The flashing annunciation tells the pilot that areas of precipitation and/or turbulence have been detected, and weather radar should be put back on the display and analyzed.

Data Display—Pushing the NAV DTA button on the CHP sequentially selects elapsed time, ground speed, time-to-go, or true airspeed for display under the letters 'ET', 'GSP', 'TTG', or 'TAS' as applicable. The last data display selected is stored and reappears at the next power up.

Ground speed and time-to-go are from the same sensor selected for the active course, and are the same color. If that sensor is VOR, the calculation occurs in the EFIS system. Otherwise, the navigation system transmits the data. No-computed-data from a valid VOR/DME source replaces the TTG or GS numeric readout with dashes in the same color as the annunciator label. Failure of the VOR/DME or LNAV data source; or detection of an inactive data bus from the data source turns off the numeric readout for TTG or GS. Failure or an invalid flag from the air data source turns off the TAS numeric readout and failure of a DCP turns off the ET numeric readout.

True airspeed shows cyan if from the on-side air data system and yellow if from the cross-side system.

Ground speed and true airspeed display range is 0-999 knots with 1 knot resolution, and time-to-go display range is 0 to 399 minutes with 1 minute resolution.

Elapsed time is shown in minutes and seconds up to 59:59 and then in hours and minutes up to H9:59. Above H9:59, the display is dashed. Sequentially pushing the ET button on the CHP controls the reset/start/stop cycle of the timer. The elapsed time display is cyan unless DCP control has been transferred in which case the display changes to yellow. Sequentially pushing an installer supplied switch controls the reset/start/stop cycle of the timer. The first push of the switch starts the timer. The second push stops the count and the next push resets the timer to zero. Once the timer has been started, other data may be selected for display via the NAV DTA button on CHP without disturbing the count.

The -002 and later versions DPU and MPU, in combination with appropriate LNAV systems, can show LNAV-generated TTG and GS data. In this mode the data is annunciated FTTG and FGS to identify LNAV-generated data. For this data mode to occur, the DME/VOR sensor must be turned on, be valid and in the no-computed-data mode or dashes will be displayed. All other functions of FTTG and FGS are the same as described for the TTG and GS annunciators.

Wind Display—Relative wind speed (digitally in knots) and direction (via an arrow), if available as an input from a compatible LNAV, is displayed full time to the left of the lubber line.

The wind display is white if from the on-side LNAV and is yellow if from the cross-side LNAV. If the on-side LNAV is not valid, the wind display automatically switches to the cross-side system. The wind display is blanked if data is not available from either LNAV.

Second Course Display—Second course arrow selection is made using the 2ND CRS button on the DCP. Sequentially pushing the 2ND CRS button steps through the available sensors. If the 2ND CRS button is pushed and held for approximately 1/2 second, the second course arrow turns off.

When the second course arrow has been selected for display, it appears in cyan as a single-line course arrow with a solid arrowhead. The center portion of the course arrow is the second course lateral deviation bar. The deviation displayed is always angular and is never linearized. No deviation scale or to-from indicator is provided for the second course arrow. The navigation sensor selected for the second course is annunciated in cyan immediately above the navigation sensor annunciation for the active selected course arrow.

If the second course navigation sensor is incompatible with the heading sensor (true or magnetic), the navigation sensor annunciator flashes until a compatible selection is made. If the sensor is an LNAV and magnetic variation is available, the LNAV data is corrected and no flashing occurs.

second course VOR/LOC, TCN, or RNV sensors—When the second course is a VOR/LOC, TCN, or RNV, the on-side CHP's selected CRS knob controls the course setting if the CRS knob is not being used for a different sensor. If the CRS knob is being used for the active selected course, the second course arrow is set to the bearing with the deviation zeroed.

If the DME associated with the displayed VOR, TCN, or RNV is put into hold, a yellow 'H' is added to the right of the numeral in the second course navigation sensor annunciation. If the DME associated with the displayed VOR, TCN, or RNV fails, a yellow 'D' is added to the right of the numeral in the navigation sensor annunciation.

When the second course is a VOR or TCN and a localizer frequency has been tuned, the sensor annunciation changes to 'LOC1' or 'LOC2'.

second course LNAV sensors—One of two second course related lateral navigation sensors may also be displayed. These are: PLOC (LNAV pseudo localizer) and FLOC (LNAV controlled localizer).

If pseudo ILS approach data is received from the LNAV selected as the active sensor, the EFIS-86C uses this data in place of real ILS data. This is annunciated by forcing the second course sensor annunciation to be 'PLOC#' where '#' is the active LNAV reference number. The second course arrow is driven by data from the LNAV and overrides any pilot selected second course. This second course data should coincide with the actual runway heading. When the pseudo localizer is within capture range, the second course PLOC replaces the active sensor, and the only remaining deviation is PLOC.

The LNAV controlled localizer is selected on the LNAV's CDU and is displayed on the EHSI when the LNAV is selected as the active sensor. The operation is the same as the pseudo ILS described above except that the annunciation is 'FLOC#'.

Vertical Deviation Display—A scale and pointer on the right side of the EHSI indicates deviation from a selected vertical path. The scale consists of two white dots above and below a yellow center index. The pointer is green from an on-side sensor and yellow if from a cross-side sensor. The aircraft is below the selected vertical path if the pointer is deflected upward.

The output from one of five vertical deviation sensors may be displayed. The sensors are GS, MGP (microwave glidepath), PGS (LNAV pseudo glideslope), FGS (LNAV controlled glideslope), and VNAV (vertical navigation). One of three scale types may be displayed. The conditions governing these are as follows:

- GS = VOR/LOC selected, LOC tuned and not back course.
- MGP = MLS selected.
- PGS = LRN selected and pseudo ILS capture initiated.
- FGS = LRN selected and ILS capture initiated.
- VNAV = VNAV enabled and no GS, MGP, PGS, or FGS display.
- None = None of the above

GS and MGP vertical deviation—For GS and MGP, each of the inner scale dots represents a 1/4-degree displacement and each of the outer scale dots represents a 1/2-degree displacement. The deviation pointer shape is a dual triangle, green for on-side and yellow for cross-side. When GS is the sensor and back localizer mode is detected, the glideslope scale is removed from view and 'B/C' is annunciated to the right of the active course navigation sensor annunciator.

PGS and FGS vertical deviation—When PGS or FGS is the selected vertical deviation sensor, the letters 'PGS' or 'FGS' are annunciated vertically to the right of the top two dots on the vertical deviation scale. This annunciation coincides with the PGS and FGS annunciation on the EADI for LNAV pseudo glideslope and LNAV controlled glideslope respectively.

The shape of the deviation pointer is the same as for GS and MGP in all cases, except for -002 and later DPU and MPU. In this case the pointer symbol shows as a diamond on its side with a horizontal center line.

If a compatible LNAV is selected and is in approach mode, one-dot deviation from the selected vertical path represents 100-foot displacement. LNAV approach mode is annunciated by the white letters 'APPR' located just above left center of the display.

VNAV vertical deviation—The VNAV deviation pointer is a green star-shaped symbol for the on-side sensor and yellow star-shaped symbol for cross-side. The vertical deviation pointer represents the center of the VNAV track and the center of the scale represents the airplane position relative to the VNAV track.

VNAV deviation scaling varies with DPU/MPU configuration status as follows:

- For -001 version DPU/MPU, each of the inner deviation scale dots represent 200 feet of displacement and each of the outer dots represent 400 feet of displacement from the vertical flight path.
- For -002 and -003 version DPU/MPU, each of the inner deviation scale dots represent 500 feet of displacement and each of the outer dots represent 1000 feet of displacement from the vertical flight path.

- For the -004 DPU/MPU version, if the LNAV is using GPS vertical navigation data, the FMS will send the EFIS a scaling word which can range from 1024 feet per dot enroute to 1.00 foot per dot in approach.
- For all LNAVs set to FMS approach mode, each of the inner deviation scale dots represent 100 feet of displacement and each of the outer dots represent 200 feet of displacement from the vertical flight path. The white annunciator 'APPR' shows above the left two dots of the lateral deviation scale.

Lateral Deviation Bar—The center portion of the course arrow is the lateral deviation bar which represents the centerline of the selected course. This bar moves laterally with respect to the course arrow and the white deviation dots to indicate deviation from the selected course.

If a localizer frequency is tuned, VOR/LOC is selected on the DCP, and the selected course is more than 105 degrees from the lubber line, the annunciation 'B/C' (back course) appears to the right of the active course navigation sensor annunciator. When 'B/C' is annunciated, the course deviation automatically reverses to provide the correct left/right deviation for back course approaches.

LIN/ANG/XTK and B/C (Back Course) Annunciations—When flying VOR, TCN, or RNV with the deviation linearized, 'LIN' is annunciated. Linear deviation optimizes computation while tracking a VOR, TCN, or RNV course, especially close in to the station. EFIS always linearizes deviation whenever DME is present and not in DME hold, unless an external switch is actuated to defeat linearization and display angular deviation ('ANG' is annunciated).

Linear deviation is turned off if a localizer frequency is tuned in either VOR or APPR mode. Angular deviation is used in approach mode, but is not annunciated. 'XTK' (cross-track deviation) is annunciated when the LNAV is flying an offset.

When VOR, TCN, or RNV is the active sensor and the deviation changes from 'LIN' to 'ANG' (such as would occur if the DME is lost or DME hold is selected), the 'ANG' annunciator flashes for 10 seconds before becoming steady. The flashing can be eliminated by pushing the master warn reset.

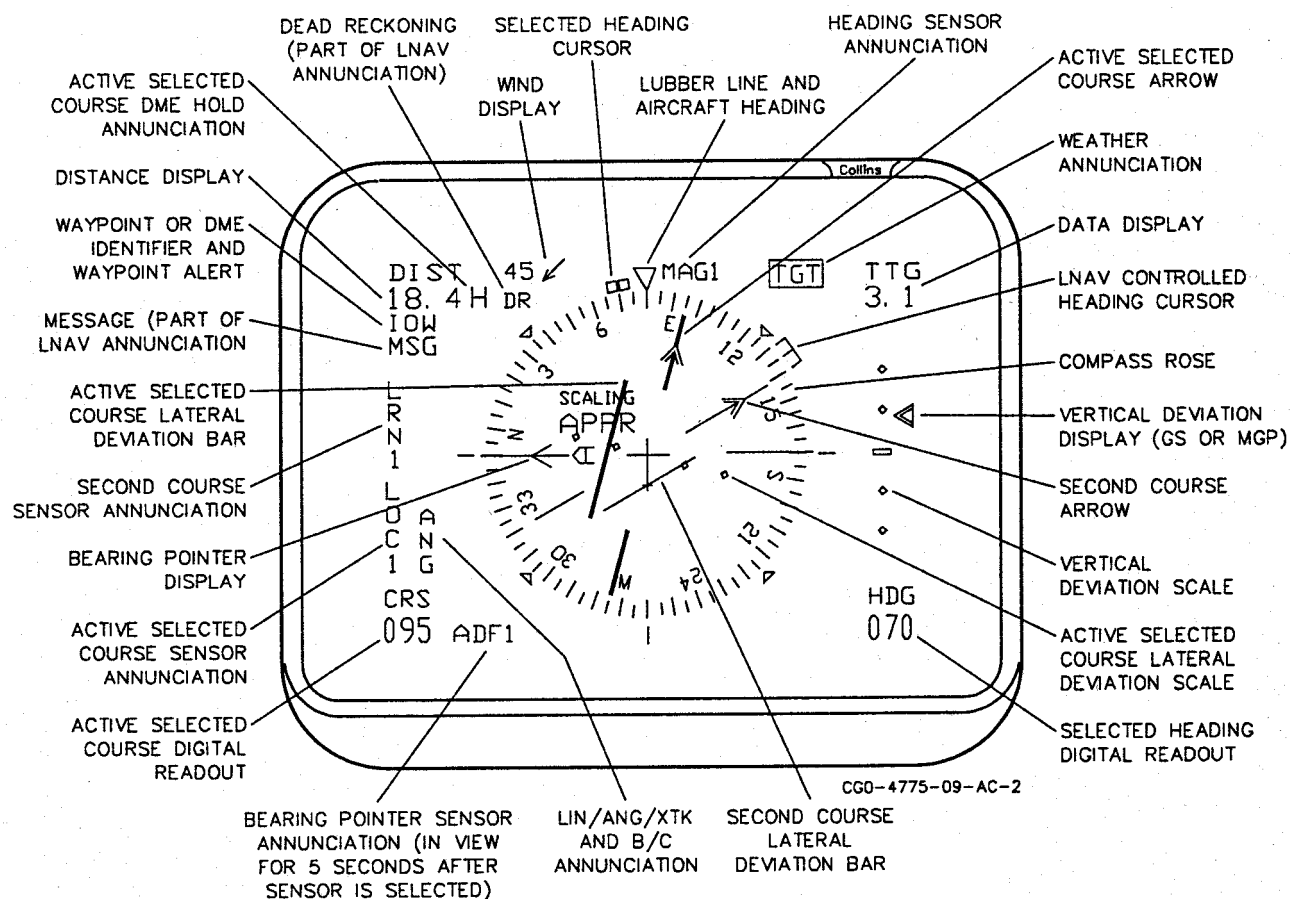
'B/C' (back course) is described in the preceding lateral deviation bar paragraphs.

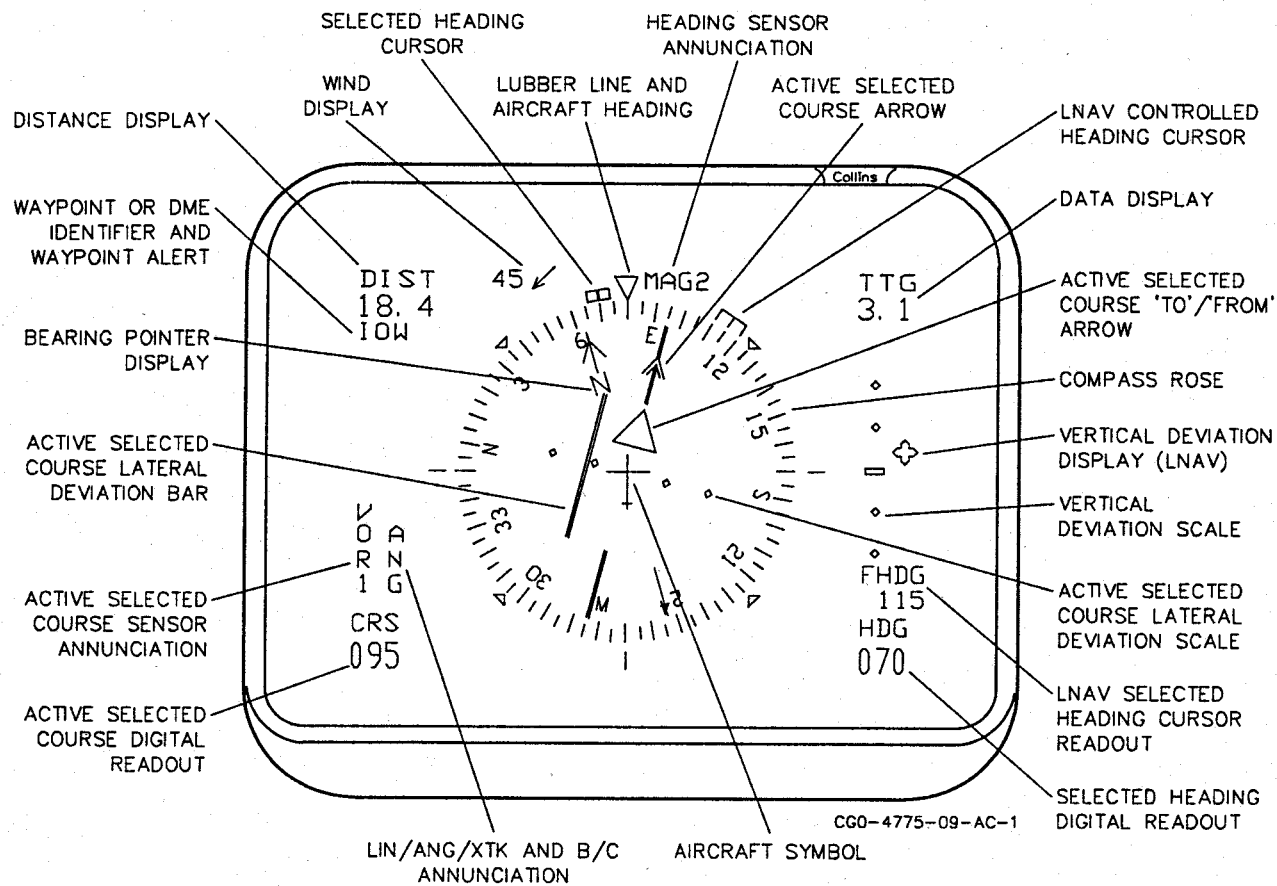
Waypoint or DME Identifier and Waypoint Alert—The station ident of the selected DME or the ident of the first waypoint is shown directly below the digital distance display. When the ident is for a waypoint, it flashes to indicate waypoint alert.

Distance Display—The distance to the VOR, TCN, RNV, or waypoint selected for display by the active course arrow is shown digitally below the letters 'DIST'. Resolution is 0.1 nmi below 100 nmi, and 1 nmi above 100 nmi. When DME hold has been selected, the letter 'H', in yellow, appears to the right of the digits.

NOTE

The displays shown below and on the next page are for illustrative purposes only. All of the pointers, symbols, annunciators, etc. may not appear together on an actual display.





EHSI Displays and Annunciators in the Sector Format

Compass Sector Display—The expanded compass sector displayed in the sector format is similar to the full compass card displayed in the rose format except the EHSI displays a ± 40 -degree compass segment at the top of the display and the aircraft symbol is placed at the bottom center of the display. The compass sector, active selected course arrow, second course arrow, heading cursor, and bearing pointer rotate about the intersection of the wings and fuselage of the aircraft symbol.

All other functions of the compass display (colors, annunciation of mag or true heading, directional gyro mode, on-side or cross-side heading sensor, heading flag, and comparator warning) are identical to the rose format.

Selected Heading Display—Selected heading is shown by the location of two adjacent magenta rectangles (heading cursor) with respect to the compass sector when on scale.

When selected heading is off scale, the heading cursor disappears and is replaced by a small magenta arrow *above* the appropriate end of the compass sector closest to the selected heading value. As the selected heading passes through 180 degrees from the present selected heading, the small magenta arrow moves *above* the other end of the compass sector and reverses direction so that it always points in the direction in which the selected heading is the closest. The digital selected heading readout in the lower right hand corner always remains in view and displays the valid selected heading unless the DCP is flagged.

All other functions of the selected heading display are the same as described for the rose format.

Active Selected Course Display—Active selected course is shown by a solid double bar course arrow related to the compass segment with a deviation bar near the aircraft symbol. A deviation scale which rotates with the course arrow is also shown. The deviation bar and scale can be wholly or partially out of view for some selected courses.

As the course arrow passes off scale, it disappears and is replaced by a small magenta arrow *below* the appropriate end of the compass sector closest to the selected course value. As the selected course passes through 180 degrees from the present selected course, the small magenta arrow moves *below* the other end of the compass sector and reverses direction so that it always points in the direction in which the selected course is the closest. The digital selected course readout in the lower left hand corner always remains in view and displays the valid selected course.

All other functions and colors of the selected course display are the same as described for the rose format.

Bearing Pointer Display—All functions and colors of the bearing pointer display are identical to the rose format except that the pointer rotates around the relocated aircraft symbol.

Vertical Deviation Display—All functions and colors of the vertical deviation display are identical to the rose format.

Range Display—When weather radar is not selected via the RDR button on the DCP, range is selected by pulling the DH/RNG knob on the DCP out to the PULL RANGE position, and then turning the knob. Range is displayed as a dashed cyan range arc located half-way between the compass sector and the aircraft symbol. Half of the full scale range is annunciated at the right hand end of the range arc. Selectable full-scale ranges are 5, 10, 25, 50, 100, 200, 300, and 600 nmi. Half-scale annunciated ranges are 2.5, 5, 12.5, 25, 50, 100, 150, and 300 nmi.

To/From Display—To-from is indicated by the letters 'TO' or 'FR' appearing in the lower part of the display to the right of the aircraft symbol. When LNAV is the selected course and the LNAV outputs a waypoint alert, the 'TO' or 'FR' annunciator flashes.

Weather Radar Display—Adding a weather radar display is accomplished by having a weather detection or mapping mode selected on the WXP-850(), and then pushing the RDR button on the DCP. Pushing the RDR button again removes the weather radar display.

With RDR mode selected on the WXP, detectable weather appears as one of four colors: green, yellow, red, and magenta (least reflective to most reflective), plus the black screen itself.

Various annunciations from the weather radar system are displayed at both ends of the range arc which is a solid arc in weather radar mode. The arc is located at midrange and half-scale range is annunciated at the right hand end of the arc. Full-scale range extends to the perimeter of the compass segment. In weather radar mode, the range is selected using the RANGE control on the WXP.

If the WXP is turned off or loss of data from the WXP is detected when weather radar information is being displayed, the range arc and weather displays are removed and the range defaults to 300 nmi.

Second Course Display—A second course may be selected using the 2ND CRS button on the DCP as described for the rose format. The second course is displayed in cyan and its sensor is annunciated in cyan above the active selected course sensor.

When the second course is VOR, an octagon-shaped cyan symbol is placed in proper rho-theta position with respect to selected range and aircraft heading. The station ident, if available, is displayed next to the symbol. If the second course sensor is an RNV, the symbol is star shaped, and for TCN, a VORTAC symbol is used.

If the DME associated with a displayed VOR, TCN, or RNV is put into hold or if the DME data is invalid, the VOR, TCN, or RNV symbol is removed from view and a yellow 'H' (for hold) or a yellow 'D' (for DME invalid) is added to the right of the numeral in the second course navigation sensor annunciation.

When the second course is a VOR or TCN and a localizer frequency is tuned, no symbol is displayed and the sensor annunciation shows 'LOC1' or 'LOC2'.

If the second course sensor is an LNAV or RNV, the 'to' waypoint is displayed in cyan as a star-shaped symbol shown in the proper rho-theta position with respect to selected range and aircraft heading. The waypoint number or identifier, if available, is also shown in cyan with the number or identifier being next to the symbol. If the LNAV is compatible, the waypoint symbol flashes for waypoint alert.

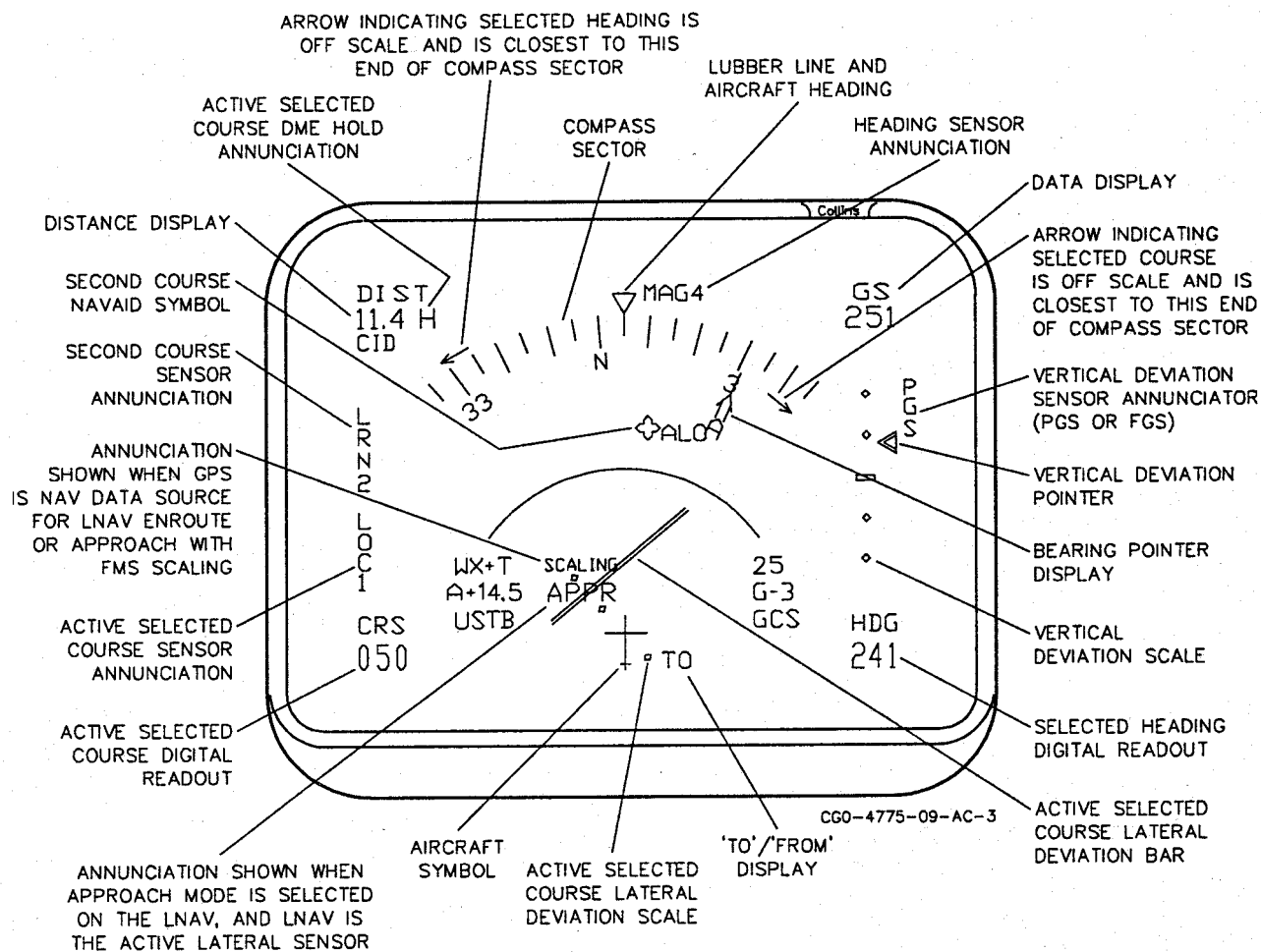
Second course related PLOC and FLOC operate the same as described for the rose format.

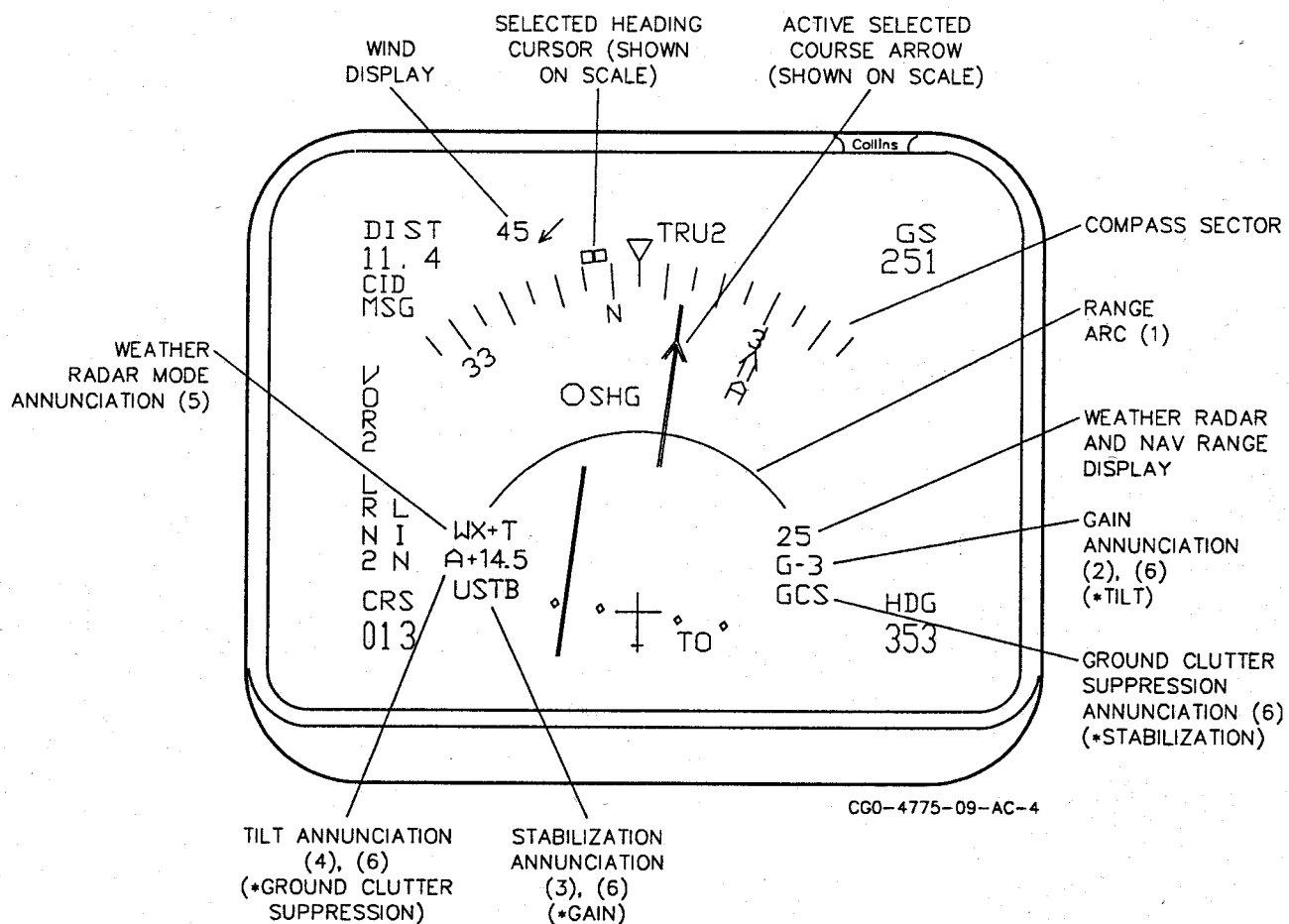
If the second course navigation sensor is incompatible with the heading sensor (true or magnetic), the navigation sensor annunciator flashes until a compatible selection is made. If the sensor is an LNAV and magnetic variation is available, the LNAV data is corrected and no flashing occurs.

Peripheral Displays—All functions and colors of the distance display, data display, weather annunciation, and navigation annunciation are the same as described for the rose format.

NOTE

The displays shown below and on the next page are for illustrative purposes only. All of the pointers, symbols, annunciators, etc. may not appear together on an actual display.





NOTES:

(*) THIS SYMBOL DENOTES AND IDENTIFIES -003 AND LATER VERSION DPU/MPU ANNUNCIATIONS.

(1) RANGE ARC IS WHITE FOR MAP MODE AND CYAN FOR ALL OTHER MODES. THE RANGE ARC IS SOLID WHEN RADAR IS SELECTED AND VALID, AND DASHED WHEN RADAR IS NOT SELECTED AND NAV MODE IS SELECTED.

(2) ANNUNCIATIONS ARE G+3, G+2, G+1, BLANK, G-1, G-2, OR G-3.

(3) NO ANNUNCIATION WHEN ANTENNA IS STABILIZED.

(4) NORMAL TILT ANNUNCIATION IS + OR -15 DEGREES WITH THE NEAREST QUARTER DEGREE ALSO SHOWN. AUTOTILT ANNUNCIATION IS IDENTICAL, ONLY THE DIGITS ARE PRECEDED WITH AN 'A'.

(5) ANNUNCIATED MODES IN THIS LOCATION ARE 'OFF', 'STBY', 'TEST', 'TGT', 'MAP', 'WX', 'WX+T', AND 'TURB'. WHEN 'TGT' MODE IS SELECTED, THE RANGE ARC BECOMES DASHED, AND ALL RADAR INFORMATION EXCEPT THE RANGE, THE MODE, AND THE 'TGT' AND/OR 'TRB' ANNUNCIATIONS ARE REMOVED FROM THE DISPLAY. ALL MODES EXCEPT 'TGT' ALTERNATE WITH 'HOLD' WHEN WEATHER RADAR HOLD IS SELECTED. ALL MODES ALTERNATE WITH 'FAULT' IF A FAULT IS DETECTED. MODE COLOR IS CYAN IF UNSLAVED. IF SLAVED, MODE ANNUNCIATION TURNS YELLOW AND AN 'X' PRECEDES THE MODE ANNUNCIATION.

(6) COLORS ARE CYAN IF UNSLAVED, AND YELLOW IF SLAVED.

EHSI Displays and Annunciators in the Sector with Map Format

Compass Sector Display—All functions and colors of the compass display for the sector with map format are the same as described for the sector format.

Active Selected Course Display—When the active selected course is VOR, an octagon-shaped symbol is placed in proper rho-theta position with respect to selected range and the aircraft symbol. The station ident is displayed next to the VOR symbol on the side nearest the vertical centerline of the display. A selected course line is drawn through the symbol with its position controlled by the applicable CRS knob on the CHP. This line is solid on the 'to' side of the symbol and dashed on the 'from' side. If the symbol is off scale with respect to selected range and/or bearing, the course line is drawn with an arrow pointing toward the station, and the ident is displayed on the line. The navigation sensor is annunciated at the lower left side of the display. The VOR symbol, course line, station identifier, and navigation sensor annunciation are green if selected from the on-side DCP or yellow if selected from the cross-side DCP. If the active selected course is RNV, the symbol is star shaped, and for TCN, a VORTAC symbol is used.

If a localizer frequency is tuned, the VOR symbol, ident, and course line are removed from view and the sensor annunciation shows 'LOC1' or 'LOC2'.

If DME hold is selected, the VOR symbol, ident, and course line are removed from view. The sensor annunciator still displays the selected sensor, and DME hold is indicated by a yellow 'H' placed to the right of the distance display.

When the active NAV sensor is an LNAV, a flight plan string of star-shaped waypoints, VORTAC-shaped NAVAIDS, and/or airport-shaped symbols (up to five symbols) are located in their proper rho-theta position with respect to the selected range and the aircraft symbol. The symbols are identified by numbers or idents placed next to the symbol and are connected by track lines as output by the LNAV system. An arrow is drawn on the track line pointing at the 'to' waypoint which may be a waypoint, a NAVAID, or an airport as determined by the LNAV and the selected flight plan. If the 'to' waypoint is off scale, an arrow is drawn on the track line pointing at the 'to' waypoint, and 'WPT #' or waypoint ident is displayed on the line. The 'to' waypoint symbol flashes to indicate waypoint alert.

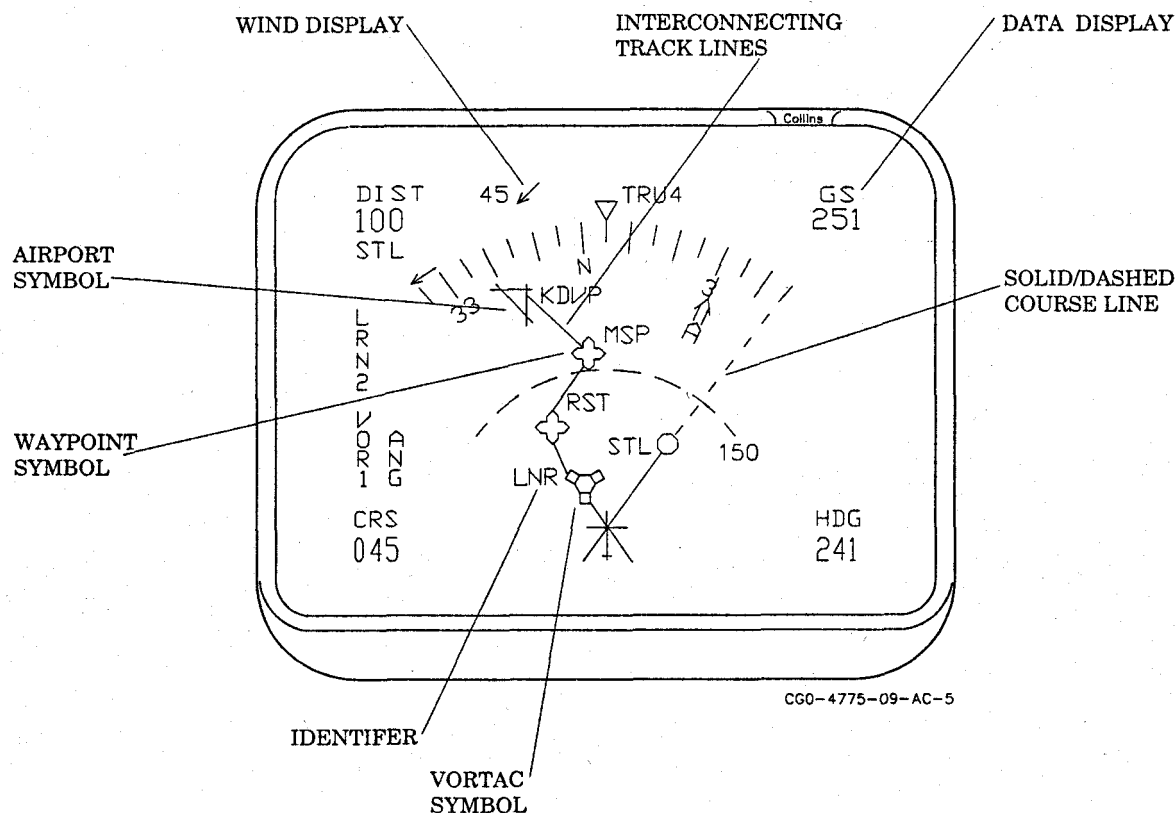
In -003 and later version DPU and MPU configurations, operations are as described above except that up to five waypoints and/or intersections, with their identifiers, can show on the display with connecting desired track lines.

CAUTION

The airport symbols that may be displayed in the sector with map format are for display purposes only and do not in any way show the actual runway orientation of the airport represented by the symbol. The pilot should always refer to approved navigation charts or approach plates for the actual runway orientation of any airport.

LNAV approach modes are not displayed on the sector with map format. If an LNAV approach mode is selected, the track lines are removed from the display.

If the navigation sensor is incompatible with the type of heading (true or magnetic), the sensor annunciator flashes until a compatible selection is made. If the sensor is an LNAV and magnetic variation is available, the LNAV data is corrected and no flashing occurs.



Second Course Display—A second course may be selected using the 2ND CRS button on the DCP as described for the rose format. The second course is displayed in cyan, and its sensor is annunciated in cyan above the active selected course sensor.

When the second course is a VOR, an octagon-shaped symbol is placed in proper rho-theta position with respect to selected range and station bearing, and a cyan course line is drawn through it. The station ident is displayed next to the symbol. The course line is rotated to the selected course angle and is solid on the 'to' side of the symbol and dashed on the 'from' side. If the station is off scale, the course line is drawn with an arrow pointing to the station, and the station ident is displayed on the line. If the second course sensor is an RNV, the symbol is star shaped, and for TCN, a VORTAC symbol is used.

When the second course is a VOR/LOC, TCN, or RNV, the on-side CHP's selected CRS knob controls the course setting if the CRS knob is not being used for a different sensor. If the CRS knob is being used for the active selected course, the second course arrow is set to the bearing with the deviation zeroed.

If a localizer frequency is tuned, no symbol, ident, or course line is displayed and the sensor annunciator shows 'LOC1' or 'LOC2'. If MLS is selected, no symbol or course line is displayed and the sensor annunciator shows 'MLS1' or 'MLS2'.

If the DME associated with a displayed VOR, TCN, or RNV is put into hold or fails, the symbols and connecting lines are removed from view and a yellow 'H' (for hold) or a yellow 'D' (for fail) is added to the right of the numeral in the sensor annunciation.

When the second course is an LNAV, a flight plan string of star-shaped waypoints, VORTAC-shaped NAVAIDS, and/or airport symbols (up to five symbols) are located in their proper rho-theta position. The symbols are identified by 'WPT #' or ident placed next to the symbol. The symbols are connected by track lines as output by the LNAV system. An arrow is drawn on the track line pointing at the 'to' waypoint. If the 'to' waypoint is off scale, the track line is still displayed with the 'WPT #' or ident shown on the line. The 'to' waypoint symbol flashes to indicate waypoint alert.

In -003 and later version DPU and MPU configurations, operations are as described above except that up to five waypoints and/or intersections, with their identifiers, can show on the display with connecting desired track lines.

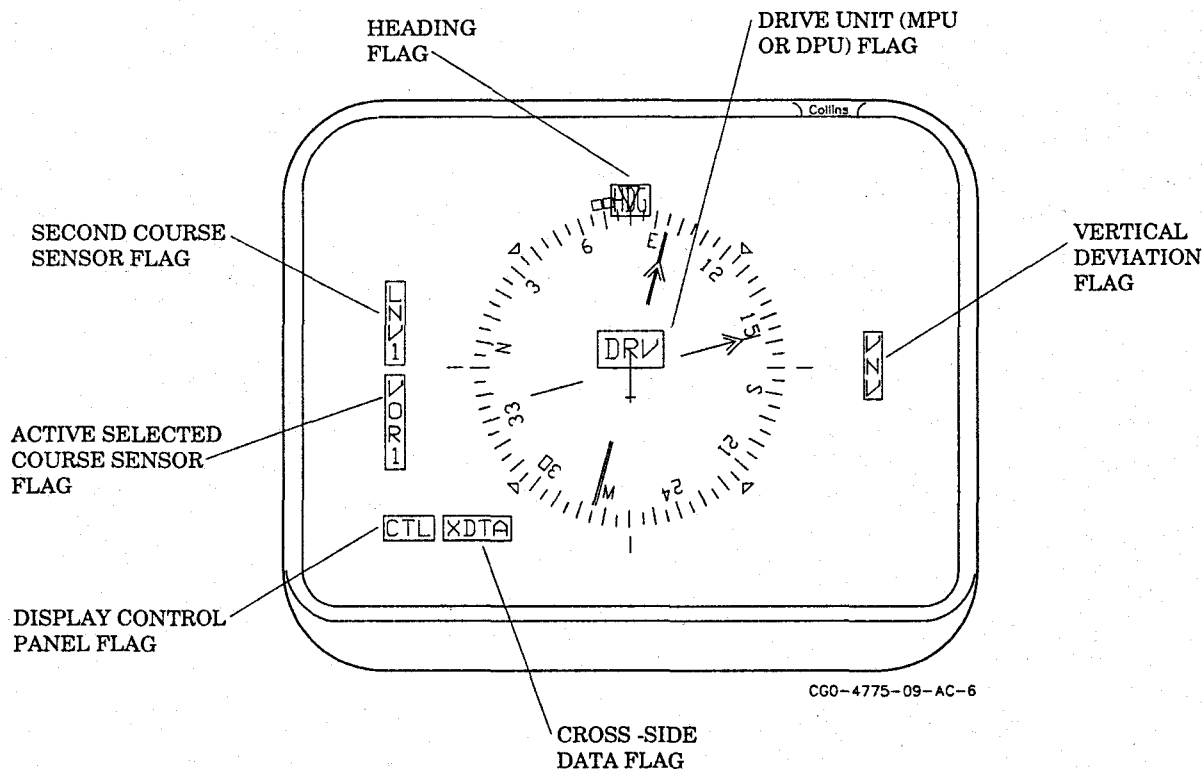
If the second course navigation sensor is incompatible with the heading sensor (true or magnetic), the navigation sensor annunciator flashes until a compatible selection is made. If the sensor is an LNAV and magnetic variation is available, the LNAV data is corrected and no flashing occurs.

Other MAP Format Displays—All functions and colors of the selected heading display, bearing display, range display, vertical deviation display, distance display, data display, weather annunciation, and navigation annunciation are the same as described for the sector format.

EHSI Warning Flags

NOTE

Unless otherwise noted, when an EHSI warning flag appears, it flashes for 10 seconds and then becomes steady. The flashing can be eliminated by pushing the master warn reset.



Distance Flag—If invalid distance data is detected from a DME or LNAV, the distance digits are replaced with dashes that are the same color as the active course display. If no computed data is received from the DME, the distance display blanks. Dashes remain displayed if no computed data is received from an LNAV. Loss of signal from either the DME or LNAV also causes the distance display to blank.

Heading Flag—If a failure of the heading system occurs, a red box with the letters 'HDG' inscribed appears superimposed on the lubber line. If the heading synchro monitor detects a failure, the HDG flag also appears. Nothing is removed from the EHSI when a HDG flag appears.

Drive Unit (MPU or DPU) Flag—If an MPU or DPU fails, the non-flashing letters 'DRV' appear in red centered on the display. If the EHSI is being driven by the MPU, the flag is 'MPU FAIL'. Nothing is removed from the EHSI when a DRV flag appears.

Data Flags—If invalid time-to-go, or ground speed is detected, the digits are replaced with dashes that are the same color as the active course display. If invalid true airspeed or elapsed time is detected, the digits are removed from the display. If the digital bus monitor detects an inactive bus, the data or dashes are removed from the display.

Wind Flag—If the on-side LNAV supplying wind data is invalid, the display automatically switches to the cross-side LNAV and turns yellow. If wind data from the cross-side LNAV is also invalid, the wind display is blanked.

Vertical Deviation Flags—If a failure of the glideslope or MLS glidepath receiver is detected, a red box with the letters 'GS' or 'MGP' inscribed appears in place of the scale and pointer. If a VNAV flag is detected, a red box with the letters 'VNV' inscribed appears in place of the scale and pointer. If an LNAV pseudo glideslope failure is detected, a red box with the letters 'PGS' inscribed appears in place of the scale and pointer. If an LNAV controlled glideslope flag is detected, a red box with the letters 'FGS' inscribed appears in place of the scale and pointer.

Display Control Panel Flag—If a failure of the display control panel occurs, a red box with the letters 'CTL' inscribed appears in the lower left corner replacing the digital selected course readout. The selected heading and NAV data displays are also removed from view. The EFIS-86C continues to operate in the mode that was active prior to the DCP failure.

Cross-Side Data Flag—If a failure of the cross-side data bus occurs, a red box with the letters 'XDTA' inscribed appears in the lower left of the EHSI. Data from the cross-side sensors is no longer available, and any display driven by cross-side data is flagged.

Bearing Pointer Flag—If a bearing pointer sensor failure occurs, the bearing pointer is removed from view.

Active Selected Course Flag—If a navigation sensor failure is detected while it is selected for the active selected course, the active selected course sensor annunciator becomes red and boxed, but does not flash. The course deviation bar and scale, to/from display, station identifier, NAV data display, wind display, and the bearing pointer are removed from view. If they are being displayed, NAVAID symbols and connecting lines are also removed from view.

If a failure is detected by a synchro or bus parameter monitor, the corresponding flag appears and the lateral deviation bar and scale are removed from view.

If LNAV approach mode is selected and a NAV sensor failure is detected, the sensor annunciation turns red, becomes boxed. For a sensor failure that allows the LNAV to remain in approach mode,

LNAV flags ('PGS', 'FGS', 'PLOC', or 'FLOC') also appear beside the vertical and lateral deviation scales. LNAV approach annunciation remains in view. For a sensor failure that cancels the approach mode, the vertical and lateral deviation scales are removed from view; no flags are shown; and there are no reversions back to the previously selected vertical sensor.

Second Course Flag (Rose Format)—If a navigation sensor failure is detected while it is selected for the second course, the second course sensor annunciator becomes red and boxed. The second course arrow deviation bar is removed from view if the failure was from the internal VOR, TCN, RNV, or LNAV synchro monitor or bus parameter monitors.

Second Course Flag (Sector and Sector with Map Formats)—If a navigation sensor failure is detected while it is selected for the second course, the NAVAID symbol (sector format) or NAVAID symbols and connecting lines (sector with map format) are removed from view and the second course navigation sensor annunciator becomes red and boxed.

EHSI Comparators and Master Warn

Comparator monitoring is performed in each DPU and MPU in the EFIS-86C system. On-side data is read and stored by each DPU and the MPU. Each DPU sends its information to the cross-side DPU so that each of the processors has the information required to perform the comparison function. When the MPU is driving the EHSI, it provides the comparator function in place of the failed DPU.

EHSI Comparator Warnings

NOTE

When an EADI comparator warning occurs, it appears nonflashing. The messages disappear when the external comparator reset switch is actuated, if the comparator error is no longer present.

In addition to the comparator warnings displayed on the EHSI, the EFIS-86C provides individual heading and ILS comparison outputs from each DPU and from the MPU when in drive transfer. These comparator warning outputs are used to turn on remote mounted heading and ILS annunciators (not part of the EFIS-86C system)

Comparator Reset—Comparator reset operates the same as described for the EADI.

Master Warn and Master Warn Reset—The master warn and master warn reset operates the same as described for the EADI.

Heading Comparator

Enable Logic—A comparison of the two headings is made when:

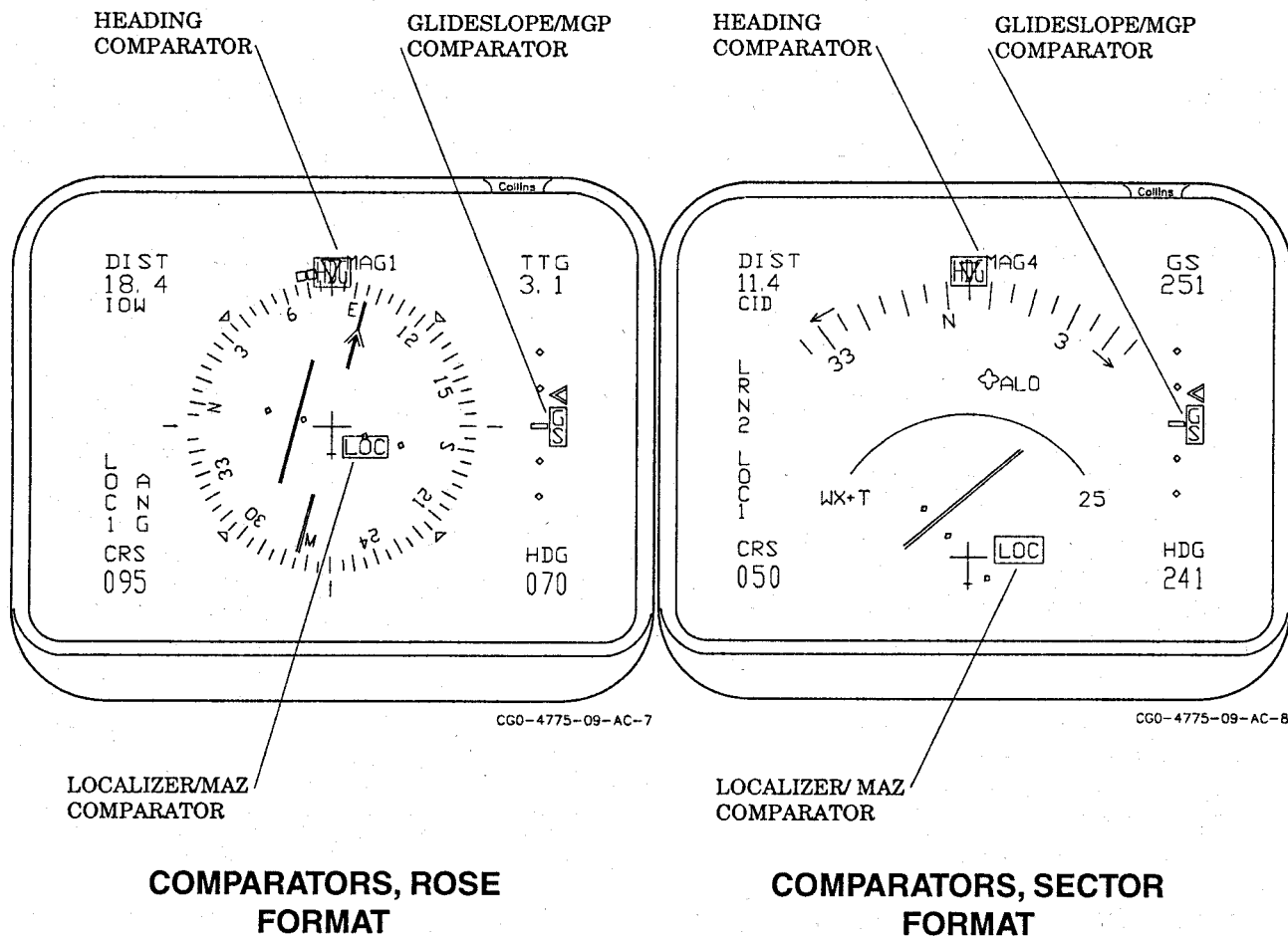
- Neither side has a HDG flag
- Both sides are using the same heading reference (magnetic or true)
- No cross-side data failures are detected
- Attitude is valid and roll is less than 20°

NOTE

DG mode does not inhibit the compare function if both headings are magnetic.

Comparison Limits—A heading comparator warning occurs when the enable conditions are met and the two headings differ by more than $6^\circ + 1/2$ of the absolute value of the roll angle.

Annunciation—When a heading comparator warning is detected, the letters 'HDG', boxed and in yellow, appear superimposed on the lubber line. The external 'HDG' comparator annunciator (if installed) and the master warn comparator also turn on.



EHSI Category II Excessive Deviation Warnings

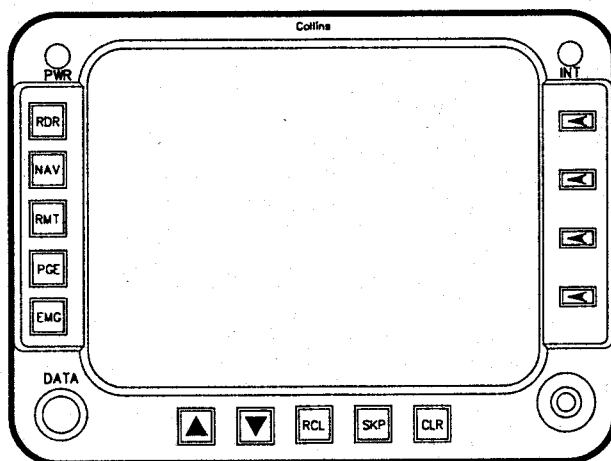
In the EFIS-86C system, there are no special visual cues that appear on the EHSI to indicate CAT II excessive deviation warnings. All CAT II excessive deviation warnings appear on the EADI. Refer to the 'EADI Comparator Warnings' paragraphs for operational details on EADI comparators and EADI CAT II excessive deviation warnings.

MFD-85B DISPLAY FORMATS

The MFD-85B Multifunction Display (MFD) is a flight planning unit used to increase the operational capabilities of the EFIS-85C(14)/86C(14) system. The MFD-85B is used in EFIS-85C(14)/86C(14) installations that do not have a TCAS system or in which the TCAS system is not connected to the EFIS system. Installations in which a TCAS system is connected to the EFIS-85C(14)/86C(14) system use the MFD-85C described later in this book. Standard functions shown on the MFD-85B include:

- weather radar
- three selections of pictorial navigation maps ("heading up", "north up with aircraft centered", and "north up maximum view")
- an additional fourth selection (map plan) if a compatible LNAV is installed
- and page data for checklists.

The MFD can also show information from compatible LNAV sources in remote data display modes. Also, an extended data display mode can show additional detailed inflight information, from a compatible LNAV source, about navigation sensor data and data for several modes of flight (cruise, approach and CAT II checklists). In addition, diagnostic information for use by maintenance personnel can be shown on the MFD.



CGO-3098-40-AC-1

MFD-85B Multifunction Display

Overview of Standard Functions

Weather Radar—The radar display mode shows detected precipitation from the TWR-850 Turbulence Weather Radar System on the MFD. Weather radar mode, range, and other radar system operating parameters are selected on the WXP-850().

Weather radar data can show on the MFD in a dedicated display mode or as part of a composite display of weather radar and navigation information. The navigation portion of the composite display shows only the “heading up” map display mode. If any other NAV mode map is selected (“north up aircraft centered”, “north up max view” or “map plan”), the display changes to the “heading up” map when weather radar is added to form the composite display of navigation information and radar data.

The Collins TWR-850 Turbulence Weather Radar System consists of the RTA-852, -854, or -858 Receiver-Transmitter-Antenna, a WXP-850A or -850B Weather Radar Panel, and an EFIS system for the weather radar display. Weather radar data from the TWR-850 is processed by the WXP and shown on the MFD and EHSI displays.

The WXP processes the radar data, controls the radar operating modes, and sets the display range. Data output from the WXP to the EFIS system for display on the MFD includes:

- four color weather (green, yellow, red, and magenta, on a black screen)
- detected turbulence and turbulence alert annunciation
- ground map (MAP display mode)
- PAC Alert annunciation (Path Attenuation Correction, yellow arcs at the perimeter of the display)
- GCS (Ground Clutter Suppression)
- target alert annunciation.

Some WXPs also feature slave switching for use in installations with two radar displays (such as both EHSIs, or an MFD and an EHSI) and two WXP controls. Other WXP features include autotilt and sector scan.

“PAC Alerts” show on the display to indicate that the radar beam, in the area indicated, has encountered a target with rainfall rates heavy enough to greatly attenuate the radar beam. In this situation, targets behind that rainfall may not be detected and shown on the display. A “PAC Alert” shows as a yellow arc or band at the outer perimeter of the radar display. The portion of the radar’s scan for the width of the yellow arc and between the arc and the intervening rainfall is the area in which the radar may not detect targets.

Navigation Information—One of four pictorial navigation maps may be selected from the NAV menu as the presentation for the navigation display mode. The four maps are:

- heading up
- north up with aircraft centered
- north up max view
- map plan (map plan is available only in installations with a compatible LNAV system)

All of the navigation maps are capable of simultaneously showing the location of:

- two VOR/DME stations
- two waypoint strings of up to 3, 5, or 6 waypoints each (depends on the LNAV and the aircraft wiring)
- up to seven closest NAVAIDS (the number available depends on the LNAV installed)
- and up to seven closest airports (the number available depends on the LNAV installed).

NOTE

The airport symbols shown in the navigation map display formats are for display purposes only and *do not* in any way show the actual runway headings or the orientation of the airport represented by the symbol. Always refer to the published navigation charts or approach plates for the actual runway headings and the airport orientation.

Navigation information can show on the MFD in a dedicated display mode or as part of a composite display of navigation information and weather radar data. (Weather radar must be set to a weather detection mode to show weather.) The navigation portion of the composite display shows only the "heading up" map display mode. If any other navigation map is selected ("north up aircraft centered", "north up max view" or "map plan"), the navigation display changes to the "heading up" map when weather radar is added to form the composite display of navigation and radar data. When the radar mode is turned off, the MFD changes back to the previously selected navigation map.

Anytime the radar display mode shows on the MFD, as part of a composite display mode or in the dedicated radar mode, the display range is set by the WXP. In the dedicated navigation display mode, the display range is set by the line advance or line reverse keys at the bottom of the MFD front panel. The range set for the dedicated navigation display mode is stored in memory and is recalled when the dedicated navigation display mode is selected.

Page Data—100 pages of user information may be stored in nonvolatile memory in the MFD. This data may be arranged in chapters and can be designated as either normal PGE (page) or EMG (emergency) data. All 100 pages can be input and modified by the user with the handheld RDP-300 Remote Data Programmer or the CEU-85 or CEU-85A Checklist Entry Unit. The CEU-85 is used in conjunction with an Apple® II plus, IIe, or IIc enhanced computer with a Collins Pro Line II Interface Card installed and checklist processing disk software. The CEU-85A is used in conjunction with an IBM® or compatible personal computer and the master checklist processing program.

Overview of Additional Functions

When used with a full featured, compatible LNAV, the MFD provides the following additional functions.

Waypoint Definition—When the navigation display mode shows on the MFD as part of a composite display mode or by itself, the MFD joystick may be used to move a waypoint cursor around the map display. Move the joystick in the direction required to place the cursor in the desired position on the display relative to the aircraft symbol and selected display range. When the joystick is moved, the word "ENTER" shows in green next to the third line select key from the top and a bearing and distance annunciator shows in green in the left center of the display. The "ENTER" line key, transfers the bearing and distance data for the waypoint cursor to a compatible LNAV system.

Remote Data—A compatible LNAV system can give pages of navigation information to the EFIS system for display on the MFD in remote (RMT) data display modes. There are three RMT sources available in installations with early versions of DPU and MPU. Later versions have only two RMT sources. The different pages of data (and the colors of the displayed data) are controlled by the control unit for the LNAV, and no control operations are required on the MFD following selection of the remote port.

Weather Radar (RDR) Display Mode

Push the RDR button to alternately turn on and off the weather radar display mode on the MFD. In the RDR display mode, the display mode annunciator "RDR" shows in green in the upper left corner of the display.

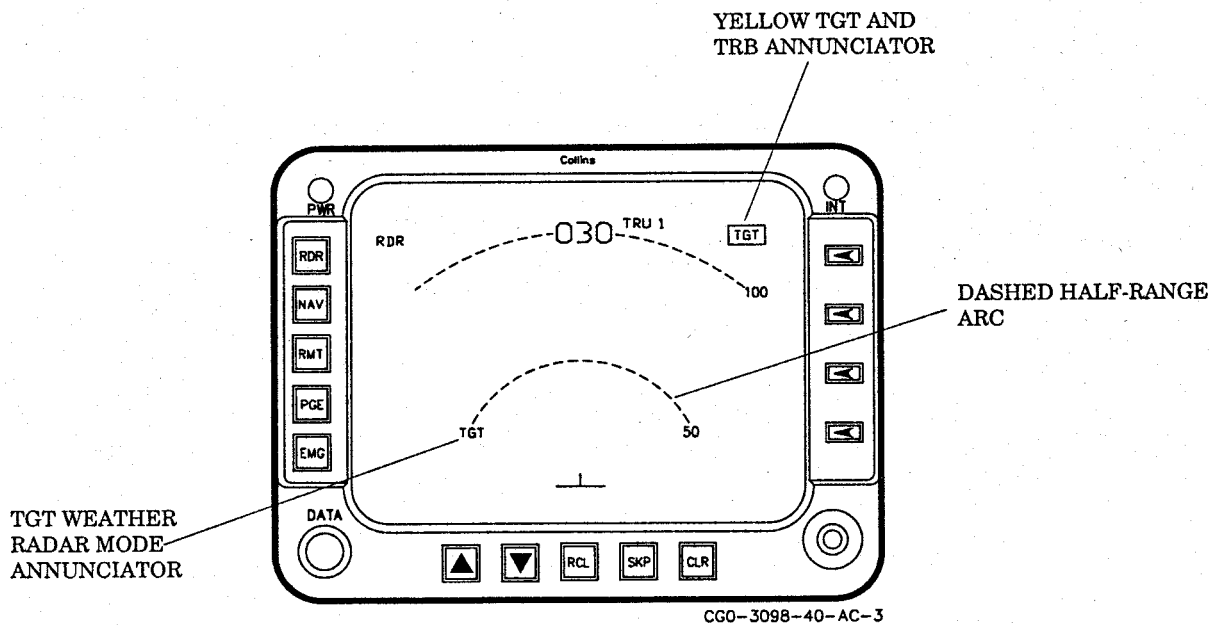
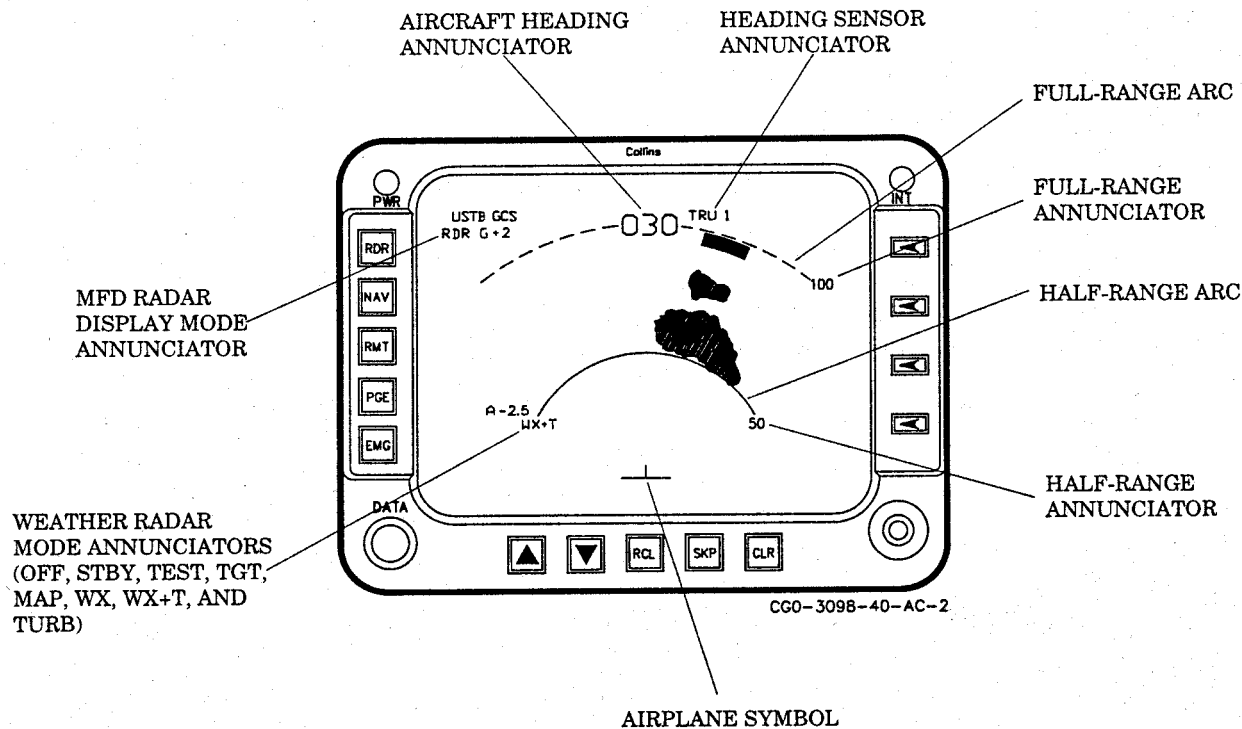
In the dedicated RDR display format the MFD shows a white aircraft symbol centered at the bottom of the display, and a digital heading read-out centered at the top of the display. The heading type and sensor number (MAG#, TRU#, or DG#) is annunciated to the right of the digital heading display. A dashed range arc shows at full range, and a solid range arc shows at half of the selected range. Range is selected by the "RANGE" control on the WXP. The selected full scale range annunciator shows at the right-hand end of the full-range arc, and the half-range annunciator shows at the right-hand end of the half-range arc. The full-range arc and both range annunciators show in cyan (the annunciators change to yellow when slaved operation is selected). The half-range arc shows in white in the radar ground mapping mode (MAP) and in cyan in all other modes. PAC alert arcs show in yellow along the inside edge of the full range arc.

The weather radar display area extends from the aircraft symbol to the full-range arc. Weather intensity shows on the MFD in four colors on a black background. The minimum rainfall level shows in green, the next level in yellow, the next in red, and the heaviest level shows in magenta. Magenta is also used to show areas of detected precipitation-related turbulence when the WXP is set to the WX+T or TURB mode and the range is set to 5, 10, 25, or 50 nautical miles.

Weather radar mode annunciators (OFF, STBY, TEST, TGT, MAP, WX, WX+T, and TURB) show in cyan at the left end of the half-range arc in the RDR display mode, and in the upper left corner of the MFD when the RDR display mode is not selected. In the slaved mode the radar mode annunciators show in yellow preceded with the letter X (i.e., XWX+T, XMAP, XTURB, etc...).

When the radar target alert (TGT) mode is selected on the WXP, the MFD shows the mode annunciator "TGT" in cyan at the left end of the half-range arc in the RDR mode or in the upper left corner when not in the RDR mode. (Some early EFIS versions show the cyan "TGT" annunciator in the upper left corner of the MFD in all display modes.) Also, a second "TGT" annunciator shows in yellow in a yellow box in the upper right corner of the display in all MFD display modes. The half-range arc shows as a dashed cyan arc and detectable weather does not show on the display. Display range is controlled by the WXP when it is set to the TGT mode, even though detected weather does not show on the display. The WXP controls the display range in all radar operating modes when the MFD is set to show radar data (RDR display mode).

Target alert notifies the pilot of weather targets that are producing rainfall rates greater than 0.5 inch (12.7 millimeters) per hour and/or of turbulence consisting of precipitation areas where the wind velocity shifts in excess of 16.40 feet-per-second (5 meters-per-second). The WXP target alert mode provides a target alert "window" from 7- to 200-nmi range (50 nmi maximum range for turbulence detection) and ± 15 degrees of dead ahead, regardless of the range selected on the WXP. When selected on the WXP, the TGT mode is active in all MFD display modes. The yellow "TGT" annunciator in the upper right corner of the display flashes when a *precipitation* target is detected within the target alert "window". The flashing yellow box shows the letters "TRB" in place of TGT when an area of *precipitation related turbulence* is detected within the target alert "window". If areas of precipitation and turbulence are both detected within the target alert "window", the flashing annunciation alternates between "TGT" and "TRB" at a faster rate. A radar mode on the WXP that shows weather on the MFD (WX+T or WX), must be selected to see and analyze the detected weather.



Additional radar annunciators shown on the MFD in cyan (yellow in slaved mode) are:

- USTB (antenna unstablized)
- GCS (Ground Clutter Suppression)
- G+1, G+2, G+3, G-1, G-2, and G-3 (transmitter-receiver gain settings other than calibrated)
- +15.5, -15, A+15, A-10.2, etc... (antenna tilt and autotilt settings other than 0).

The USTB annunciator shows on the display when the STB button on the WXP is in the out position. In early versions of the EFIS-85C(14)/86C(14) systems the USTB annunciator shows at the right end of the half-range arc. In later versions the USTB annunciator shows in the upper left corner of the display. Detection of an attitude fault by the radar system shows the USTB annunciator flashing.

WARNING

Continuous operation with GCS turned on is not recommended, because some precipitation returns may also be reduced in intensity or eliminated from the display.

The GCS annunciator shows on the display when the GCS button is set to the in position. Early systems show the GCS annunciator at the left end of the half-range arc to the right of the radar mode annunciator. Later systems show the GCS annunciator in the upper left corner of the display to the right of the USTB annunciator. GCS reduces the intensity of ground returns so that many of them disappear from the display to make it easier to view the precipitation returns.

The gain annunciator shows on the display when the GAIN knob on the WXP is set to any position other than CAL (calibrated). Gain annunciators show to the right of the RDR display mode annunciator in the upper right corner of the display in all systems. Higher gain settings are shown with the + symbol in the annunciator (i.e., G+2) and lower gain settings are shown with the - symbol (i.e., G-1). Normal operation is with the GAIN knob on the WXP set to the CAL position and no annunciator shown on the display.

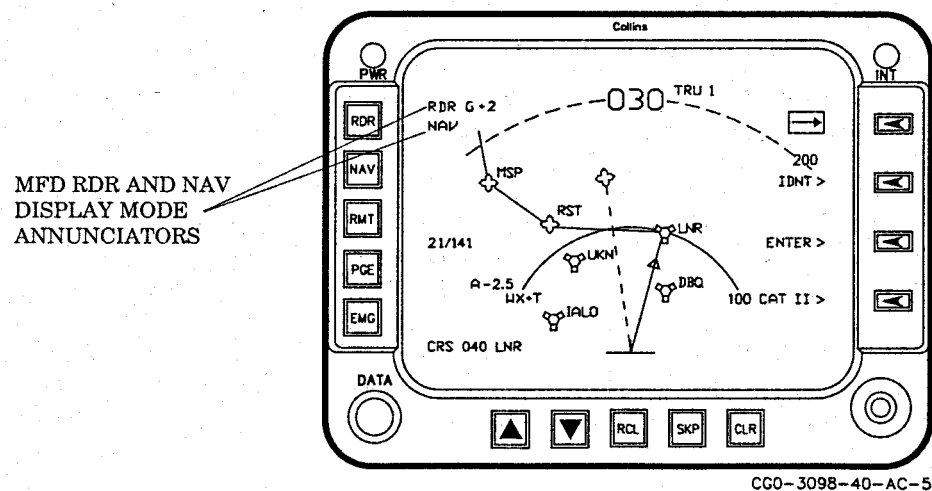
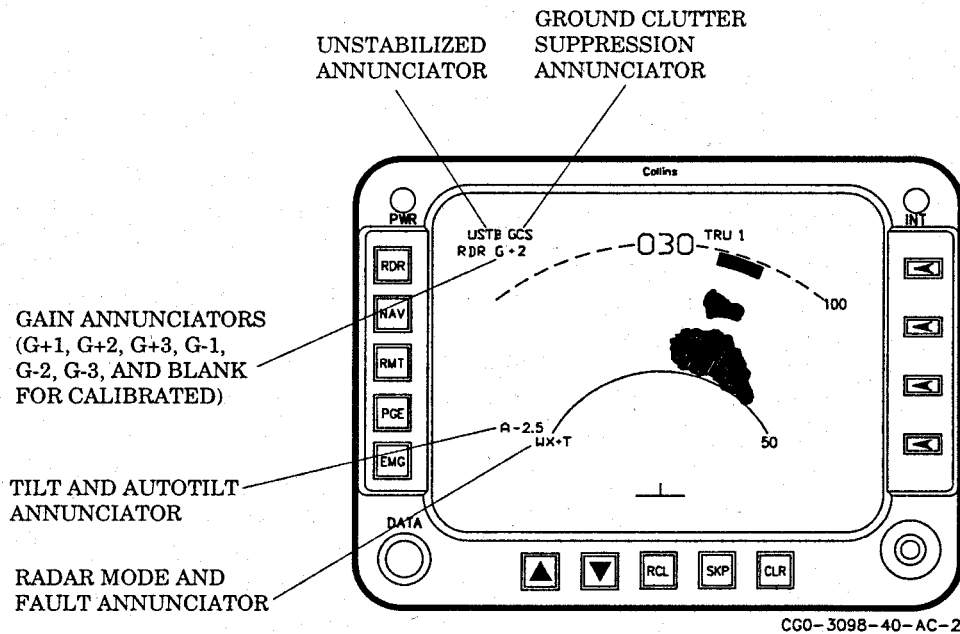
In early systems the tilt/autotilt annunciator shows on the display at the right end of the half-range arc. In later systems the tilt annunciators shows at the left end of the half-range arc. Up tilt is shown by the + symbol in the annunciator (i.e., +10.7) and down tilt is shown with the - symbol in the annunciator i.e., -12.5). When the AUTO switch on the WXP is pulled out, the tilt annunciator shows the letter A in front of the tilt setting (i.e., A+5.2).

With the MODE knob on the WXP set to the "MAP" mode, the MFD shows ground contours in four colors cyan, green, yellow, and magenta (least reflective to most reflective) and the half-range arc as a solid white arc. The "MAP" mode annunciator shows in cyan (yellow in slaved mode) at the left end of the half-range arc.

Setting the WXP MODE knob to the OFF position, or detection of invalid data or loss of signal from the WXP, removes the solid half-range arc from the display, sets the display range to 300 nmi, and shows the radar system mode annunciator "OFF".

Detection of a fault in the radar system shows the yellow annunciator "FAULT" flashing alternately with the selected weather radar mode annunciator. The fault annunciator has priority over any selected weather radar mode. If the fault is in the form of an interrupted communication link between the RTA (receiver/transmitter/antenna) and the WXP, the "FAULT" annunciator shows nonflashing in cyan. Transient faults can be cleared by momentarily setting the WXP to the STBY mode, then re-setting it to the desired operating mode. If the annunciator "FAULT" remains on the display after the reset, then a malfunction of the weather radar system is indicated, and the radar system should be turned off (WXP mode knob set to OFF).

The NAV display mode can also be turned on with the RDR display mode to show a composite display of both radar data and navigation information on the display.



Example of NAV and RDR Composite Display Mode

Navigation (NAV) Display Mode

Display Features and Elements

The following paragraphs describe the features and display elements that are the same in all NAV mode map displays. Following them are descriptions of the NAV MENU display page and features and display elements specific to each of four NAV mode map displays.

Push the NAV button to alternately turn on and off the navigation display mode on the MFD. In the NAV display mode, the display mode annunciator "NAV" shows in green in the upper left corner of the display. One of four map displays, selected from a NAV menu, shows on the MFD in the NAV mode. The four maps are:

- "heading up"
- "north up with aircraft centered"
- "north up maximum view"
- and "map plan" (available only in installations with a compatible LNAV).

In addition to the dedicated NAV display mode, a composite display mode of both the NAV and RDR modes can show on the MFD. When the MFD is in the NAV and RDR composite display mode, only the "heading up" map shows for the NAV portion of the display and the display range is controlled by the WXP. Range is always controlled by the WXP when the RDR display mode shows on the MFD.

Line key annunciators show adjacent to the line keys along the right side of the display. Push the line key with the green boxed arrow shown next to it (top line key), to select the NAV menu. Push the line key with the "IDNT>" annunciator shown next to it (second line key from the top), to turn on and off the display of airport and navaid identifiers. The IDNT annunciator shows only when airport and/or navaids are set, in the NAV menu, to show on the display. When the joystick is moved, the annunciator "ENTER" shows next to the third line key. Push the ENTER key to add a joystick positioned waypoint to a flight plan string from a compatible LNAV. Push the line key with the "CAT II>" annunciator next to it (bottom line key) to change the MFD directly to the CAT II checklist with a "yellow" condition of the extended data in the RMT display mode. The CAT II> annunciator shows on the display only when the system is set for for a CAT II approach. Push the NAV button to return to the previously selected display mode.

Display range is set with the line advance (▼) and line reverse (▲) buttons at the bottom of the MFD in all of the dedicated NAV mode map displays. The range arc and/or rings show as dashed cyan arcs or rings. If the RDR mode is selected to show on the display with the NAV mode (composite display mode), the display range is controlled by the WXP and the half-range arc shows as a solid cyan arc. Range is always controlled by the WXP when the RDR mode shows on the display.

When VOR stations are selected, in the NAV menu, they show on the display as octagon-shaped symbols in the proper range and bearing position with respect to the aircraft heading and the selected display range. A course line is drawn through the selected VOR station symbol and its orientation is controlled by the CRS knob on the corresponding CHP. If DCP control is transferred, or if the DCP course is allocated to another navigation sensor, the VOR course line on the MFD is removed from the display. In installations with later versions of DPU and MPU, the VOR1 course line shows on the MFD only when it is selected as the active or preset navigation sensor on the pilot's EHSI and the VOR2 course line shows only when it is selected as the active or preset navigation sensor on the copilot's EHSI.

When a localizer frequency is tuned, no symbol shows on the display for the station and the sensor annunciator shows LOC1 or LOC2 and the selected course. The colors for these annunciators are green for the left side (pilot's) and yellow for the right side (copilot's).



TCN and RNV stations show similarly to VORs except they do not show identifiers. RNVs show as a star-shaped symbol and TCNs show as a VORTAC symbol. RNV and TCN sensors are selected from the LNV1/LNV2 line of the MFD NAV menu when those sensors are available.

A selected LNAV flight plan string shows on the MFD as star-shaped waypoint symbols with track lines in the proper range and bearing positions. (Some LNAV installations may show waypoint symbols as star-shaped, VORTAC, or airports.) A total of either 3, 5, or 6 waypoints can show on the display. The maximum number is determined by the LNAV type and the aircraft wiring. When turned on, identifiers show next to the LNAV waypoints. For LNAV installations that show a maximum of 3 waypoints, the identifiers show as "WPT #" (# is replaced with the number 1, 2, or 3). For LNAV installations that show a maximum of 5 waypoints, alphanumeric idents show next to the waypoint symbols. For LNAV installations that show a maximum of 6 waypoints, the identifiers show as "TO+#" (# is replaced with the number 1, 2, 3, 4, 5, or 6).

NOTE

Airport symbols that may show on the display (in installations with a compatible LNAV system) are for display purposes only and *do not* in any way represent the actual runway headings or the orientation of the airport.

The LNAV sensor annunciator, a numeric course read-out for the next "to" waypoint, and the waypoint's ident for the pilot's side LNAV, show in the lower left corner below the VOR1 course annunciation. Similarly, the copilot's side LNAV sensor annunciator, course read-out, and waypoint ident show below the VOR2 course annunciation. LNV1 (pilot's) data shows in white and LNV2 (copilot's) data shows in cyan. If the next waypoint is off screen, the course line shows on the display with an arrow showing the direction to that station.

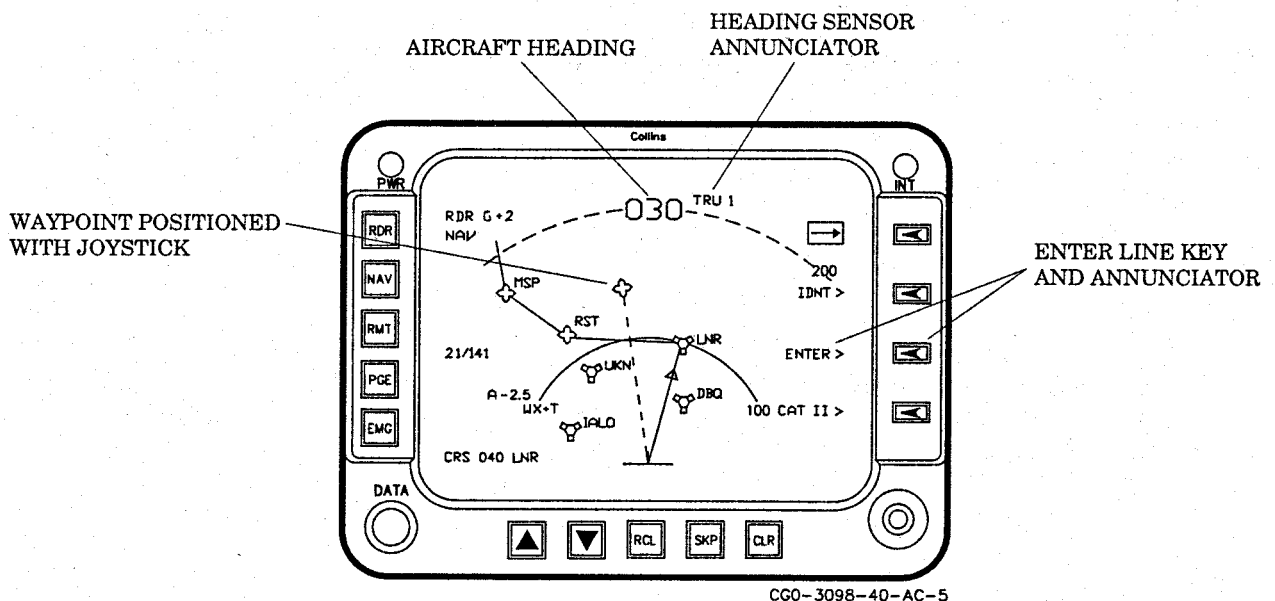
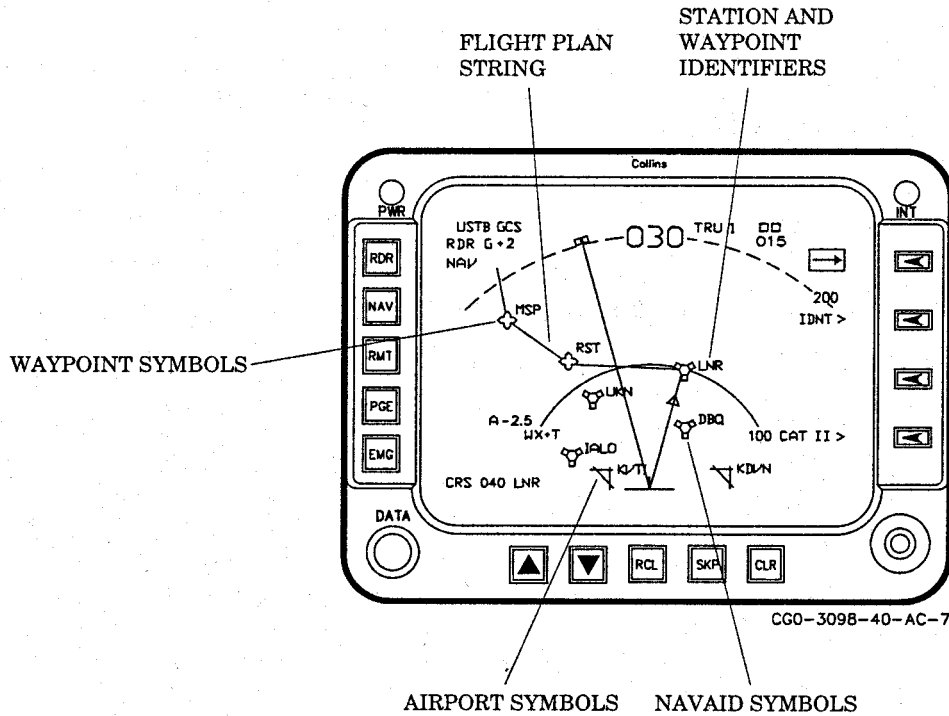
When the distance to a waypoint on the flight plan is greater than the selected range of the display (600 nmi), a line is drawn as if the waypoint was off screen. Breaks in the flight plan can be displayed as can a course line without a waypoint at its end if the information is available from a compatible LNAV. The "to" waypoint symbol flashes to indicate waypoint alert from a compatible LNAV.

In system installations with a compatible LNAV, up to seven each of background NAVAIDS and closest airports show on the display when selected in the NAV menu. The alphanumeric idents for these symbols can be turned on or off by alternately pushing the second line key with the "IDENT" annunciation shown next to it.

To position a waypoint for use in a flight plan string of a compatible LNAV system, select the LNAV system to show on the display (from the NAV menu) and use the joystick to position a waypoint on the display. As soon as the joystick is moved, the third line key (from the top) shows the annunciator "ENTER"; a waypoint shows at the airplane symbol moving in the direction commanded by the joystick; and a numerical heading/distance read-out for the waypoints position shows in the left center of the display. The waypoint, "ENTER" annunciator, and data read-out stay on the display for up to 30 seconds after the waypoint is last moved and then turn off. Push the ENTER line key any time prior to the 30-second time-out to load the waypoint into the compatible LNAV system.

A magenta selected heading display (cursor, selected heading line, and digital selected heading read-out) show on the MFD when selected in the NAV menu. The position of the selected heading cursor and line is controlled by the on-side CHP HDG knob and PUSH SYNC button.

The heading type and sensor number ("MAG#", "TRU#", or "DG#") are annunciated to the right of the digital heading read-out in the NAV mode heading up map and to the right of the N in all other NAV maps. If any selected navigation sensor is incompatible with the type of heading shown on the display (magnetic or true), the NAV sensor annunciator flashes until a compatible selection is made. Installations with later versions of MPU and DPU, show the "MAG#" annunciator only when both the pilot and copilot select the same magnetic sensor on both sides. When this occurs, the "MAG#" annunciator shows in yellow on both sides.



NAV MENU Display

The NAV menu is used for selecting display modes and various navigation sensors and features that show on the MFD. To show the NAV menu on the display, set the MFD to show the NAV display mode (either the dedicated NAV mode or the composite mode of RDR and NAV) and then push the top line key with the green boxed arrow annunciator. The NAV menu shows the display mode annunciators "<RADAR", "<HEADING UP", "<NORTH UP-A/C CNTR", "<NORTH UP-MAX VIEW" and "<EMERGENCY" on the left side of the display and NAV display mode feature annunciators "VOR1/VOR2>", "LRN1/LRN2>", "NAVAIDS ↔#>", "AIRPORTS ↑#>", and "□ □ HDG ↓" on the right side of the display (# = numbers from 0-7). Each of the annunciators points to its associated button or line key. Annunciators LRN1 and LRN2 are examples only. The actual annunciators shown in a given installation depends on the equipment installed in the aircraft and the aircraft wiring. The annunciators available in no particular order are:

- LNV1, LNV2, or LNV3
- RNV1, RNV2, or RNV3
- FMS1, FMS2, or FMS3
- TCN1, TCN2, or TCN3
- INS1, INS2, or INS3
- VLF1, VLF2, or VLF3
- GPS1, GPS2, or GPS3
- LRN1, LRN2, or LRN3

Push a button or line key to select a desired display mode or NAV display feature. Annunciators for selected modes and features show in green and unselected show in white. For the "VOR1/VOR2>", "LRN1/LRN2>" line keys, each push of the key cycles through a selection sequence of: number 1 sensor, number 2 sensor, number 1 and 2 sensors, and no sensors. For the "NAVAIDS ↔#>", "AIRPORTS ↑#>", and "□ □ HDG ↓" line keys, each push alternately turns on and off the display of the associated NAV data. Any combination of sensors and NAV data from the right side of the NAV menu can be selected to show on the display. For selection of a display mode shown on the left side of the NAV menu, push the button for the desired display mode. The MFD changes directly to that display mode. Only one display mode can be selected from the NAV menu at any time.

In the NAV menu the joystick selects the number of nav aids and airports that show on the MFD in the NAV display mode. The ↔ portion of the nav aids annunciator indicates to move the joystick left or right to set the number of nav aids that can show on the display. The ↑ portion of the airports annunciator indicates to move the joystick up or down to set the number of airports that can show on the display. The number following the arrows is the current setting. From a compatible LNAV up to seven nav aids and seven airports can show on the MFD in the NAV display mode. Refer to the preceding note regarding the airport symbols. If the LNAV is not compatible, the "NAVAIDS ↔#>" and "AIRPORTS ↑#>" annunciators do not show in the NAV menu.

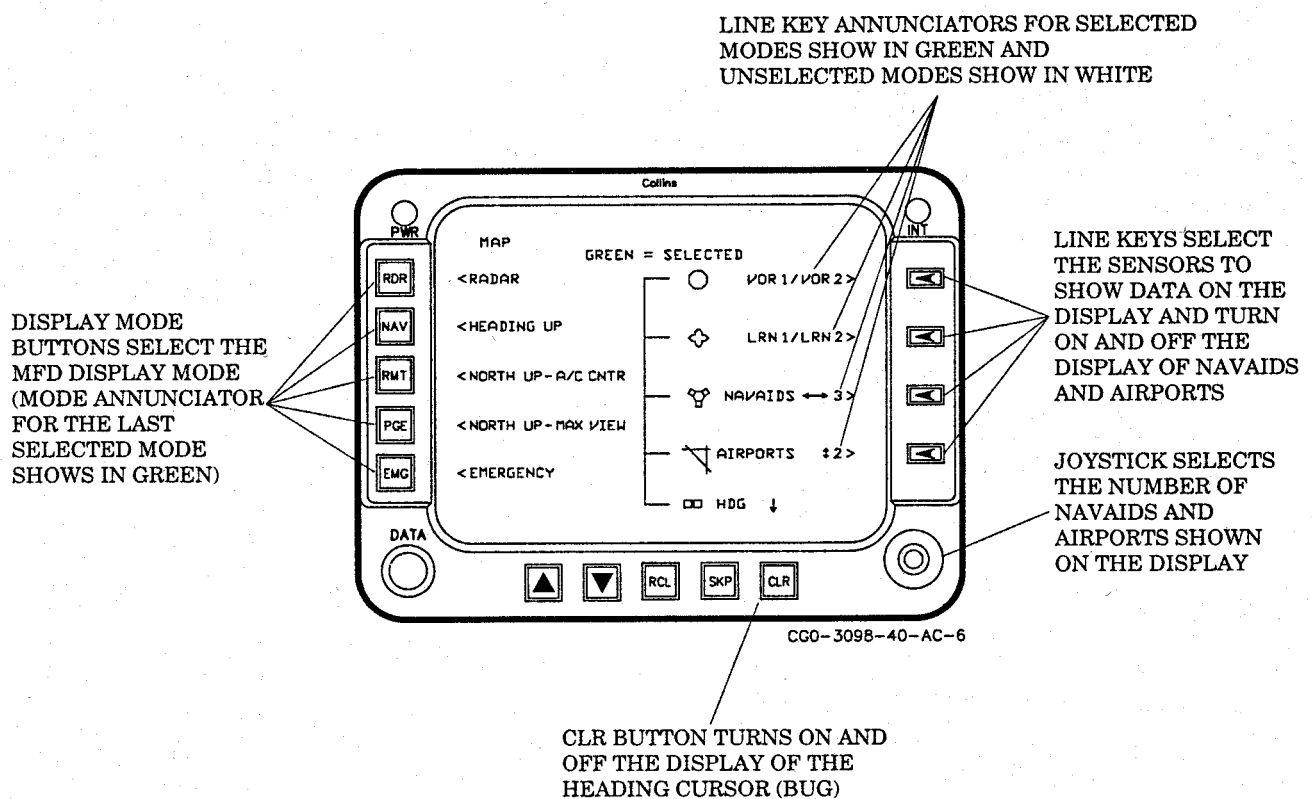
NOTE

Airport symbols that may show on the display (in installations with a compatible LNAV system) are for display purposes only and *do not* in any way represent the actual runway headings or the orientation of the airport.

Push the NAV menu “ $\square\square$ HDG $\downarrow$ ” (CLR) button at the bottom of the MFD, to alternately turn on and off the magenta selected heading line, heading cursor, and numerical selected heading read-out shown in the NAV and RDR display modes.

Push the NAV menu “RADAR” button to change the MFD directly to the composite RDR and NAV “heading up” display mode. Turning off the RDR mode with the RDR button changes the MFD back to the NAV mode showing the last selected map. To turn off the RDR mode from the NAV menu push any NAV map display button.

Push the NAV menu “EMERGENCY” button to change the MFD directly to the emergency page data checklist. Push the NAV button to return the display to the previously selected mode.



NAV Menu Display Mode

Heading Up Map Display

NOTE

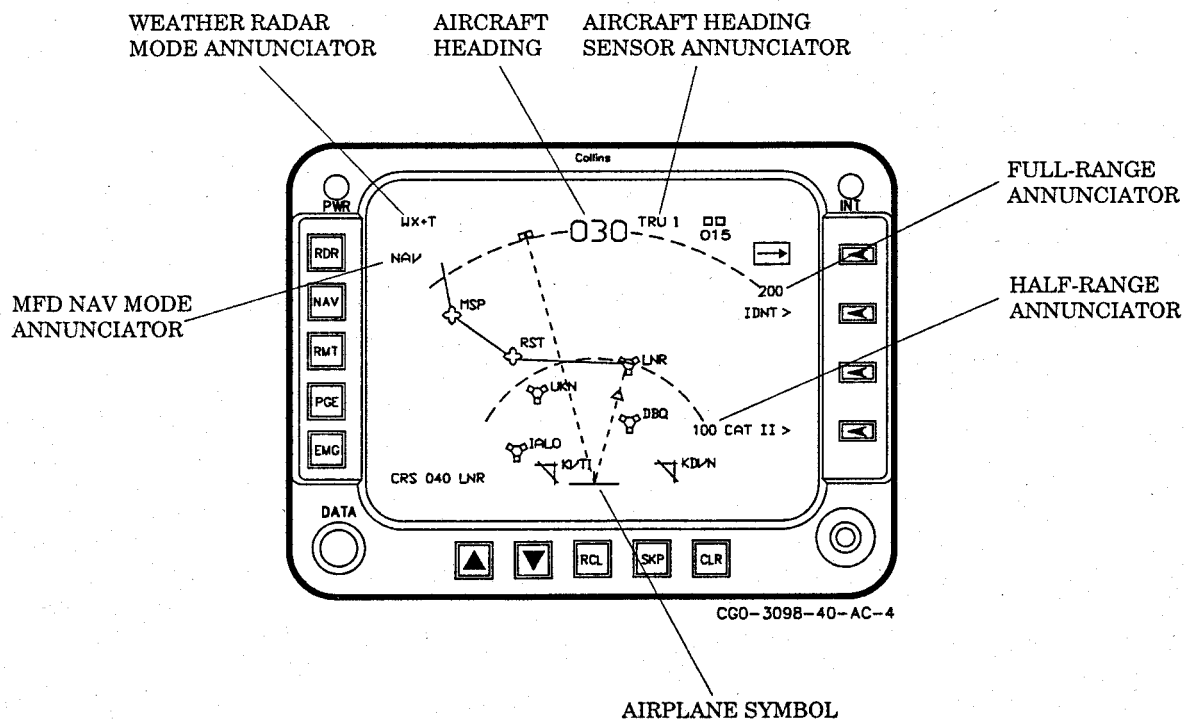
Only the display features and elements unique to the heading up map are described in this section. Features and elements common to all map displays are described in the beginning of the NAV display format section.

Push the "HEADING UP" button in the NAV menu, to show the NAV mode heading up map display.

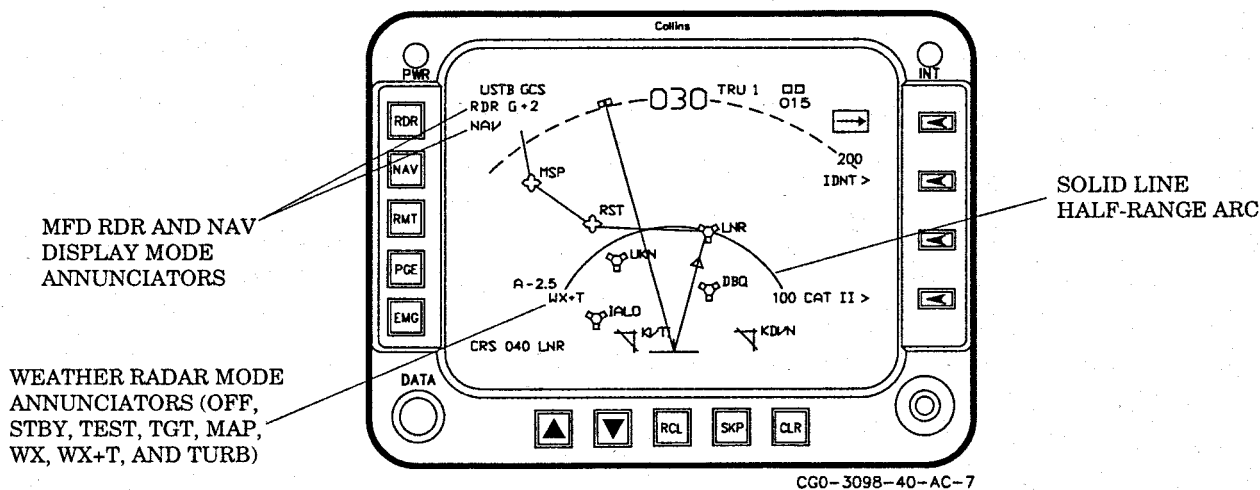
In the NAV mode "heading up" map, the display shows an aircraft symbol at the bottom center of the display; a digital heading read-out at the top center of the display; and two dashed range arcs, one at full-range (top of the display) and one at half-range (middle of the display). Range for each arc is annunciated at the right end of each arc and the full range arc annunciator is the selected display range.

Push the RDR button to show weather radar with the NAV mode heading up map (RDR and NAV composite display mode) in the form selected on the WXP. When the RDR mode shows on the display, with or without the NAV mode, the half-range arc shows as a solid arc. Anytime the RDR mode shows on the MFD, it shows in the heading up map. If any other NAV mode map is showing on the display, it is changed to the heading up map when RDR is selected. Turning off the RDR display mode returns the display to the previously selected NAV map.

When the NAV and RDR modes are turned off and no other mode is selected the MFD shows only the heading read-out; dashed full range arc; the weather radar operating mode annunciator in the upper left corner; and if turned on in the NAV menu, the magenta selected heading cursor, line, and the read-out in the upper right corner of the display.



NAV Mode Heading Up Map Display



RDR and NAV Mode Composite Display

North Up With Aircraft Centered Map Display

NOTE

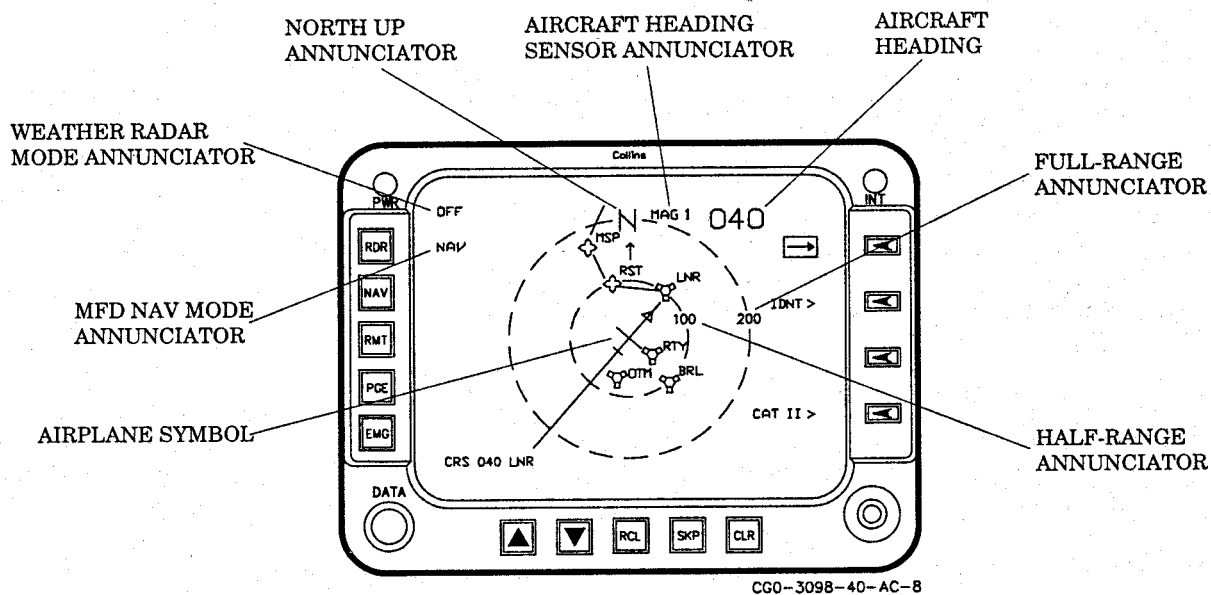
Only the display features and elements unique to the "north up with aircraft centered" map are described in this section. Features and elements common to all map displays are described in the beginning of the NAV display format section.

Push the "NORTH UP-A/C CNTR" button in the NAV menu, to show the NAV mode "north up with aircraft centered" map display.

In the NAV mode "north up with aircraft centered" map, the display shows a white aircraft symbol in the center surrounded by two dashed cyan range rings; the letter N with an arrow (↑) below it pointing at the top center of the display; and a numeric heading read-out in the upper right corner of the display. The top center of the display is always true or magnetic (if a magnetic compass is selected) north. The white aircraft symbol in the center of the display rotates as the heading of the aircraft changes. The nose of the aircraft symbol and the numeric heading read-out show the aircraft's heading. Waypoints, nav aids, and airports move around the display to maintain the proper range and bearing position with respect to the aircraft symbol and selected display range. The inner dashed range ring is the half-range ring and the outer ring is the full-range ring. Annunciators centered on the range rings on the right side of the display show the range for each ring.

Push the RDR button to show the RDR mode with the NAV mode (composite mode) on the MFD. The display changes to the "heading up" map with weather radar information shown in the form selected on the WXP. A second push of the RDR button turns off the RDR mode and returns the MFD to the previously selected "north up with aircraft centered" map.

When the NAV and RDR modes are turned off and no other mode is selected the MFD shows only the north up N and arrow; the heading read-out; the dashed full range arc; the weather radar operating mode annunciator in the upper left corner; and if turned on in the NAV menu, the magenta selected heading cursor, line, and the read-out in the upper right corner of the display.



NAV Mode North Up Aircraft Centered Map Display

North Up Maximum View Map Display

NOTE

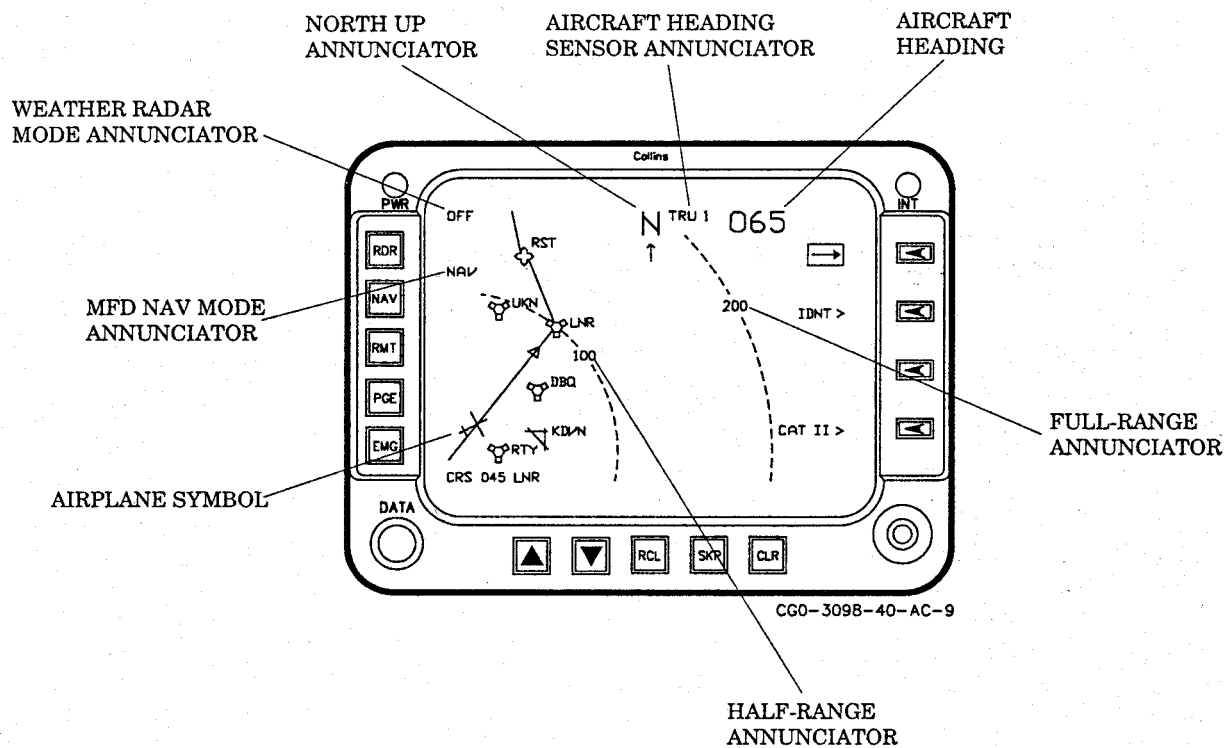
Only the display features and elements unique to the "north up maximum view" map are described in this section. Features and elements common to all map displays are described in the beginning of the NAV display format section.

Push the "NORTH UP-MAX VIEW" button in the NAV menu, to show the NAV mode "north up maximum view" map display.

In the NAV mode "north up-maximum view" map, the display shows a white aircraft symbol; two cyan dashed range arcs one for half-range and one for full-range; range annunciators on each range arc; the letter N with an arrow (↑) below it pointing at the top center of the display; and a numeric heading read-out in the upper right corner of the display. True or magnetic (if a magnetic compass is selected) north is always at the top center of the display. The aircraft symbol is positioned on an imaginary circle near the edge of the display so that the greatest amount of display area is shown in front of the aircraft. Orientation of the aircraft symbol is such that it moves around the edge of the display on the imaginary circle so that current track of the aircraft always shows the nose of the aircraft symbol pointing toward the center of the display. Waypoints, nav aids, and airports also move around the display to maintain the proper bearing and range position with respect to the aircraft symbol and selected display range. The range arcs are always centered on the aircraft symbol.

Push the RDR button to show the RDR mode with the NAV mode (composite mode) on the MFD. The display changes to the "heading up" map with weather radar information shown in the form selected on the WXP. A second push of the RDR button turns off the RDR mode and returns the MFD to the previously selected "north up maximum view" map.

When the NAV and RDR modes are turned off and no other mode is selected the MFD shows only the north up N and arrow; the heading read-out; the dashed full range arc; the weather radar operating mode annunciator in the upper left corner; and if turned on in the NAV menu, the magenta selected heading cursor, line, and the read-out in the upper right corner of the display.



NAV Mode North Up Maximum View Map Display

Plan View Map Display

NOTE

Only the display features and elements unique to the "plan view" map are described in this section. Features and elements common to all map displays are described in the beginning of the NAV display format section.

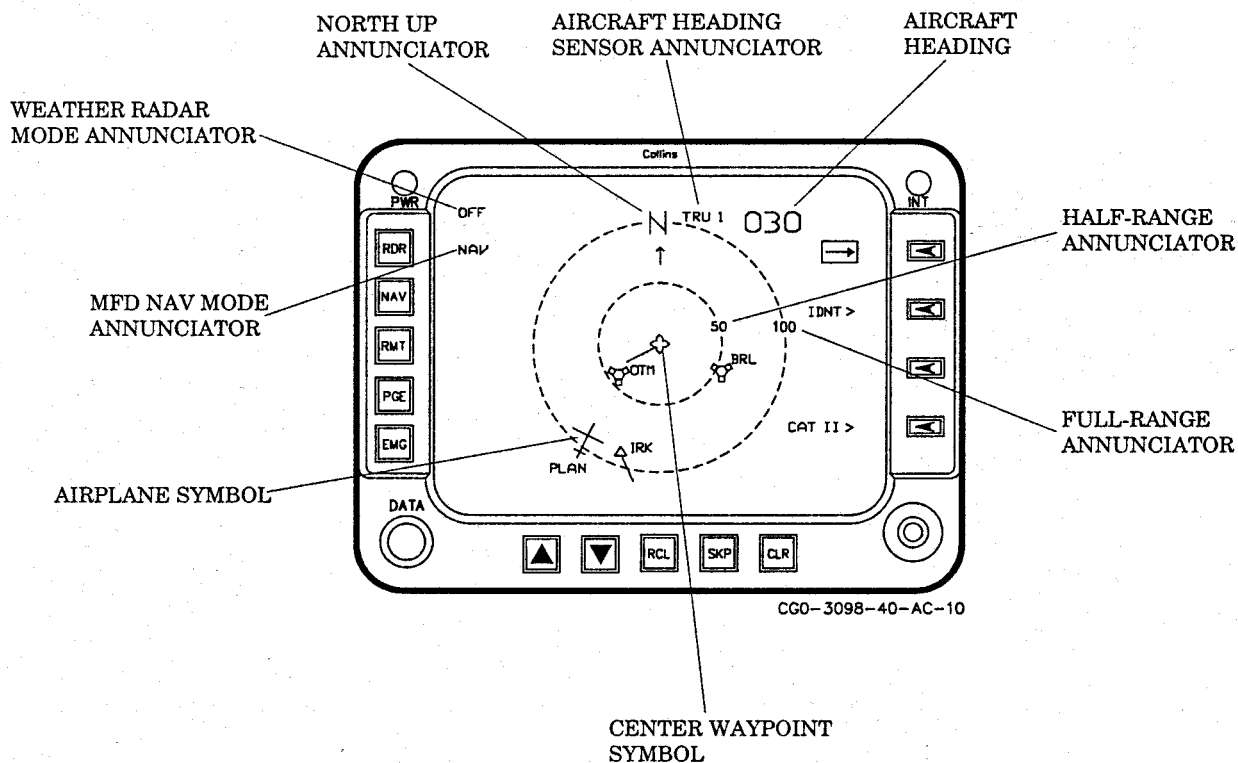
Select the plan view display mode on a compatible LNAV and set the MFD to the NAV display mode (NAV button) to show the "plan view" map.

The information shown on the "plan view" map is centered on a waypoint that is received from a compatible LNAV. Only parameters from the LNAV and the available display features (waypoints, NAVAIDS, and airports) selectable from the right side of the NAV menu show on the display. In dual LNAV installations, if both LNAVs are selected on the MFD, only one plan view shows on the display at a time (pilot's side LNAV has priority).

In the NAV mode "plan view" map, the display shows two dashed cyan range rings centered around a waypoint in the center of the display; the letter N with an arrow (↑) below it pointing at the top center of the display; and a numeric heading read-out in the upper right corner of the display. The inner dashed range ring is the half-range ring and the outer ring is the full-range ring. Annunciators centered on the range rings on the right side of the display show the range for each ring. True north (magnetic north is not used in the "plan view" map) is always at the top center of the display. A white aircraft symbol shows on the display only when the aircraft's position is within the selected display range with respect to the waypoint at the center of the display. Orientation of the aircraft symbol is the same as the aircraft's track with respect to the waypoint at the center of the display. The airplane symbol moves around or across the display following the aircraft's progress with respect to the waypoint at the center of the display. VOR, TCN, and RNV stations do not show on the display in the plan view format unless they are part of the LNAV data. VOR, TCN, and RNVs cannot be selected in the NAV menu.

Joystick waypoint entry operation is limited in the plan view. If the aircraft symbol is beyond the selected display range and not shown on the display, the joystick is disabled. When the aircraft symbol is within the selected display range and shows on the display, then the joystick operates normally as previously described.

Push the RDR button to show the RDR mode with the NAV mode (composite mode) on the MFD. The display changes to the "heading up" map with weather radar information shown in the form selected on the WXP. A second push of the RDR button turns off the RDR mode and returns the MFD to the previously selected "plan view" map display.



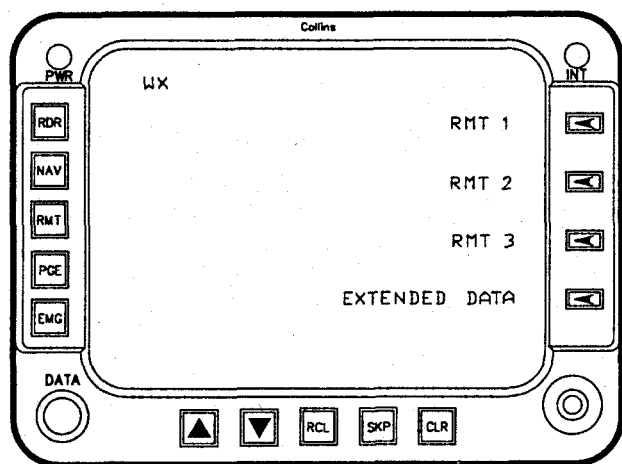
LNAV Plan View Map Display

Remote Data (RMT) Display Mode

Push the RMT button to show the RMT menu next to the display line keys. The menu shows "RMT 1>", "RMT 2>", "RMT 3>", and "EXTENDED DATA>". Weather radar mode and target alert status show on the MFD in all of the remote data display modes (TGT mode must be selected on the WXP in order to enable the target alert mode).

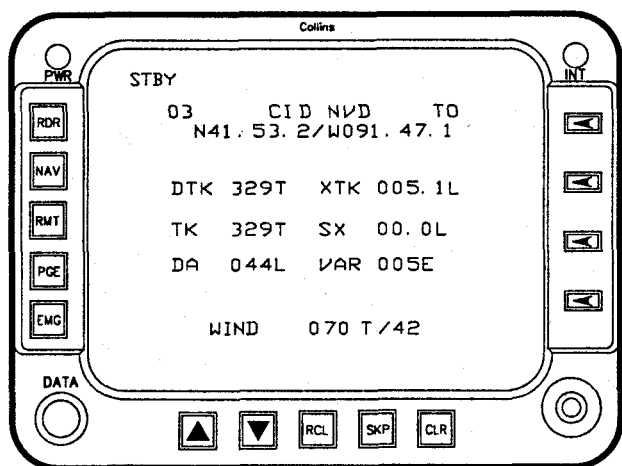
RMT

Push a "RMT #" line key to show the remote data for that remote source on the display. The data shown depends on the operating mode(s) and capabilities of the selected remote source. If no remote source is connected or if the selected source does not output remote data, the message "NO DATA FROM REMOTE DEVICE #X" shows on the display after approximately 5 seconds.



CGO-3098-40-AC-12

RMT Format, RMT Menu



CGO-3098-40-AC-13

Typical RMT Data Display

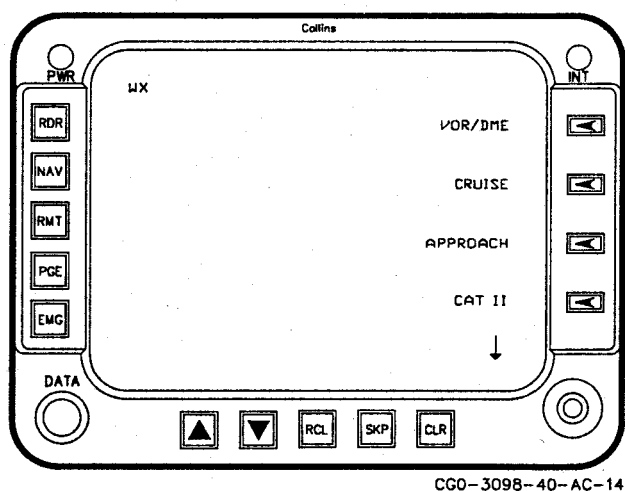
Extended Data

The "EXTENDED DATA" display formats show several pages of data useful to pilots or maintenance personnel. These are:

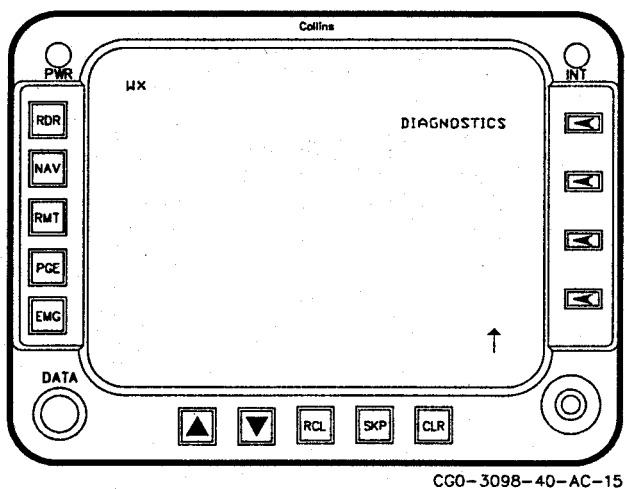
- VOR/DME
- CRUISE
- APPROACH
- CAT II
- DIAGNOSTICS.

Push the line key for "EXTENDED DATA" on the right side of the MFD to show a line-key menu with the items listed above. The "CAT II" menu item shows only when CAT II is selected from either side. A downward pointing arrow in the lower right corner of the display indicates that there is more data associated with the extended data "menu" page. This arrow shows which direction to push the joystick (down in this case) to display the next page of menu data. Move the joystick down to show the next menu page. The menu item "DIAGNOSTICS" shows on the display. Note that the arrow in the lower right corner of the display now points upward indicating the direction to move the joystick to show the previous menu page. The operating procedure for "EXTENDED DATA" is as follows:

1. Push the line select key next to the desired menu title ("VOR/DME", "CRUISE", "APPROACH", "CAT II", or "DIAGNOSTICS") to show the first page of data for that title on the display.
2. If there is more data associated with that title, an arrow shows in the lower right corner of the display indicating the direction (up or down) to move the joystick to show the next page of data.
3. Move the joystick left or right to show the first page of the previous or next title (of the "EXTENDED DATA" menu).
4. Push any one of the top three line select keys on the right side of the MFD to return to the "EXTENDED DATA" menu from any of the data pages.
5. Push any of the display mode buttons on the left side of the MFD to exit the extended data mode and change the MFD directly to the selected display mode (TFC, RDR, NAV, RMT, PGE, or EMG).
6. Weather radar mode and target alert annunciators show on the top line of the MFD display even when it is set to show extended data (TGT mode must be selected on the WXP to enable the target alert feature).



EXTENDED DATA MENU—PAGE 1



EXTENDED DATA MENU—PAGE 2

Examples of extended data displays are shown on the following pages. These data displays are for illustrative purposes only. The extended data available in any given installation may differ from the examples shown. Where applicable, the sensor supplying the data is identified. Data shown is from pilot side (left side) sensors unless otherwise identified. Display legends show in white and variable data shows in green. If a sensor's data is not available, green dashes show in place of the data.

Extended Data, Examples of VOR/DME Pages

ACTIVE PAGE

WX-WX+T		PAGE 1 OF 3		VOR/DME	
		VOR1	DME1		
		CID			
FREQ	114.10	114.10	HOLD		
BRG TO	123.0°	DIST	235.5 NM		
		GSP	250 KTS		
		VOR2	DME2		
		IOW			
FREQ	116.20	116.20	HOLD		
BRG TO	110.5°	DIST	230.7 NM		
		GSP	252 KTS		
		↓			

VOR #1 AND DME #1
SENSOR DATA

VOR #2 AND DME #2
SENSOR DATA

CGO-3261-03-AC-1

DME 1 PAGE

WX-STBY		PAGE 2 OF 3		VOR/DME	
ACTIVE 1	IDENT	CID			
	DIST	120.0 NM			
	FREQ	114.10 MHZ	HOLD		
PRESET 1	IDENT	ALO			
(MLS)	DIST	175.5 NM			
	FREQ	112.20 MHZ			
REMOTE 1	IDENT	DSM			
	DIST	25.7 NM			
	FREQ	117.5 MHZ	↑		

DME #1

CGO-3261-03-AC-2

DME 2 PAGE

WX-TEST		PAGE 3 OF 3		VOR/DME	
ACTIVE 2	IDENT	ALO			
	DIST	175.5 NM			
	FREQ	112.20 MHZ			
PRESET 2	IDENT	DSM			
(MLS)	DIST	25.7 NM			
	FREQ	117.50 MHZ			
REMOTE 2	IDENT	CID			
	DIST	120.0 NM			
	FREQ	114.10 MHZ	HOLD↑		

DME #2

CGO-3261-03-AC-3

Extended Data, Examples of Cruise Pages

CRUISE PAGE 1

WX-WX
PAGE 1 OF 6
CRUISE

ADC #1 → TAS 450 KTS GS 390 → LNAV #1

WIND 44°/10 KTS (TRU)

CALCULATED → SAT -15 °

TAT -29 °C

ISA -7 °C

LNAV #1 → TIME TO WPT 01:32 ETE 13:32 ETA

TIME TO DEST 06:15 ETE 18:15 ETA

TO WPT ID CIDARPT

BRG/DIST 79°/147 NM

CROSS TRACK 12.5 NM L OF A/C ↓

CGO-3260-06-AC-1

CRUISE PAGE 2

WX-STBY
PAGE 2 OF 6
CRUISE

LNAV #1 → MAG HDH 123°

TRUE HDG 120°

MAG VAR 3° WEST

DESIRED TRACK 123° (TRU)

ACTUAL TRACK 125.5° (TRU)

DRIFT ANGLE 2.5° LEFT

CROSS TRACK 12.5 NM L OF A/C

↑

CGO-3260-06-AC-2

CRUISE PAGE 3

CALCULATED → WX-WX+T

PAGE 3 OF 6
CRUISE

LNAV #1 → TIME TO WPT 01:32 ETE 13:32 ETA

TIME TO DEST 06:15 ETE 18:15 ETA

TO WPT ID CIDARPT

BRG/DIST 79°/147 NM

MESSAGE WPT ALRT DR

APPROACH XTRK WARN

NEXT WPT ID ALOVOR

↑

CGO-3260-06-AC-3

Examples of Cruise Pages (continued)

CRUISE PAGE 4

WX-MAP
PAGE 4 OF 6
CRUISE

PRES POS N 42°12.1'
W117°33.1'

CID N 42°12.2'
W118°15.1'

SLIX N 38°29.1'
W132°29.5'

FSD N 27°49.0'
W133°36.3'

↑

CGO-3260-06-AC-4

LNAV #1

CRUISE PAGE 5

WX-WX
PAGE 5 OF 6
CRUISE

AIR DATA
ALTITUDES: 12500 FT BARO
12300 FT PRESSURE
10000 FT PRESELECT

SPEEDS 250 KTS IAS
450 KTS TAS
350 KTS VMO
390 KTS GS
0.783 MACH
-650 FPM VS

↑

CGO-3260-06-AC-5

ADC #1

LNAV #1

CRUISE PAGE 6

WX-OFF
PAGE 6 OF 6
CRUISE

VNAV

ACT VERT SPEED -800 FPM
SEL OFFSET -16 NM
SEL VERT PATH -3.1°
ACT VERT PATH -2.8°
PRESSURE ALT 10500 FT
AIMPOINT ALT 8000 FT

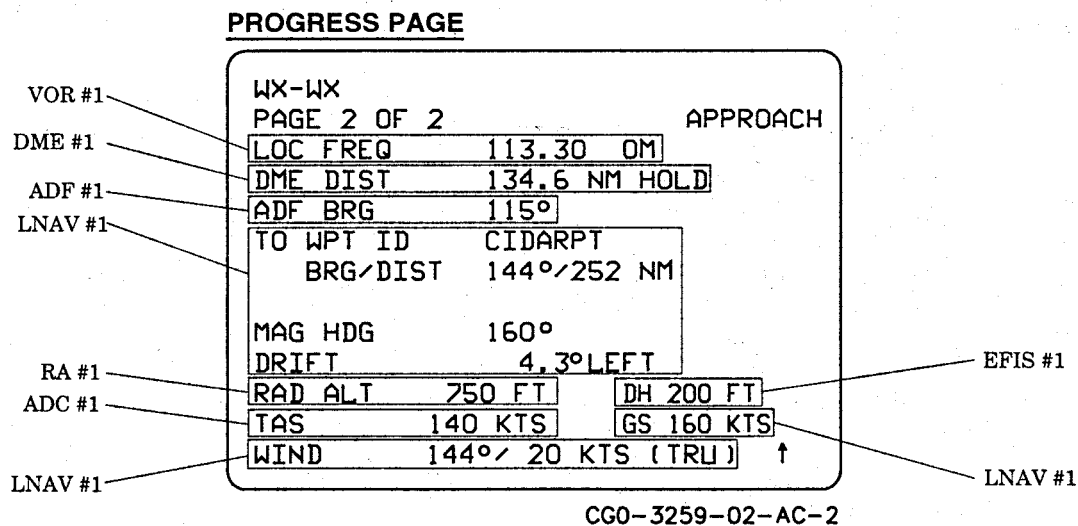
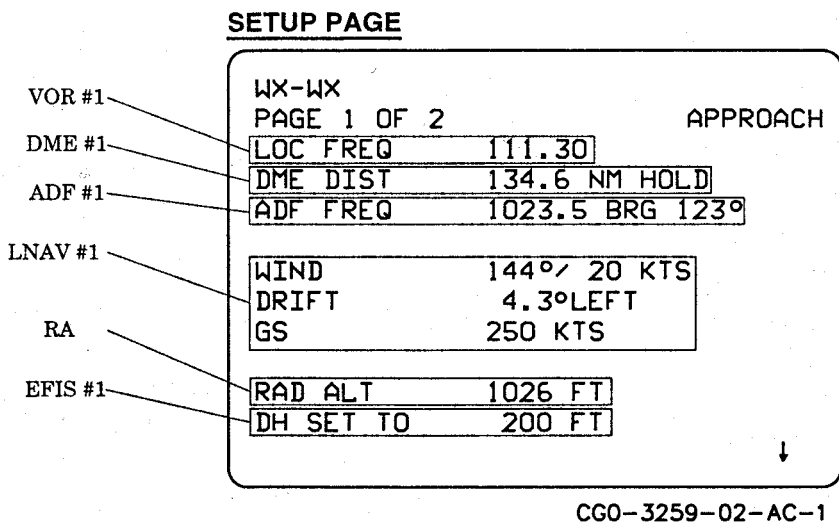
↑

CGO-3260-06-AC-6

ADC #1

LNAV #1

Extended Data, Examples of Approach Pages



Extended Data, Examples of CAT II Pages

PAGE 1

WX-TGT		TGT
PAGE 1 OF 3		CAT II
PILOT	MODE A,A	COPLT
OK	ONSD LOC ACTIVE	OK
OK	MAG HEADING SELECTED	OK
OK	ONSD ATTITUDE	OK
OK	ONSD HEADING	OK
OK	FCS NOT IN GO AROUND	OK
OK	NOT HSI MAP FORMAT	OK
OK	NOT BACK COURSE	OK
OK	NOT COMPOSITE	OK
OK	ONSD AIR DATA	OK
OK	CAT II SELECTED	↓OK

CGO-3258-03-AC-1

PAGE 2

WX-TGT		TGT
PAGE 2 OF 3		CAT II
PILOT	MODE A,A	COPLT
OK	LOC/GS COMPARE	OK
OK	HEADING COMPARE	OK
OK	RADIO ALT VALID	OK
OK	AIR DATA VALID	OK
OK	ATT/HDG VALID	OK
OK	LOC/GS VALID	OK
OK	DCP VALID	OK
OK	XSIDE VALID	OK
OK	AP APPR MODE SELECTED	OK
		↑

CGO-3258-03-AC-2

PAGE 3

WX-TGT		TGT
PAGE 3 OF 3		CAT II
PILOT	MODE A,A	COPLT
OK	FD VALID	OK
OK	ATT/HDG VALID	OK
OK	LOC VLD AFTER GS CAP	OK
OK	GS VLD AFTER GS CAP	OK
OK	ATTITUDE COMPARE	OK
OK	AP ENGAGED	OK
OK	LOC/GS EXCESSIVE	OK
		↑

CGO-3258-03-AC-3

Extended Data, Examples of Diagnostic Pages

The diagnostics pages are designed to aid maintenance personnel and contain no meaningful information for the pilots.

PAGE 1—CONFIGURATION STRAPPING

WX-STBY		
PAGE 1 OF 7		DIAGNOSTICS
	CONFIGURATION	
	PILOT	COPILOT
HEADING:	A42P P1	CSDB 3
ATTITUDE:	A429 P1	CSDB 3
ACCLRMTR:	A429 P1	CSDB 3
FCS TYPE:	APS-85 PA	APS-85 PA
ADF TYPE:	A429	NONE
LNAV ANN:	RNV 1	LRN 3
LNAV BUS:	A429 P1	561 ANA M
VNAV BUS:	CSDB	CSDB
RAD ALT:	ALT-55	ALT-55 ↑

CGO-3257-07-AC-1

PAGE 2—CONFIGURATION STRAPPING

WX-WX+T		
PAGE 2 OF 7		DIAGNOSTICS
	CONFIGURATION	
	PILOT	COPILOT
FLAPS:	UP	UP
STRUT:	GROUND	GROUND
VOR DEV:	LIN	ANG
LNAV TRK:	TAE+DA	DTK
XSD FLAG:	ENABLED	DISABLED
IAS XDEV:	ENABLED	ENABLED
RSNG RNWY:	ENABLED	ENABLED
HDG PANEL:	DUAL	DUAL
CTL PANEL:	DCP-85H	DCP-85H ↓

CGO-3257-07-AC-2

Examples of Diagnostic Pages (continued)

PAGE 3—PILOT'S SIDE

WX-WX		DIAGNOSTICS									
PAGE 3 OF 7											
PILOT'S SIDE CSDB											
AHRS		F3	F1	C1	D2	7A	6B	00	AB		
FCS		F3	D1	3F	58	81	92	13	95		
ADC		F3	D2	3F	08	83	92				
DME	CH1	F3	57	DA	6C	BA	A3				
DME	CH2	F3	57	DA	5C	BA	A3				
DME	CH3	F3	57	DA	4C	BA	A3				
VIR		F3	56	DA	6D	3A	A3				
VNAV		F3	E5	7A	8D	1E	D5				↑

CGO-3257-07-AC-3

PAGE 4—COPILOT'S SIDE

WX-MAP		DIAGNOSTICS									
PAGE 4 OF 7											
COPILOT'S SIDE CSDB											
AHRS		F3	F1	C1	D2	7A	6B	14	90		
FCS		F3	D1	3F	58	81	92	1D	5B		
ADC		F3	D2	3F	08	83	92				
DME	CH1	F3	57	DA	6C	BA	A3				
DME	CH2	F3	57	DA	5C	BA	A3				
DME	CH3	F3	57	DA	4C	BA	A3				
VIR		F3	56	DA	6D	3A	A3				
VNAV		F3	E5	7A	8B	1E	D5				↑

CGO-3257-07-AC-4

Examples of Diagnostic Pages (continued)

PAGE 5

WX-OFF				DIAGNOSTICS			
PAGE 5 OF 7							
BUS ACTIVITY							
		LF	RT			LF	RT
429	LRN:	OK	--	CSDB	ADC:	OK	OK
561	LRN:	--	--	CSDB	VIR:	OK	OK
429	ADF:	OK	OK	CSDB	DME:	OK	--
429	MLS:	OK	--	CSDB	FCS:	OK	OK
DIG	WXP:	OK	OK	CSDB	DCP:	OK	OK
429	AHRS:	OK	--	CSDB	VNI:	--	--
				CSDB	AHRS:	--	--
↑							

CGO-3257-07-AC-5

PAGE 6

WX-WX+T		PAGE 6 OF 7	DIAGNOSTICS
DIGITAL OUTPUTS OF FCS			
A5A5A5A5A5A5	C6634567ACD1		
C01234567890	C700123DCA6B		
C10897975661	C89292939ACD		
C20000000000	C9A12A43CD66		
C34A6755566D	AC5123400099		
C478765ACC12	B00000000000		
C56067DCABDF	F30000000000		

↑

CGO-3257-07-AC-6

PAGE 7

WX-WX+T		PAGE 7 OF 7		DIAGNOSTICS	
DIGITAL OUTPUTS TO					
DSP:				FMS:	
A5A5A5A5				A5A5A5A5A5A5	
81767700				CB8042350000	
82454300				CB8042350000	
83234000				CC8000000000	
84AC1B00				CD8A0F620000	
85561330				CDCA087B0000	
86561330				A780FD100000	
				↑	

CGO-3257-07-AC-7

Page (PGE) and Emergency (EMG) Display Modes

NOTE

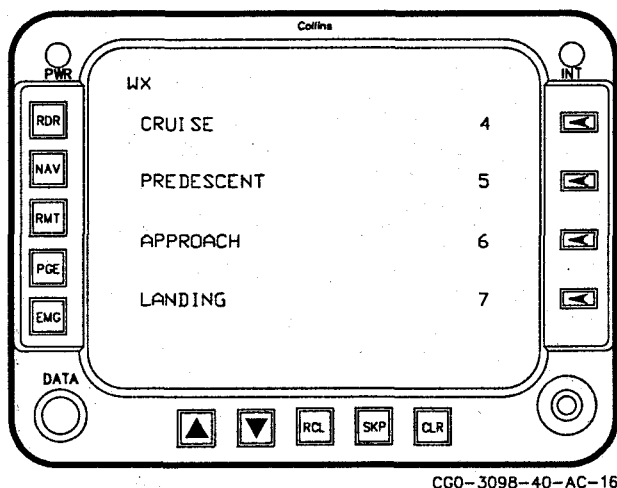
Refer to the "PROGRAMMING MFD PGE AND EMG DATA PAGES" for procedures to program the PGE and EMG display modes to show checklist information on the MFD.

Combined, there are 100 pages available for chapter title and checklist information data pages for the PGE and EMG display formats. There are no restrictions as to how the 100 pages of data are used. There can be 1 chapter of 100 data pages, 100 chapters of 1 data page each, or any combination of chapters and data pages with a total not exceeding 100. Also, the 100 pages can be divided in any quantity between the PGE and EMG display modes. Each page has a 13-line-per-page, 20-character-per-line format. The top line of each page is reserved for weather radar mode and/or target alert annunciators. The other 12 lines are available for programmed data.

Chapter title and page data information is entered into the MFD with the handheld CEU-85() Checklist Entry Unit or the RDP-300 Remote Data Programmer as described in the "PROGRAMMING MFD PGE AND EMG DATA PAGES" section of this book.

The top line (second line on the display) of the first page of a chapter is the chapter title line. Chapter titles show next to the display line keys when the PGE or EMG display mode is first selected. Chapter titles are entered into the MFD in the same manner as page data.

The PGE and EMG chapter and data pages show checklist information on the display in three colors. The line shown in cyan is the line with the cursor and is the line in the checklist that is currently being completed. Completed lines show in green and uncompleted lines show in yellow.



Typical PGE Chapter Title Menu

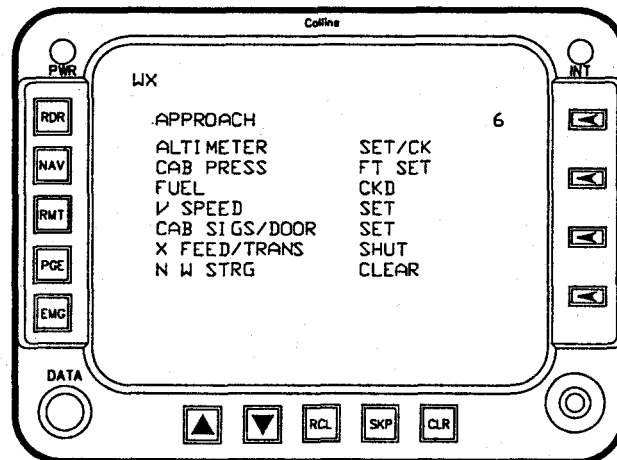
Push the PGE or EMG buttons to show a chapter title menu of the available checklist information. Up to four chapter titles show next to the line select keys on the right side of the MFD. Move the joystick upward to show additional pages of chapter titles (up to four titles at a time, if available). To return to the previous page of the chapter titles move the joystick downward. Push the line key adjacent to a chapter title to show the first data page of that chapter on the display. To exit from the PGE or EMG display modes push any other display mode button on the left side of the MFD. To switch between the PGE and EMG display modes push the button of the desired mode and the display changes directly to that mode.

Switching to the RDR, NAV or a composite of both, or RMT display modes when a page or chapter of checklist data is only partially completed, does not reset the current checklist. To return to the unfinished checklist, push the RCL button. The display changes directly back to the data page or chapter page that was showing when the PGE or EMG display mode was exited and the checklist can then be completed. PGE and EMG mode checklists are reset by completing a chapters checklist, which automatically returns you to its chapter title page, or by turning off the MFD power.

The buttons along the bottom of the MFD control the information shown on the PGE and EMG data pages. The following paragraphs describe the operation of the buttons as they are used in the PGE and EMG display modes.

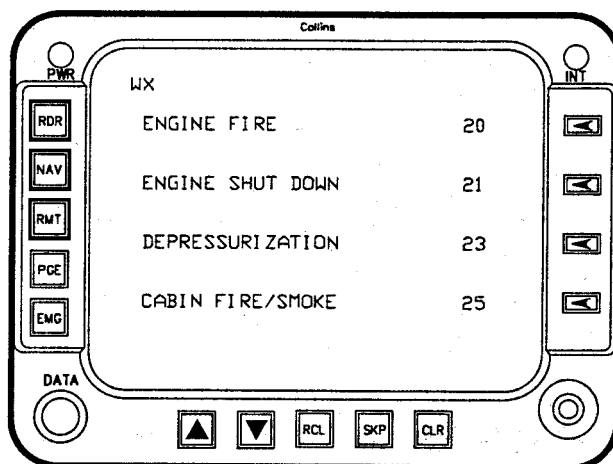
NOTE

When a chapter is first selected, the first page of that chapter always shows the first line in cyan and the rest of the lines in yellow prior to operation of any of the buttons at the bottom of the MFD.

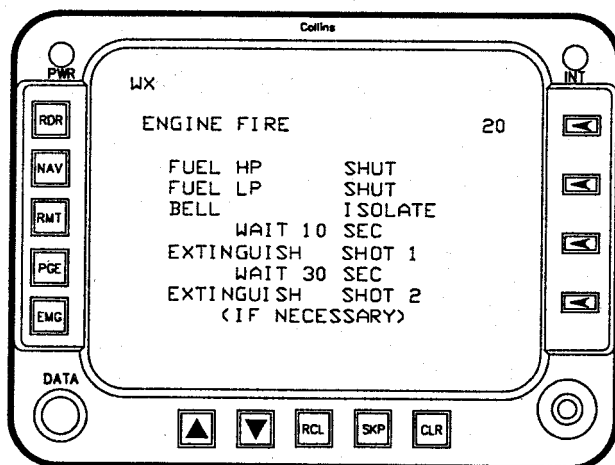


CGO-309B-40-AC-17

Typical PGE Checklist Example



Typical EMG Chapter Title Menu



Typical EMG Checklist Example

▼ **(Line Advance)**—Push the ▼ button to mark a line item as completed and move the cursor (cyan colored line) down one line. The completed line changes to green and the new line becomes cyan. When the last line on a page is marked as completed, the MFD automatically advances to the top of the next page of data, if available. When skipped lines are recalled with the RCL button, the display automatically advances to the next skipped line, regardless of where the line is located within the current chapter, when the current line (previously a skipped line) is marked as complete by pushing the line advance key.

▲ **(Line Reverse)**—Push the ▲ button to back the cursor up one line without marking the line as complete. The uncompleted line changes back to yellow and the new line (excluding blanks) changes back to cyan. If the cursor (cyan line) is the first line of a page and the line reverse button is pushed, the display backs up to the bottom line of the preceding page within the current chapter.

SKP (Skip)—Push the SKP button to move the cursor (cyan line) to the next lower line without marking the current line as completed. The unmarked line changes back to its previous color (yellow if not marked as completed, green if marked as completed) and the new line below becomes cyan.

When a line is skipped (remains yellow), it is marked as an uncompleted line and is subject to later recall as long as you do not change chapters. All skips are lost if the chapter is changed prior to completing the checklist (using the recall button). For this reason, some operators prefer to put the entire checklist in one chapter rather than breaking the checklist up into several chapters.

If the current selected line shows in cyan as a result of pushing the RCL (recall) button, pushing the SKP button skips that line without marking it as completed and automatically selects the next skipped line or the top line on the first page of the current chapter, whichever occurs first.

RCL (Recall)—Push the RCL button at any time, prior to changing chapters, to recall previously skipped lines (lines not marked as completed). Changing chapters clears all marked lines of the chapter that was left.

The recall function immediately shows the first page in the current chapter that has a skipped (yellow) line. That line changes to cyan. Push the line advance button to mark the skipped line as completed. The line changes to green and the system resumes searching for the next skipped line. If a skipped line is not to be marked as completed, push the SKP button as before. The system still identifies that line as uncompleted (yellow) and resumes searching for the next skipped line. When all of the skipped lines are marked as completed, with the line advance button, the display changes to the first page within the current chapter.

NOTE

Anytime you are in the PGE or EMG mode and the RCL button is pushed, the display changes to the top line of the first page within the current chapter and begins searching for skipped lines. Changing to another chapter cancels all skipped lines and completed checklist items of the chapter that was exited.

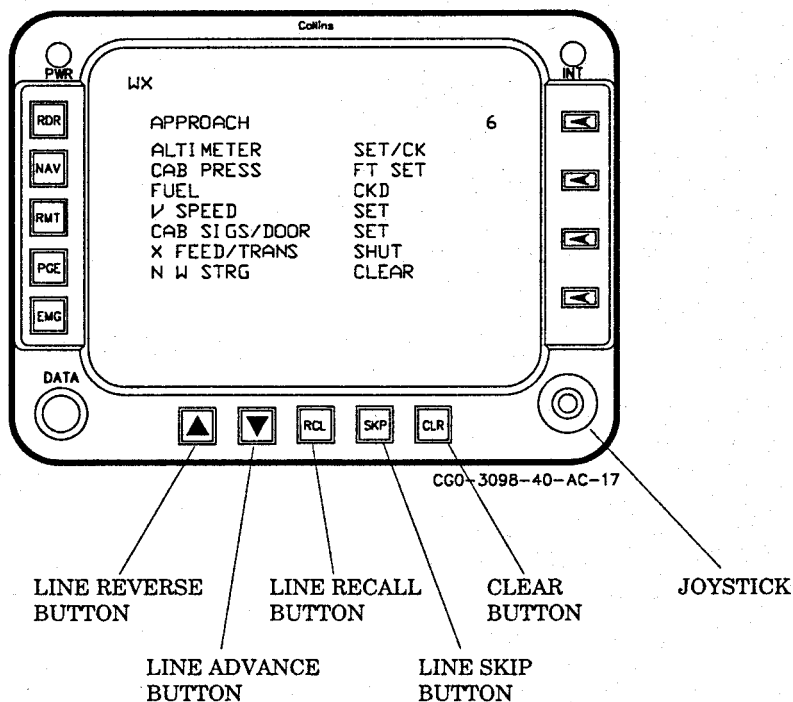
If you change to the TFC (MFD-85C), RDR, NAV or RMT display mode, from a PGE or EMG checklist chapter or data page, you can change directly back to the PGE or EMG checklist display line that was previously left (cyan colored line) by pushing the RCL button.

CLR (Clear)—Push the CLR button to cancel the current checklist and reset all lines of all pages of the current chapter back to yellow. This is also the same as the initial power up condition of all pages (all lines yellow).

NOTE

Changing chapters cancels all skipped lines and completed checklist items of the chapter that was exited.

Joystick—Move the joystick downward to show the next page or upward to show the prior page within the current chapter. Move the joystick to the right to show the first page of the next chapter or to the left to show the first page of the prior chapter.



Warning Flags and Comparators

NAV Flags—If a failure of a selected navigation sensor is detected, the annunciation in the lower left (left side course) or lower right (right side course) changes to “VOR”, “LNV”, or any other selected sensor, and turns red. The associated NAVAID symbols and connecting lines are removed from the display.

If any selected navigation sensor is incompatible with the type of heading being displayed (magnetic or true), the appropriate navigation sensor annunciator flashes until a compatible selection is made.

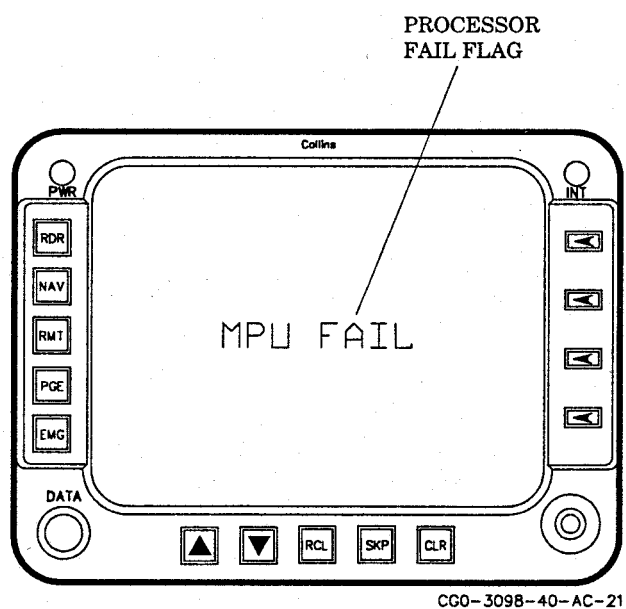
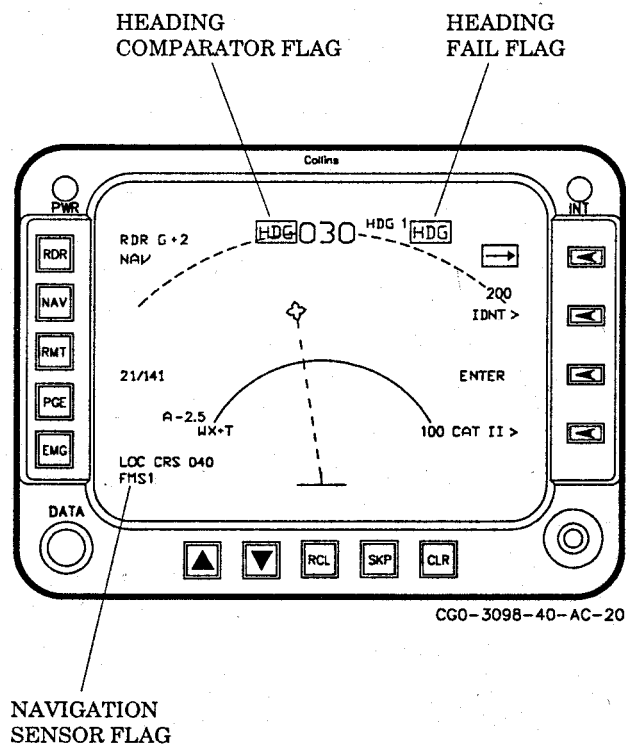
Heading Flag—If a failure of the heading sensor is detected, a red box with the letters “HDG” inscribed shows flashing to the right of the heading type annunciation. It flashes for approximately 10 seconds then stays on. To stop the flashing prior to the 10-second time out, push the master warn reset.

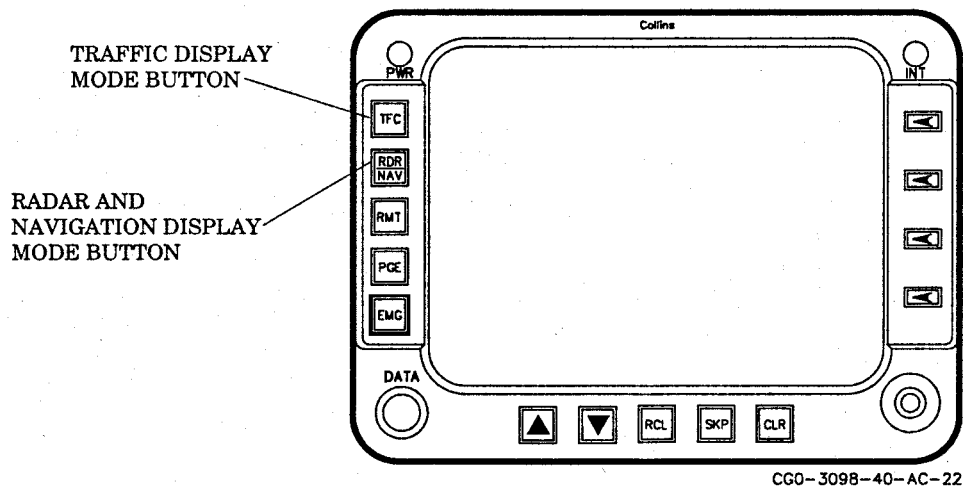
Heading Comparator—If the MPU detects a heading comparator warning, a yellow box with the letters “HDG” inscribed shows flashing to the left of the digital heading read-out or left of the north-up annunciation in the north-up maps. It flashes continuously until the cockpit mounted master warn reset button is pushed. When the master warn reset button is pushed and released, the flag stops flashing and stays on until the comparator error is reset. To reset the comparators, push and release the cockpit mounted comparator reset button. If the comparator warn error is still present, the flag turns on and stays on (non-flashing). If the comparator warn error is corrected, the flag turns off and stays off until another error is detected.

NOTE

In some EFIS-85C(14)/86C(14) installations, the yellow HDG comparator flag does not flash at any time. It turns on and stays on until the detected comparator error is corrected and the comparators are reset.

MPU FAIL Flag—Failure of the MPU shows the “MPU FAIL” flag in red in the middle of the display. After approximately 10 seconds all data on the display is removed, except the MPU FAIL flag.





MFD-85C Multifunction Display

MFD-85C DISPLAY FORMATS

The MFD-85C Multifunction Display (MFD) is a flight planning unit used to increase the operational capabilities of the EFIS-85C(14)/86C(14) system. The MFD-85C shows all the same display formats in the same manner as the MFD-85B. In addition, the MFD-85C can show a Traffic Alert and Collision Avoidance System (TCAS) traffic display. Operation of the MFD-85C is also the same as the MFD-85B, except that the MFD-85C has a TFC button to turn on and off a dedicated TCAS traffic display mode and an $\frac{\text{RDR}}{\text{NAV}}$ button to turn on and off both the radar and navigation display modes. In this chapter, only the MFD-85C display modes and operational differences from the MFD-85B are described. Refer to the MFD-85B Display Formats chapter for descriptions and operation of MFD-85C display modes not described here.

Overview

TCAS Traffic Display—On the MFD-85C the TCAS traffic display is a standard function and is used to show a pictorial presentation of traffic around the aircraft. The traffic display is an aid to the pilot for visually acquiring the traffic.

The pictorial presentation shows the traffic's:

- type
- range
- bearing
- vertical speed (shows only that the vertical speed is 500 ft/min or greater or less than 500 ft/min)
- direction of vertical speed
- and relative or absolute altitude.

Not all the above data may be shown for every traffic aircraft encountered. The data available to the TCAS system from the traffic aircraft and the TCAS and transponder system's operating modes determine what traffic shows on the display and the data shown with it. The data available from traffic aircraft depends on the traffic's installed equipment (TCAS system or transponder type) and its selected operating mode(s).

The traffic display can also show on the MFD as part of a composite display with the RDR and/or NAV display modes. However, TCAS display options can be selected or changed only in the dedicated TFC display mode.

RDR and NAV Displays—RDR and NAV display modes are selected from the same button on the MFD-85C. They can be selected as individual dedicated display modes or as a composite display of both modes. In addition, the TFC display mode can be selected from the line key menus of both the RDR and NAV display modes. In this way, the MFD can show a composite display of any combination of the RDR, NAV, and TFC display modes.

RMT, PGE, and EMG Displays—Displays for the RMT, PGE, and EMG display modes on the MFD-85C are the same as the MFD-85B, except that the radar operating mode annunciators shown in the top left corner of the display include the prefix "WX-" to differentiate the radar annunciators from TCAS annunciators (i.e., WX-WX, WX-WX+T, WX-TURB, etc...). Also, in these modes the TCAS annunciator "TRAFFIC" shows in the top right corner of the display when TCAS has detects TA or RA traffic. The annunciator shows in yellow for TA and red for RA traffic.

Traffic (TFC) Display Mode

Push the TFC button to alternately turn on and off the TCAS traffic display mode on the MFD. In the TFC mode, the display mode annunciator "TFC" shows in green in the upper left corner of the display.

Traffic Display

The TFC display mode shows:

- an airplane symbol representing your own aircraft
- three range rings or arcs (only two in the 40-nmi display range)
- four types of TCAS traffic symbols with the associated altitude and vertical speed data
- a TCAS line key menu (when turned on)
- and various TCAS annunciators and flags.

Each of the TFC display mode elements and features are described in the following paragraphs.

Airplane Symbol—The "own aircraft" airplane symbol shows in white in the lower center of the display. The symbol shows your aircraft's position with respect to traffic aircraft and the selected display range. Position of the airplane symbol on the display is such that there is about one-half the display distance below the symbol as compared to the display distance above the symbol. The symbol always points at the top center of the display and the nose of the airplane symbol represents the nose of your aircraft. Range rings surround the airplane symbol to aid in determining the distance from your aircraft to a traffic aircraft.

Display Range and Orientation—Display range is shown by three white range rings/arcs centered around the aircraft symbol and a white range annunciator in the upper right corner of the display. Display ranges available are 5, 10, 20, and 40-nmi. The first range ring, outward from the aircraft symbol, shows as dots around the symbol. Each of the dots is located at one of the twelve clock positions relative to the nose of the airplane symbol. In all display ranges, except the 40-nmi range, the dotted range ring shows a fixed range of 2 nmi from the airplane symbol. In the 40-nmi range, the dotted range ring is not shown on the display. The middle range ring, called the half-range ring, shows as a dashed line and is always one-half the distance of the selected display range. The outer range arc, called the full-range arc, shows as a solid line with marks on the line at the clock positions. The solid line range arc shows the maximum distance of the selected display range for which TCAS shows traffic on the display. At initial power-up the display range is set to 10-nmi.

Display orientation is centered on the airplane symbol and the display rotates around the symbol as the aircraft is maneuvered. The nose of the airplane symbol represents the nose of your aircraft and the top center of the display is always the twelve o'clock position relative to your aircraft.

Traffic Symbols—TCAS traffic symbols show for nearby traffic, within the selected range and altitude mode, that reply to TCAS interrogations. The symbols on the traffic display show the bearing, distance, relative or absolute altitude (if available), and as appropriate, the vertical speed of the traffic. Each of the four types of TCAS traffic show a unique symbol on the traffic display as described in the following:

NOTE

Altitude data shows with a traffic symbol only if that traffic's transponder replies with altitude data.

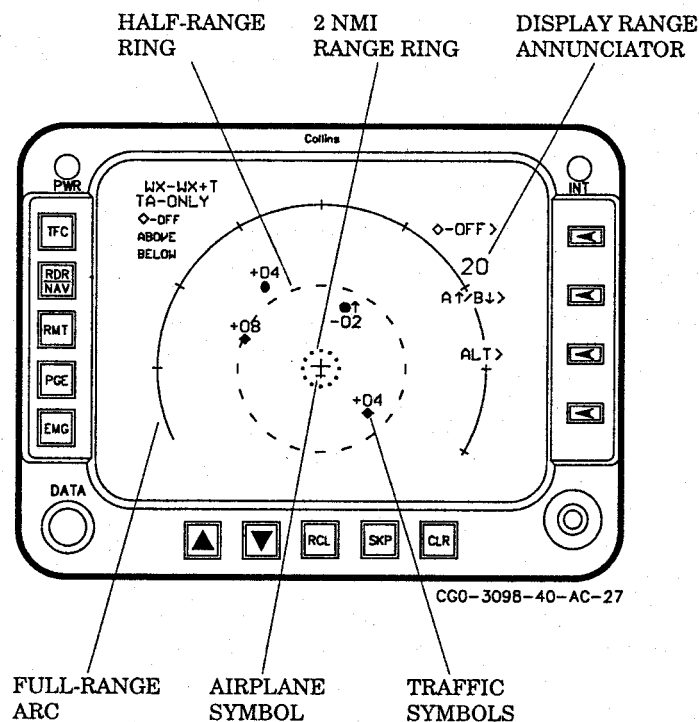
- +02
■ Resolution Advisory (RA) traffic shows as a solid red square.
- ↑
-20 Traffic Advisory (TA) traffic shows as a solid yellow circle.
- ◆↓
+10 Proximate Traffic (PT) shows as a solid cyan diamond.
- +10
◇ Other Traffic (OT) shows as an open cyan diamond.

The relative or absolute altitude and the direction of vertical speed show in the same color as the associated traffic symbol. The altitude data shows above the traffic symbol for traffic at an altitude above your aircraft and below the symbol for traffic below your aircraft. Vertical speed direction arrows show to the right of the traffic symbols.

Relative altitude data shows as two numbers preceded by a + or - sign. The number on the left shows thousands of feet of altitude and the number on the right shows hundreds of feet of altitude (i.e., +22=2200 ft above the aircraft and -02 = 200 ft below the aircraft).

Absolute altitude data shows as three numbers. The number on the left is ten thousands of feet, the middle number is thousands of feet, and the right number is hundreds of feet (i.e., 230=23,000 ft, 234=23,400 ft).

A vertical speed direction arrow shows for traffic with an actual (not relative) vertical speed equal to or greater than 500 ft/min. For climbing traffic the arrow points upward (↑) and for descending traffic the arrow points downward (↓). No arrow shows for traffic climbing or descending at an actual vertical speed of less than 500 ft/min or for traffic with no vertical speed.



TFC Line Key Menu

The TFC line key menu shows automatically, when the TFC mode is selected from any other display mode. To show the line key menu at any other time in the TFC mode, push the TFC button or any active line key. After approximately 5 seconds of button inactivity, the line key menu automatically turns off. In composite NAV and/or RDR display modes with the TFC display mode, the TFC line key menu is not selectable and does not show on the display. However, the options selected in the dedicated TFC display mode remain selected when the TFC mode shows as part of a composite display. To change the line key menu traffic display options you must select the TFC display mode.

TCAS display options, listed in the TFC line key menu, show along the right side of the display next to the line key that controls the option. $\diamond$ -OFF> shows next to the top line key, A $\uparrow$ /B $\downarrow$ > shows next to the second line key from the top, and ALT> shows next to the third line key from the top. Selected operating modes show the menu's annunciators in green, unselected modes show them in white, and failed modes show them in yellow.

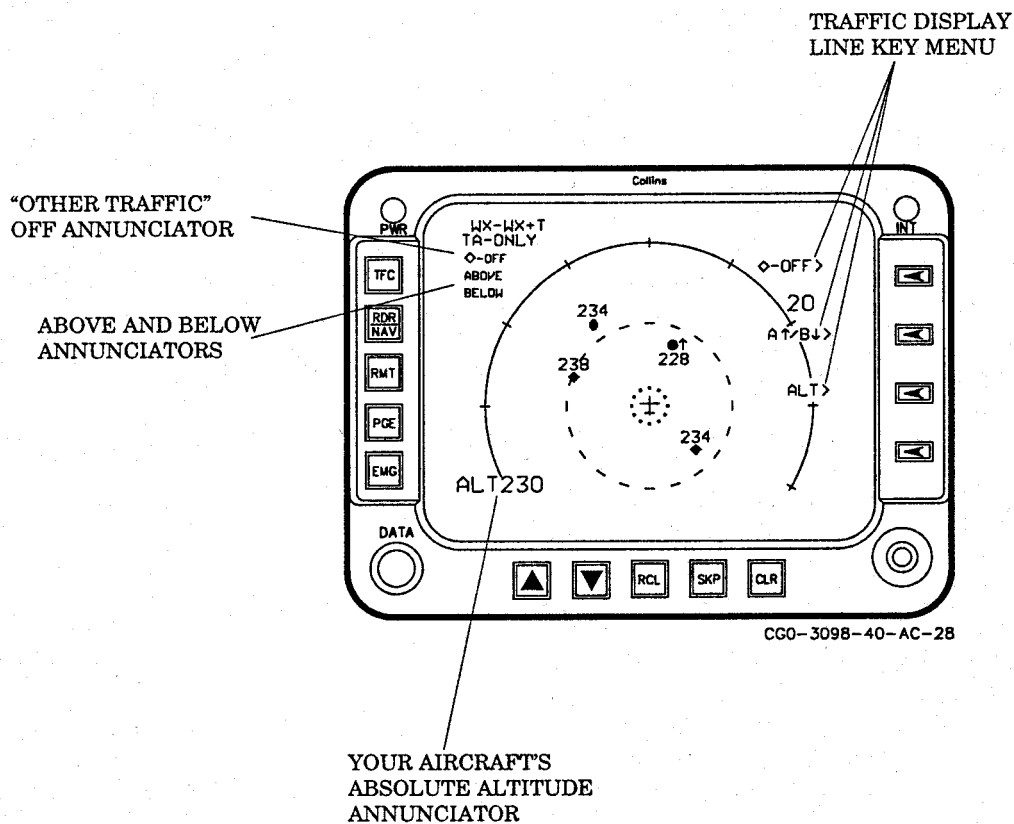
$\diamond$ -OFF>—Push the $\diamond$ -OFF line key to turn on and off the display of other traffic. When display of other traffic is turned off, the annunciator $\diamond$ -OFF> shows in white in the upper left corner of the display below the green display mode annunciators and traffic determined by the TCAS system to be "other traffic" do not show on the TFC display.

A $\uparrow$ /B $\downarrow$ >—Push the A $\uparrow$ /B $\downarrow$ line key as necessary to select the altitude volume for which TCAS shows "other traffic" on the TFC display. Each push of the button cycles through the various altitude selections. The line key annunciator A $\uparrow$ /B $\downarrow$ changes color to show the current selection. The A $\uparrow$ is for the above mode and shows the white annunciator ABOVE under the green display mode annunciators in the upper left corner of the display. B $\downarrow$ is for the below mode and shows the white annunciator BELOW under the green display mode annunciators in the upper left corner of the display. When neither the ABOVE nor the BELOW annunciators show on the display and the A $\uparrow$ /B $\downarrow$ > line key annunciator shows all in white, the TFC display is in the normal altitude mode. When both ABOVE and BELOW show on the display and the A $\uparrow$ /B $\downarrow$ > line key annunciator shows all in green, the display is in the above and below altitude mode.

ALT>—Push the ALT line key to select the altitude data type, relative (normal mode) or absolute, shown with the traffic symbols. In the absolute mode an annunciator shows in white in the lower left corner of the TFC display below the outer range ring/arc. The annunciator shows your own aircraft's baro corrected altitude in hundreds of feet with an ALT prefix (i.e., ALT750) or your own aircraft's flight level altitude with an FL prefix (i.e., FL250). The type of altitude data available to the TCAS system, flight level or baro corrected, determines the altitude data shown on the traffic display. In the relative mode (the normal operating mode) no annunciation shows.

NOTE

The flight level (FLXXX) annunciator shows only when the TCAS system does not receive baro corrected altitude data from the airdata source or during TCAS self-test. The annunciator shows on the display for approximately 15 seconds then turns off. Also, to show the loss of baro corrected altitude data, the line key menu annunciator ALT> shows in yellow instead of green.



Annunciators and Flags

The following paragraphs describe the annunciators and flags related to TCAS operations.

Display Range—A white annunciator in the upper right corner of the display next to the full-range arc, shows the selected display range in nautical miles. Display ranges available are 5, 10, 20, or 40-nmi.

TA-ONLY—"TA-ONLY" shows in the upper left corner of the traffic display when the TCAS system is set to give only traffic advisories. The annunciator shows in white except when TCAS detects traffic classified for a traffic advisory (TA). During a TA the annunciator changes to yellow and flashes from minimum to maximum brightness. Traffic classified for a resolution (RA) shows only as a TA when the TCAS system is set to the "TA-ONLY" mode. RA traffic does not show on the traffic display in the "TA-ONLY" mode.

When the TCAS traffic display shows as part of a composite display with the NAV and/or RDR display modes, the TA-ONLY annunciator shows in the upper left corner below the green MFD display mode annunciators. This annunciator also shows in the top right corner of the display when the MFD is in the RMT, FLIGHT INFO SUMMARY, EXTENDED DATA, PGE, or EMG data/checklist pages.

◇-OFF—The "other traffic" annunciator shows as a cyan diamond with the letters "OFF" in white in the upper left corner of the display. This annunciator shows that the display of "other traffic" is turned off. Only PT, TA, and RA traffic show on the display in this mode. This annunciator also shows in the upper left corner below the "TA-ONLY" annunciator when the TCAS traffic display shows as part of a composite display with the NAV and/or RDR display modes.

ABOVE/BELOW—The ABOVE and BELOW annunciators show the selected "other traffic" altitude volume. Proximate, TA and RA traffic are not affected by the ABOVE and BELOW modes and always show on the display regardless of the selected altitude volume. (Except RA traffic does not show on the display in the TA-ONLY mode.)

The ABOVE annunciator shows in white in the upper left corner of the display. This annunciator shows that TCAS is set to the above mode. Nearby "other traffic" (with transponders that reply to TCAS interrogations) from 2700 ft below to 9900 ft above your aircraft show on the traffic display.

The BELOW annunciator shows in white in the upper left corner of the display. This annunciator shows that TCAS is set to the below mode. Nearby "other traffic" (with transponders that reply to TCAS interrogations) from 2700 ft above to 9900 ft below your aircraft show on the traffic display.

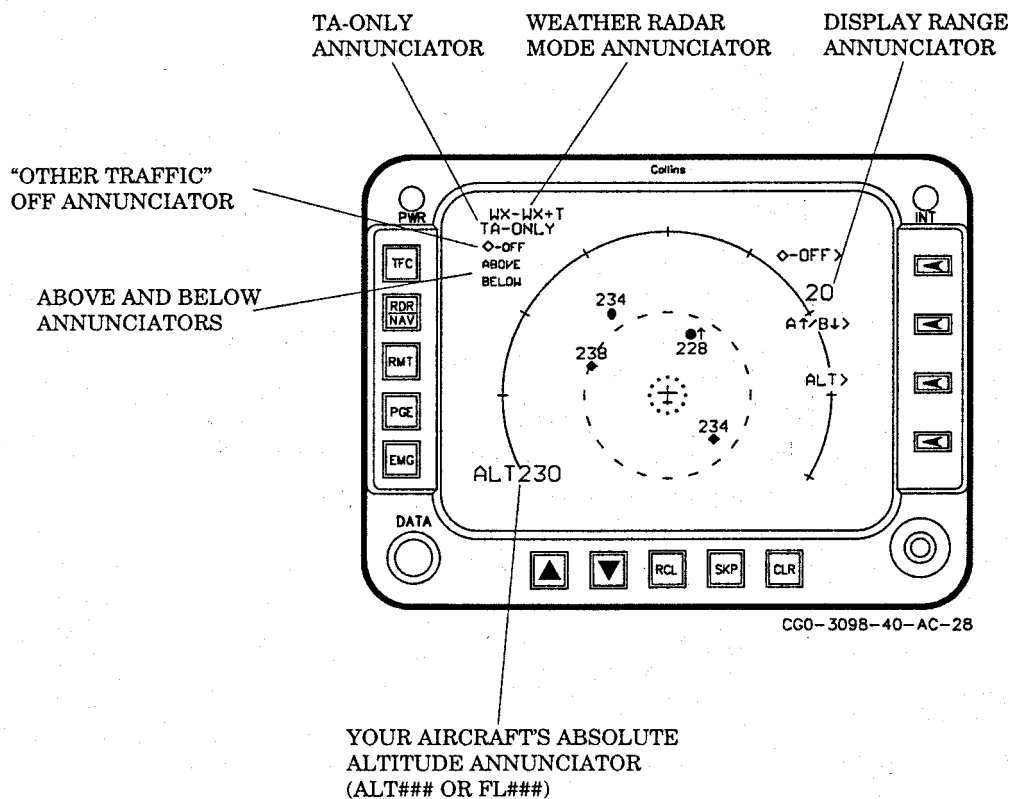
The ABOVE and BELOW annunciators may show on the display at the same time. With both the ABOVE and BELOW modes selected, nearby "other traffic" from 9900 ft below to 9900 ft above your aircraft show on the traffic display.

The ABOVE and BELOW annunciators also show in the upper left corner of the display below the "◇-OFF" annunciator when the TCAS traffic display shows as part of a composite display with the NAV and/or RDR display modes.

ALT/FL—With the TFC display set to show absolute altitude (instead of relative altitude), the annunciator ALT or FL with three numbers (i.e., ALT230 or FL230) shows in white in the lower left corner of the display at the left end of the full range arc. This annunciator shows your own aircraft's absolute altitude. ALT shows that the altitude data is baro corrected. FL shows that the altitude data is not baro corrected (flight level altitude). The left number shows ten thousands of feet, the middle number shows thousands of feet, and the right number shows hundreds of feet. When the ALT annunciator shows, the altitude data shown with the traffic symbols also changes to a three number absolute altitude read-out. Set TCAS to the relative altitude mode (the normal operating mode) to turn off this annunciator.

NOTE

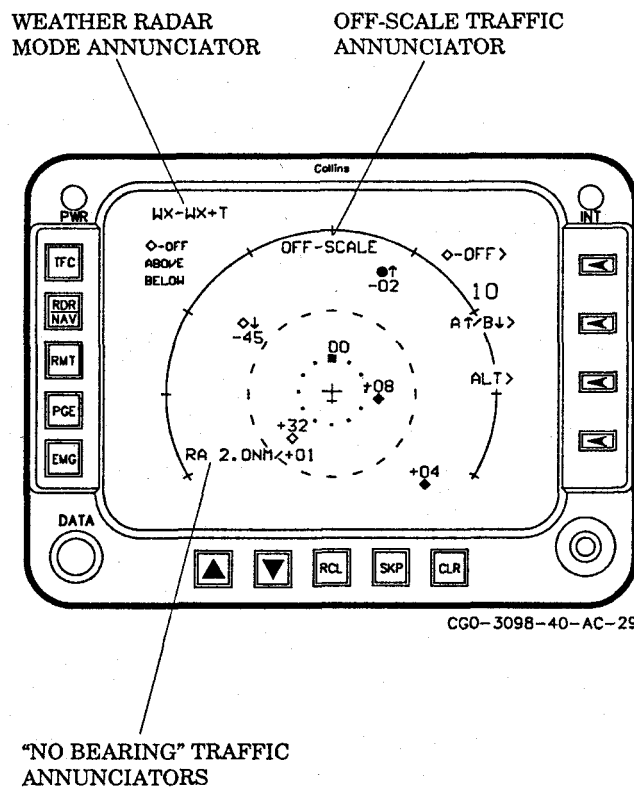
The FL annunciator shows only when the TCAS system does not receive baro corrected altitude data from the airdata source or during TCAS self-test. The FL annunciator shows on the display for approximately 15 seconds then turns off. Also to show the loss of baro corrected altitude data, the line key menu annunciator ALT shows in yellow instead of green when the FL annunciator shows in place of the ALT annunciator.



OFF SCALE—For TA and RA traffic beyond the selected display range, the annunciator “OFF-SCALE” shows at the top center of the display. The annunciator shows in yellow for TA and in red for RA traffic. If both types of traffic are detected the annunciator shows in red.

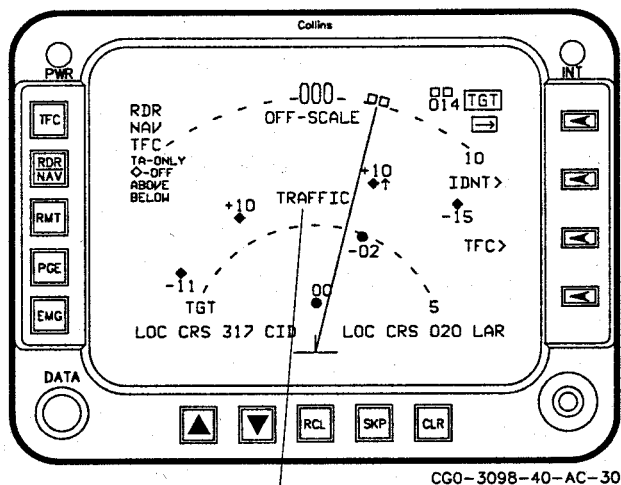
No Bearing Traffic—Up to two annunciators can show in the lower left of the display for TA and RA traffic for which TCAS cannot calculate a bearing. The annunciator gives the type (TA or RA), range, relative or absolute altitude, and direction of vertical speed of the traffic (i.e., RA 2.0NM/+01↓). TA traffic shows the annunciator in yellow and RA traffic shows the annunciator in red. Range to the traffic shows in the annunciator as a two number read-out in nautical miles with a resolution of one tenth of a nautical mile. Altitude data and vertical speed direction arrows show in the same manner as they are shown with traffic symbols.

WXP Mode—The WXP mode annunciator shows the letters WX- in preceding the WXP mode. The annunciator shows in the upper left corner of the display when the MFD is in the TCAS traffic display mode.



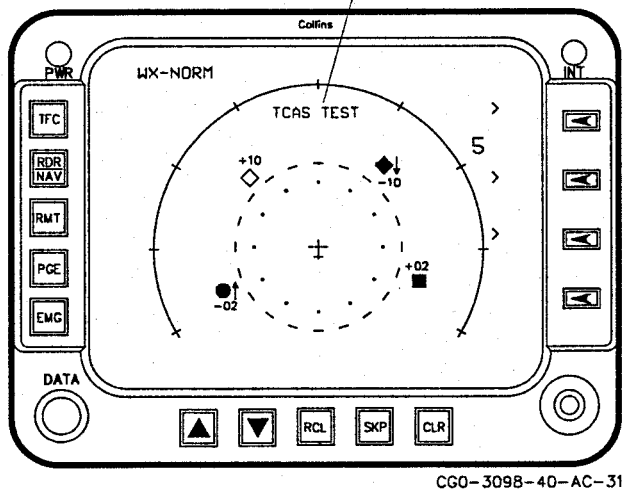
TRAFFIC—The “TRAFFIC” annunciator shows in the middle of the display in the NAV and RDR modes and in the top right corner of the display in the RMT, FLIGHT INFO SUMMARY, EXTENDED DATA, PGE, and EMG display modes. The annunciator shows on the display in yellow for TA and red for RA traffic, when the MFD is not in the TCAS traffic display mode and TCAS detects TA or RA traffic.

TCAS TEST—The “TCAS TEST” annunciator shows in white letters in the upper middle of the display in the NAV, RDR and TFC display modes. It also shows in this location for any combination of a composite display of these modes. The annunciator shows while the TCAS system self-test is in operation.



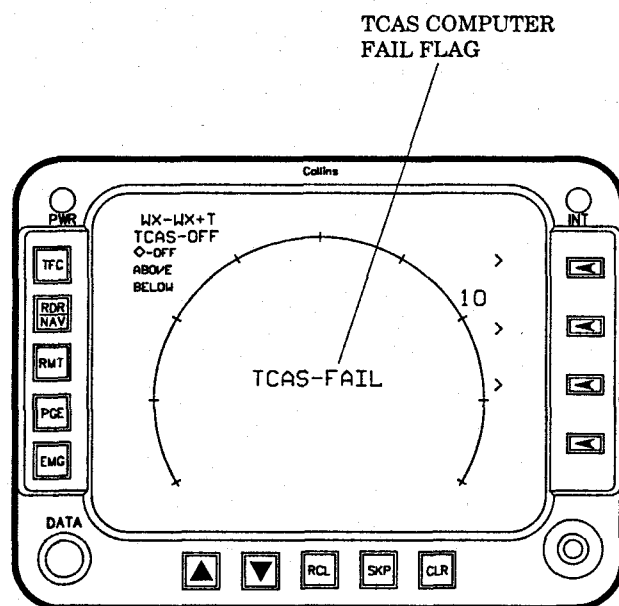
TRAFFIC
ANNUNCIATOR

TCAS TEST ANNUNCIATOR AS SHOWN
ON THE TCAS TEST DISPLAY FORMAT



TCAS-FAIL—The “TCAS-FAIL” flag shows in yellow in the middle of the display. This flag shows when a TCAS computer failure occurs. When this flag shows on the display, the airplane symbol, 2-nmi range ring, and half-range ring are removed from the display. Also, no traffic shows on the display and no resolution advisories are given by the TCAS system when this flag shows on the display. This flag shows in the upper left corner below the green MFD display mode annunciators, when the TFC display shows as part of a composite display with the NAV and/or RDR display modes. It also shows in the top right corner of RMT, FLIGHT INFO SUMMARY, EXTENDED DATA, PGE, and EMG display modes.

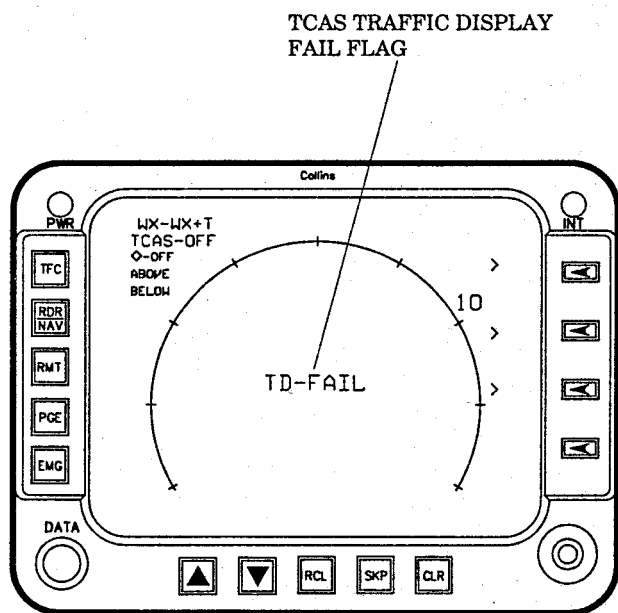
The $\diamond$ -OFF, ABOVE, and BELOW annunciators that show in the upper left corner and the ALT/FL annunciator that shows at the lower left end of the full-range arc do not turn off (if on) when the TCAS fail flag shows on the display.



CGO-3098-40-AC-32

TD-FAIL—The “TD-FAIL” flag shows in yellow in the middle of the display. This flag shows for TCAS traffic display failures. The airplane symbol, 2 nmi range ring, and halfrange ring do not show on the display when this flag shows. This flag also shows in the upper left corner below the green MFD display mode annunciators, when the TCAS traffic display shows as part of a composite display with the NAV and/or RDR display modes. This flag also shows at the top right corner of RMT, FLIGHT INFO SUMMARY, EXTENDED DATA, PGE, and EMG, display modes.

The $\diamond$ -OFF, ABOVE, and BELOW annunciators that show in the upper left corner and the ALT/FL annunciator that shows at the lower left end of the full-range arc do not turn off (if on) when the TD fail flag shows on the display.



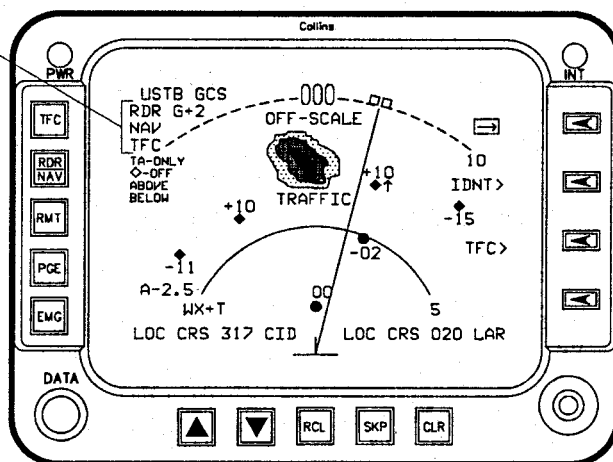
CGO-3098-40-AC-33

Composite Display

In the NAV and RDR display modes, the respective line key menus show the annunciator "TFC>" in green next to the third line key from the top. Push this line key to turn on or off the traffic display mode while in the NAV and/or RDR display modes. In this way, the MFD can show a composite display of any combination of the RDR, NAV, and TFC display modes.

When the TFC display mode shows as part of a composite display mode with the RDR and/or NAV modes, the TFC line key menu does not show on the display and no changes can be made to the currently selected traffic display options. However, the currently selected traffic display options (TA-ONLY, $\diamond$ -OFF, ABOVE, and BELOW) are still annunciated in the upper left of the display below the green display mode annunciators. To change the traffic display options, push the TFC button to turn on the dedicated traffic display then select the desired traffic display options.

MFD DISPLAY MODE
ANNUNCIATORS
(RDR, NAV, AND TFC)



CGO-3098-40-AC-35

RDR, NAV, and TFC Composite Display Mode

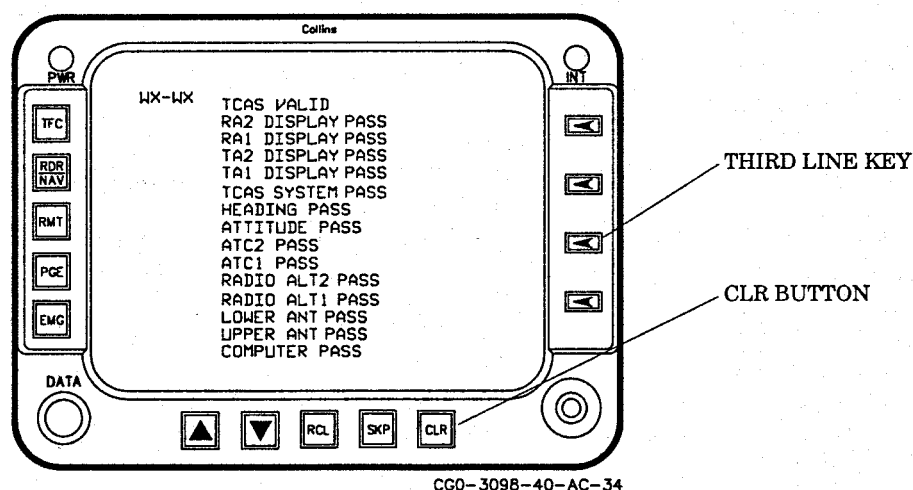
TCAS System Diagnostic Display

A TCAS diagnostic data page shows a list of the the TCAS related systems monitored by the TCAS computer and tested during the TCAS self-test routine. Valid systems show in green followed by the word "PASS". Invalid systems show in yellow followed by the word "FAIL".

If a TCAS problem is suspected and you are going to use the MFDs TCAS diagnostic data page to verify the suspected problem, it is suggested that you operate the TCAS self-test first. This is suggested, because although the COLLINS TCAS-94 system waits approximately 10 seconds before running its initialization self-test, some aircraft systems (such as heading systems) may take longer than 10 seconds to become valid at initial power-up. The TCAS computer's internal self-test monitors identify these systems as failed if they are not valid by the time the self-test operates. TCAS internal self-test monitors also indicate failures for some TCAS-related systems that are re-initialized or self-tested while in flight. TCAS computers retain self-test diagnostic information until the self-test operates again. The MFD TCAS diagnostic data page shows the TCAS system status as reported by the TCAS computer during the most recent operation of the self-test. Operating the TCAS self-test resets the TCAS diagnostics data. If TCAS-related system(s) have indeed failed, operating the TCAS self-test allows the TCAS computer to identify the failed system(s). In addition, it clears any failures recorded for TCAS-related systems that are working properly, but require longer than 10 seconds for a power-up valid or require a long initialization.

To turn on the TCAS diagnostic data page, select the TFC display mode, push and hold the CLR button, then push and hold the third line key from the top. The TCAS diagnostic data page shows on the MFD only while the buttons are pushed. Release the CLR button and the line key to turn off the diagnostic data page.

The TCAS diagnostic data page makes it easier to prepare a TCAS squawk report when a failure occurs. When preparing a TCAS squawk report, operate the TCAS self-test and then select the diagnostic data page. Include all failures shown in the list in your squawk report.



TCAS Diagnostic Data Page

Controls

Controls on the MFD-85C operate in the same manner as the controls on the MFD-85B except that the TFC button has been added, the RDR and NAV modes are controlled from the same button, and a line key menu for the TFC display mode is added. The following paragraphs describe the controls that affect the information shown in the TFC display mode.

TFC Button—Push the TFC button on the top left side to turn on the TCAS traffic display mode and the TCAS line key menu along the right side of the display. The TCAS menu turns off after approximately 5 seconds of no selection activity. With the MFD in the TCAS display mode, push the TFC button or any line key at any time to again turn on the TCAS menu.

Line Keys—In the TFC display mode a line key menu on the right side of the display, when turned on, shows the traffic display options for selection. To turn on the line key menu push any line key or the TFC button. The menu stays on for approximately 5 seconds then turns off. While the menu shows on the display, push the line key for the desired display option to select, set or turn on and off that traffic display option. Selected options show in green and unselected show in white. The following describes the operation of the EFIS system's line key traffic display options.

The top line key, with the TCAS menu identifier $\diamond$ -OFF>, turns on and off the display of "other traffic". With other traffic turned off, the annunciator " $\diamond$ -OFF" shows in the upper left corner of the display, the line key menu identifier shows in green, and TCAS does not show "other traffic" on the display. With other traffic turned on, TCAS shows a cyan open diamond traffic symbol on the display for detected "other traffic", the " $\diamond$ -OFF" annunciator turns off, and the line key menu identifier shows in white.

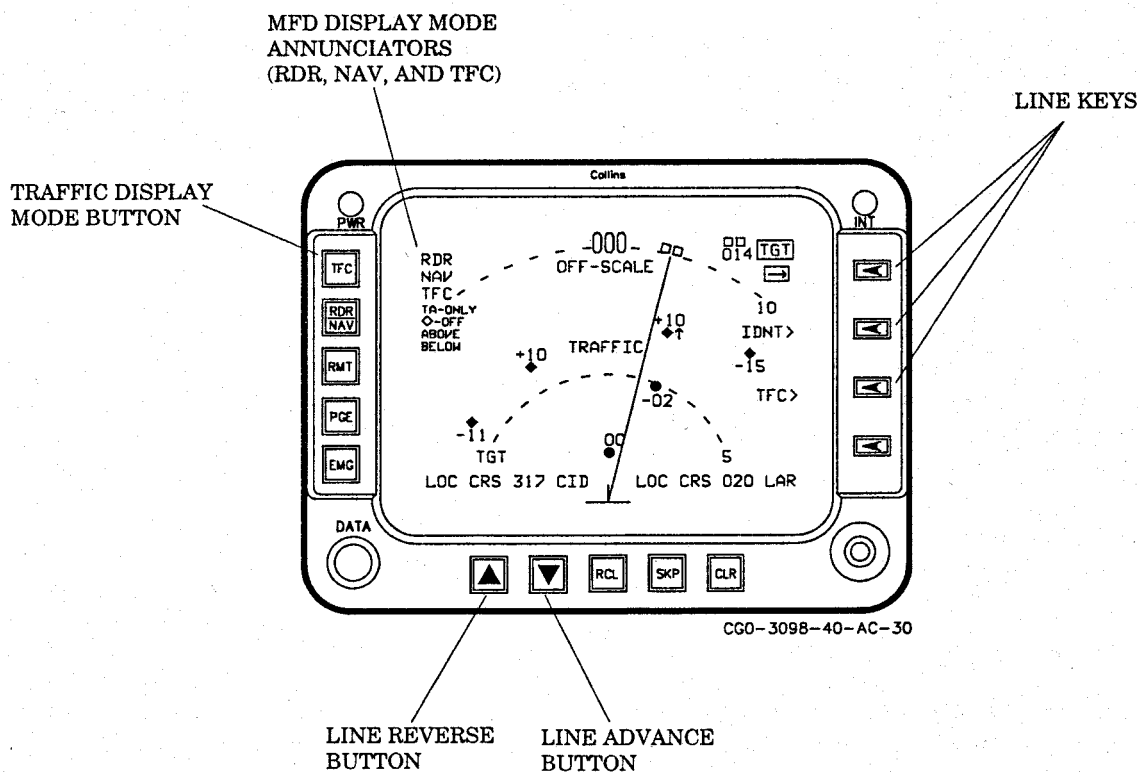
The second line key from the top, with the TCAS menu identifier A $\uparrow$ /B $\downarrow$ >, sets the altitude volume (relative to your own aircraft) for which TCAS shows "other traffic" on the display. PT, TA and RA traffic are not affected by the ABOVE and BELOW modes and always show on the display regardless of the selected altitude volume. (Except RA traffic does not show on the display in the TA-ONLY mode). Successive pushes of the button cycles through the available altitude volumes. The line key menu identifier shows selected modes in green and unselected modes in white. The ranges of the altitude modes are:

- NORMAL (2700 ft above to 2700 ft below, no annunciation on the display)
- ABOVE (9900 ft above to 2700 ft below)
- BELOW (2700 ft above to 9900 ft below)
- ABOVE and BELOW (9900 ft above to 9900 ft below).

The "ABOVE" and "BELOW" annunciators show in white in the upper left corner of the display. In the "normal" mode neither annunciator shows.

The third line key from the top, with the TCAS menu identifier ALT>, selects the type of altitude data, relative or absolute, that shows with the traffic symbols. In the relative mode, the altitude data (if available) shows as a two number read-out preceded by a "+" or "-" sign. In the absolute mode, the altitude data shows as a three number read-out. Also, in the absolute altitude mode, an annunciator at the lower left end of the full-range arc shows your own aircraft's altitude. If the altitude data is baro corrected, the annunciator shows as a three number read-out preceded by ALT (i.e. ALT240.) If the altitude data is not baro corrected or the TCAS system is in TEST mode, it shows as a three number read-out preceded by FL (i.e. FL240.)

Line Advance (▲) and Reverse (▼) Buttons (Range Selection)—The line advance and reverse buttons on the left side at the bottom of the MFD, select the traffic display range. The advance button (▲) increases the range and the reverse button (▼) decreases the range. Selected range is annunciator in the upper right corner of the TFC display. If the TFC display mode shows as part of a composite display with the RDR display mode, then range is controlled by the WXP and the line advance reverse buttons are disabled.



Radar (RDR) and Navigation (NAV) Display Modes

The MFD-85C uses essentially the same RDR and NAV display mode formats as the MFD-85B. Only the variations from the MFD-85B are described in this section. Refer to the MFD-85B Weather Radar Display Mode and Navigation Display Mode sections of the MFD-85B chapter for descriptions and operation of MFD-85C annunciators, flags, and display elements not described here.

The $\begin{smallmatrix} \text{RDR} \\ \text{NAV} \end{smallmatrix}$ button turns on and off both the RDR and NAV display modes. Each push of the button cycles through a sequence of display mode selections. The first push turns on the RDR mode. A second push turns on the NAV mode with the RDR mode to form a composite display of radar and navigation information. A third push turns off the RDR mode and leaves the NAV mode on. A fourth push turns off the NAV mode and turns on the RDR mode. At the end of the sequence, the selection cycle repeats as the button is pushed. Selection of a display mode with any other mode button (i.e., TFC, RMT, etc...) saves the currently selected RDR/NAV display mode(s) and selected display options in memory in the MFD. When the $\begin{smallmatrix} \text{RDR} \\ \text{NAV} \end{smallmatrix}$ button is pushed the last selected RDR and/or NAV display mode(s) are recalled from the memory and shown on the display.

The third line key in the line key menus for both the RDR and NAV display modes shows the annunciator "TFC" and is used to turn on and off the TFC display mode. With the $\begin{smallmatrix} \text{RDR} \\ \text{NAV} \end{smallmatrix}$ button and the TFC line key, the MFD-85C can show a composite display of any combination of the TFC, RDR, and NAV display modes. All of the annunciators, flags, and display elements for each mode, show on the display in essentially the same manner and locations as in the respective dedicated display modes. The exceptions are described in the following paragraphs.

NOTE

TCAS display options such as ABOVE, BELOW and "Other Traffic" on or off, cannot be selected or changed when the TFC mode shows as part of a composite display with the RDR and/or NAV display mode(s). Push the TFC mode button to turn on the dedicated TFC mode, to make changes to the TCAS display options.

RDR Display

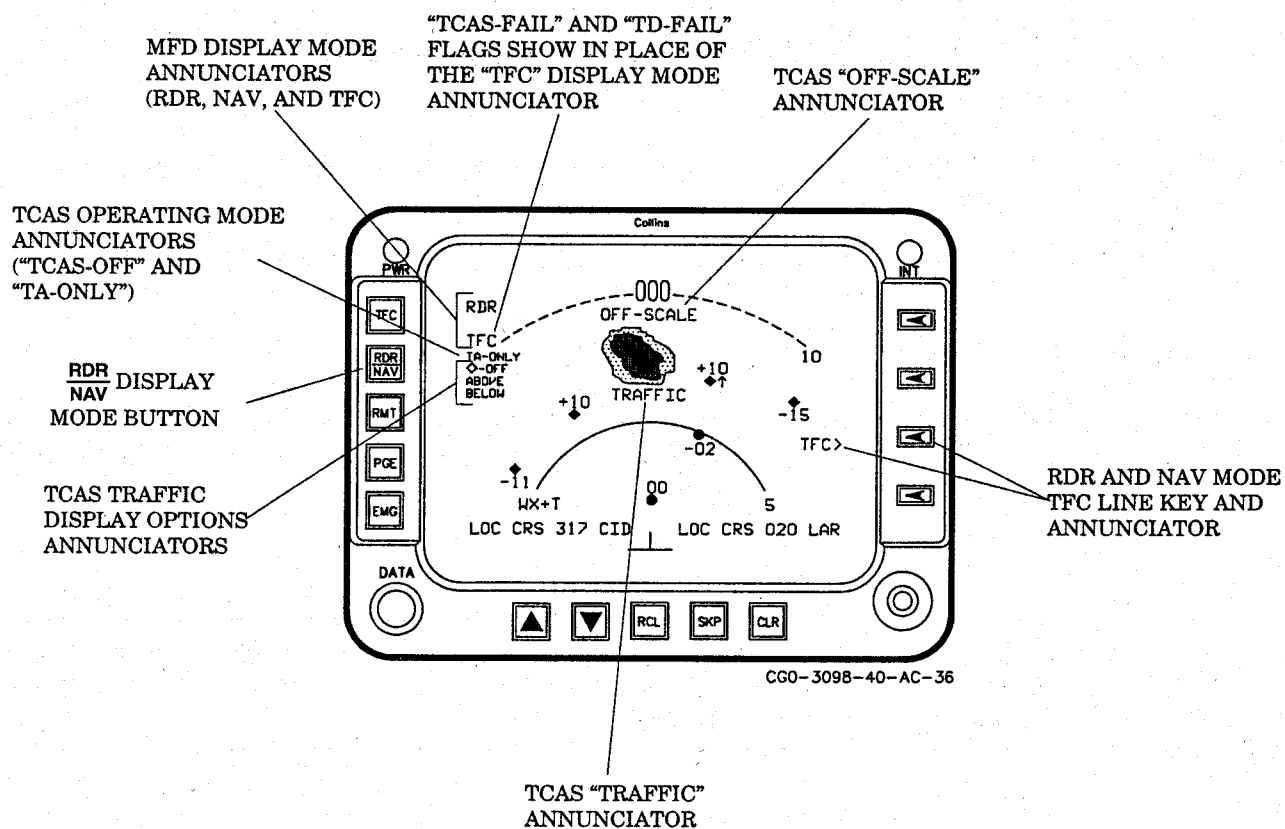
The MFD-85C RDR display mode format is the same as the MFD-85B RDR mode with the following differences:

- In the RDR mode, the line key menu shows the annunciator "TFC" on the third line key from the top. A composite display of the RDR and TFC modes shows on the MFD-85C when the TFC mode is selected with this line key.
- An additional display mode annunciator ("TFC") shows in green in the upper left corner of the display below the RDR and NAV mode annunciators, when the TFC mode is selected to show on the MFD.

In the RDR and TFC composite display mode, all of the annunciators, flags, and display elements show on the display in the same manner and locations as in the respective dedicated display modes with the following exceptions.

- TCAS mode annunciators, "TCAS-OFF" and "TA-ONLY", show below the display mode annunciators and above the TCAS display options annunciators.

- The TFC display line key menu is not available and does not show in the RDR display mode or in the composite TFC and RDR and/or NAV display modes.
- TCAS no bearing annunciators do not show in the TFC and RDR and/or NAV composite modes.
- The annunciator "TRAFFIC" shows in yellow (TA) or red (RA) in the middle of the display when TCAS detects TA or RA traffic even if the TFC mode shows as a part of a composite display with the RDR and/or NAV modes.
- Display range and range rings show in the RDR display format.
- The "TCAS FAIL" and "TD-FAIL" annunciators show in the upper left corner in place of the TFC annunciator for TCAS computer failures and TCAS display failures respectively.
- In the composite display mode of TFC, RDR and/or NAV, the TCAS display option annunciators show in the middle of the left side of the display below the MFD display mode annunciators ("RDR", "NAV", and "TFC").



RDR and TFC Composite Display Mode

NAV Display

The MFD-85C NAV display mode format is the same as the MFD-85B NAV mode with the following differences:

- In the line key menu the annunciator TFC shows on the third line key from the top in place of the ENTER annunciator unless the joystick is in use. A composite display of the NAV and TFC modes shows on the MFD-85C when the TFC mode is selected with this line key.
- An additional display mode annunciator ("TFC") shows in green in the upper left corner of the display below the RDR and NAV mode annunciators, when the TFC mode is selected to show on the MFD.

In the NAV and TFC composite display mode, all of the annunciators, flags, and display elements show on the display in the same manner and locations as in the respective dedicated display modes with the following exceptions.

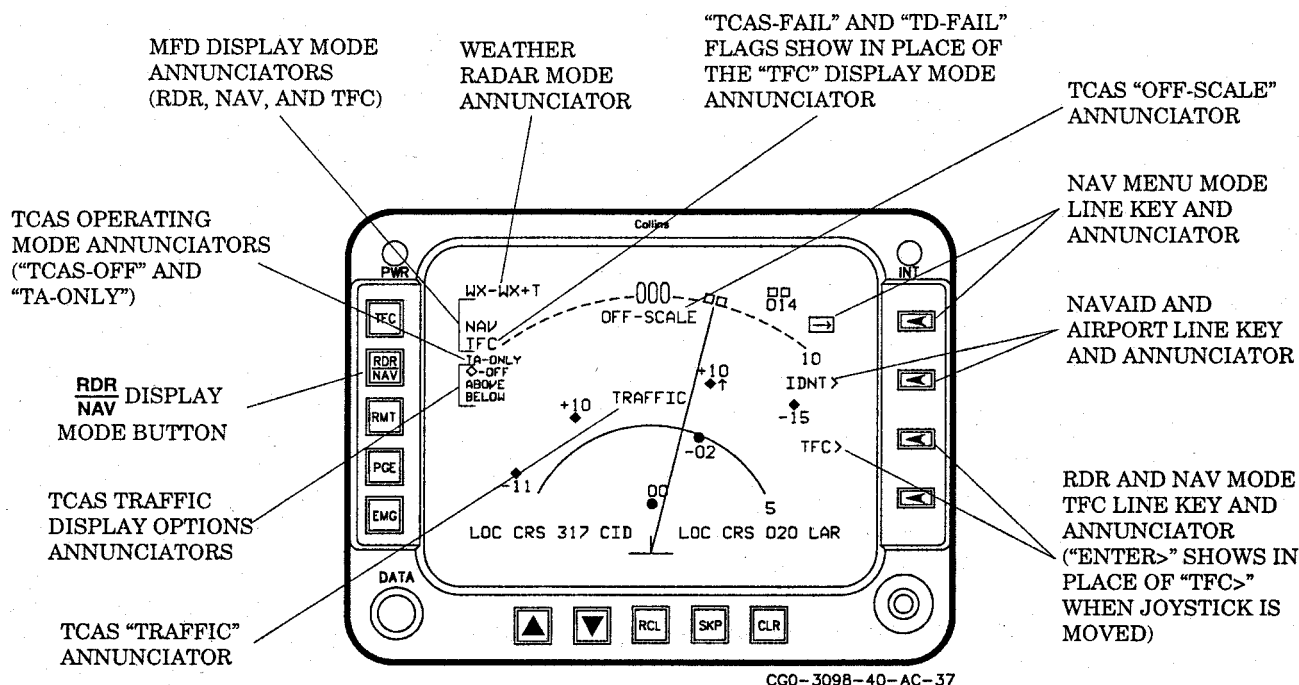
- The TCAS mode annunciators "TCAS-OFF" and "TA-ONLY" show below the display mode annunciators and above the TCAS display options annunciators.
- The TFC display line key menu is not available and does not show in the NAV display mode or in the composite display mode of TFC and NAV and/or RDR.
- The TCAS no bearing annunciators do not show in the composite display mode of TFC and NAV and/or RDR.
- The annunciator "TRAFFIC" shows in yellow or red in the middle of the display when TCAS detects TA or RA traffic, even if the TFC display shows as a part of a composite display with the NAV and/or RDR displays.
- In the composite display mode of TFC and NAV the display range and range rings show in the NAV display format. Also, Any time the RDR display mode shows as part of a composite NAV display with or without the TFC display, the display range and range rings show in the RDR display format.
- The "TCAS-FAIL" and "TD-FAIL" annunciators show in the upper left corner in place of the TFC annunciator for TCAS computer failures and TCAS display failures respectively.
- In the composite display mode of TFC, NAV and/or RDR, the TCAS display option annunciators show in the middle of the left side of the display below the MFD display mode annunciators ("RDR", "NAV", and "TFC").

NOTE

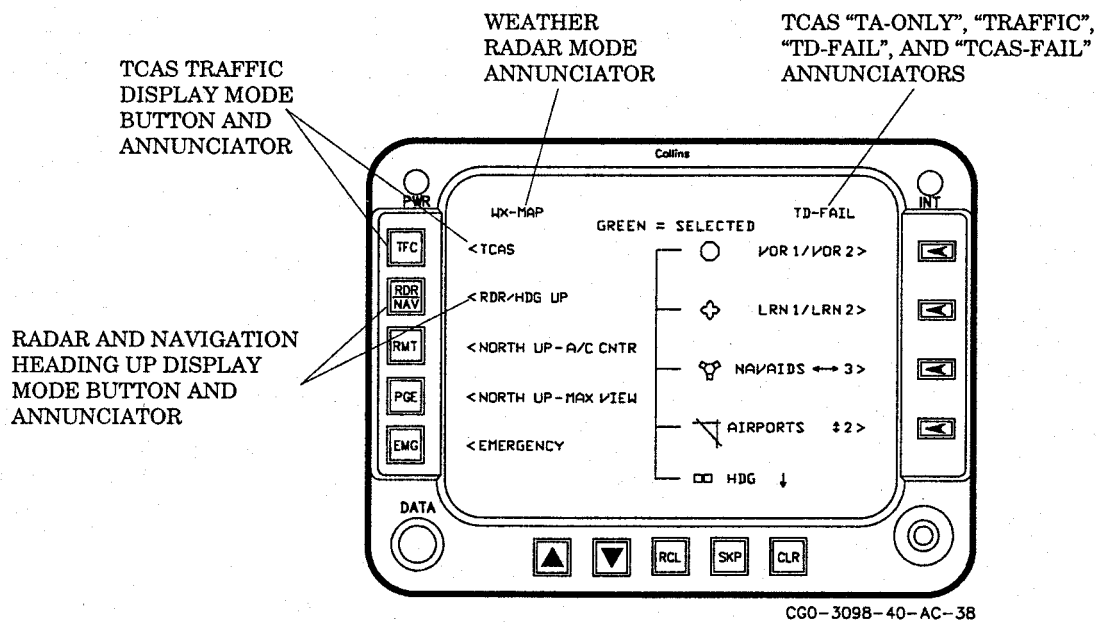
If the joystick is moved while in a NAV display mode or a composite NAV and RDR display mode, the TFC line key annunciator changes to ENTER and the line key is used to enter a waypoint into the MFD memory for transfer to a compatible LNV system.

NAV MENU Display

Operation of the MFD-85C NAV menu is the same as the MFD-85B, except for the TFC and $\frac{RDR}{NAV}$ buttons. Push the TFC button, with the display mode annunciator "<TCAS" to show the dedicated TCAS traffic display mode. Push the $\frac{RDR}{NAV}$ button, with the annunciator "<RDR/HDG UP" annunciator to show the radar and/or NAV and/or TFC display modes. From the NAV menu the $\frac{RDR}{NAV}$ button changes the display to show the mode(s) (RDR and/or NAV and/or TFC) that were selected prior to selection of the NAV menu display mode.



NAV and TFC Composite Display Mode



NAV Menu Display Mode

RMT, PGE, and EMG Display Modes

Refer to the MFD-85B section for descriptions and operation of the RMT, PGE, and EMG display modes for the MFD-85C. These display modes are the same for both the MFD-85B and -85C.

REVERSIONARY MODES OF OPERATION

EFIS-85C(14)/86C(14) systems have reversionary operating modes for the controls, displays, and display processors in the event of a failure of a system element. Control for selection of normal and reversion operating modes is provided by external, cockpit mounted reversionary switches. The following paragraphs describe the operational and display differences when the system is operated in a reversion mode.

NOTE

All external switches for reversion and drive transfer switching are installer supplied items and are not part of the EFIS system. Therefore, not all EFIS installations have reversion switching for all of the reversion capabilities. Also, switch labels and position names may not be the same as those used in the following descriptions. Refer to the appropriate aircraft manuals for specific information on the reversion operation modes specific to your aircraft.

Electronic Flight Display (EFD) Failure

EFD failures show in many ways such as:

- blank displays
- jittery or scrambled displays
- fading displays
- unstable size of the display or display elements.

Various reversionary display modes are available for the EFDs on each side of the cockpit. A DISPLAY SELECT switch sets the system to show an EADI on the top EFD and an EHSI on the bottom EFD (normal operation mode) or an EADI on either EFD (top or bottom) with the other EFD turned off or an EADI on both the top and bottom EFDs at the same time. A DISPLAY MODE switch selects either normal operation (EADI and EHSI on individual displays) or the composite display mode which shows an EADI and an EHSI together. These two switches can be operated together such that a composite display can show on either the top or bottom EFD with the other EFD turned off, or on both the top and bottom EFDs. Interconnect wiring straps, connected during installation of the EFIS system, determine the specific switching capabilities used and whether or not the composite display mode is available.

EADI Failure (top EFD)

To recover EADI display information select the lower EFD with the display select switch and select the composite display mode with the display mode switch. A composite display of the EADI and EHSI shows on the bottom EFD.

EHSI Failure (bottom EFD)

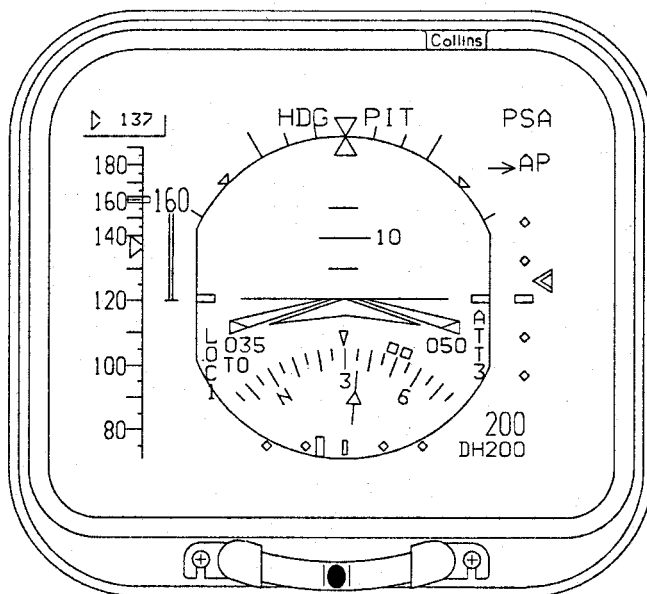
To recover EHSI display information select the upper EFD with the display select switch and select the composite display mode with the display mode switch. A composite display of the EADI and EHSI shows on the top EFD.

NOTE

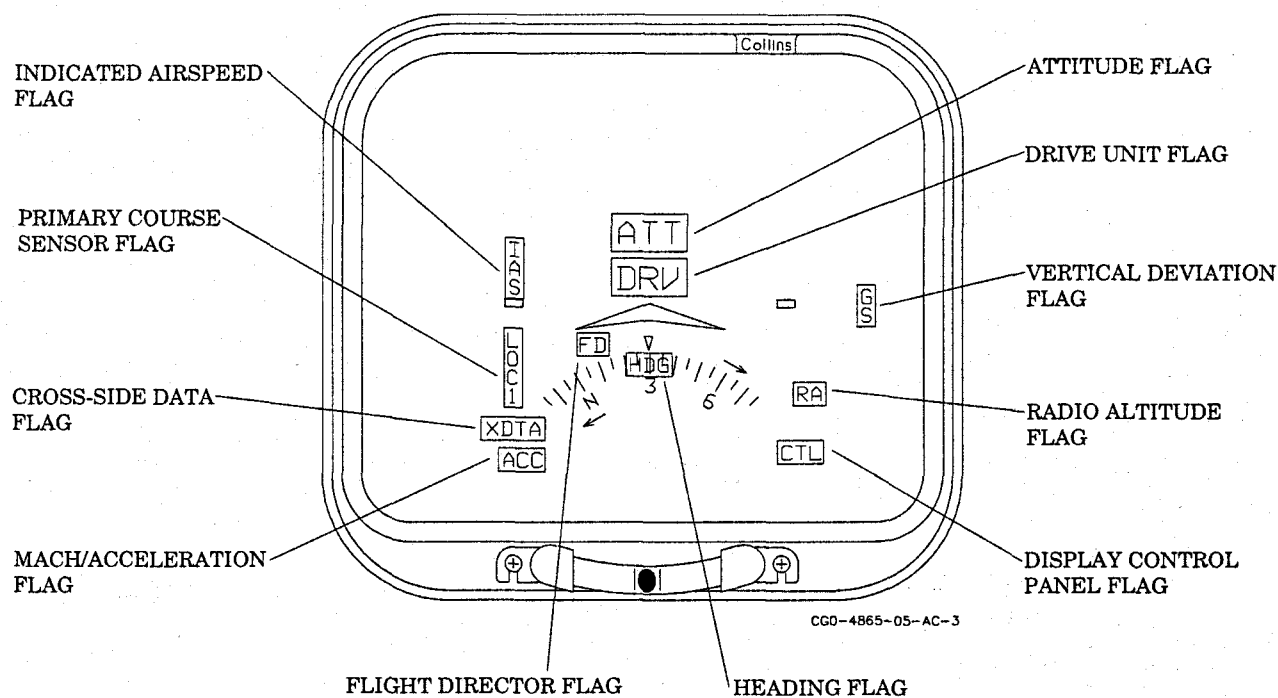
In some installations, both the display select and composite mode functions are switched together on one switch. In these installations, the composite mode is selected automatically when the display mode switch is set to other than the normal mode. The switch may be labeled as a “Composite” switch with normal mode as the center position, ADI as the upper position and HSI as the lower position.

In the composite display mode, all flags and annunciators are the same, except that they may be positioned in a different location. Refer to the figure on the following page.

In addition to moving the EADI or combining it with the EHSI in the composite display mode on either EFD, a drive transfer switch can set the MFD to show the display mode selected for the bottom EFD. If the pilot's or both the pilot's and copilot's drive transfer switches are set to the drive transfer mode, the pilot's bottom EFD display mode is shown on the MFD. If only the copilot's drive transfer switch is set to the drive transfer mode, the copilot's bottom EFD display mode is shown on the MFD. When the MFD display is in drive transfer, all MFD functions are disabled, except for the PWR button and the INT knob. Refer to the Display Processor Unit Failure paragraphs for additional information about the drive transfer modes.



EADI and EHSI Composite Display Mode



Composite Display Mode Flags

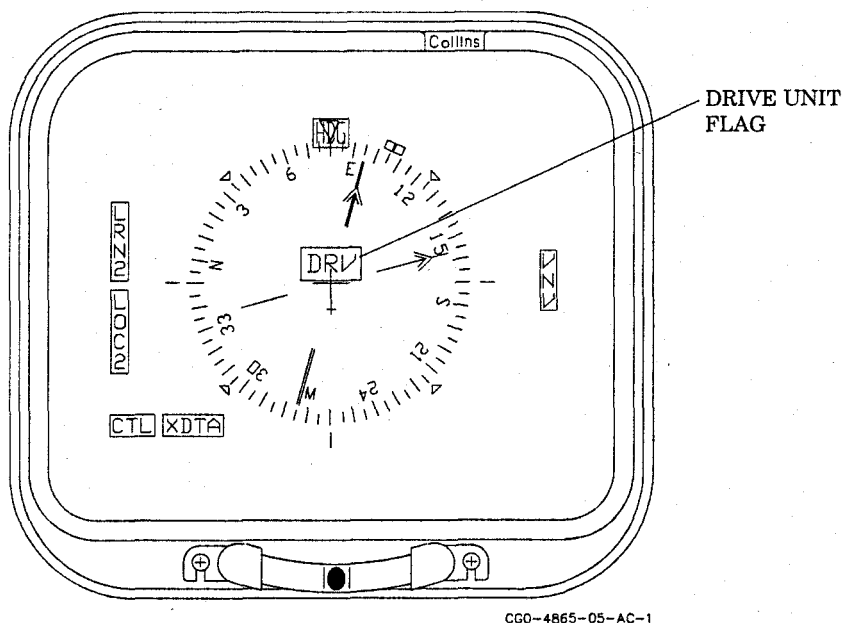
Display Processor Unit (DPU) Failure

A DPU failure shows the red boxed "DRV" annunciator in the middle of the EADI and EHSI displays. To restore operation of the displays using the MPU in place of the failed DPU, set the external, cockpit mounted drive transfer switch, on the failed side of the cockpit, to the drive transfer mode. A yellow boxed "CDRV" annunciator (center drive) shows on the EFDs set to the drive transfer mode and on the MFD. See page 133.

External, cockpit mounted drive transfer switches select the EFIS display processors (DPU or MPU) that drive the EFDs. When the switches are set to the normal operation mode, one DPU drives the EFDs on one side of the cockpit, the other DPU drives the EFDs on the other side of the cockpit, and the MPU drives the MFD display. When a DPU failure is indicated ("DRV" annunciator shows on the EADI and EHSI displays), set the drive transfer switch for that side of the cockpit to the drive transfer mode to restore operation of the EFDs. In the drive transfer mode, the MPU drives the MFD and the EFDs on the side in drive transfer mode while the DPU on the other side continues to drive its EFDs. If both sides of the cockpit are set to the drive transfer mode, the MPU drives all four of the EFDs and the MFD.

NOTE

Some installations use a single switch to select the drive transfer mode. The switch has a center position for the normal mode with a left and right position for the transfer modes. Dual drive transfer cannot be selected in these installations.

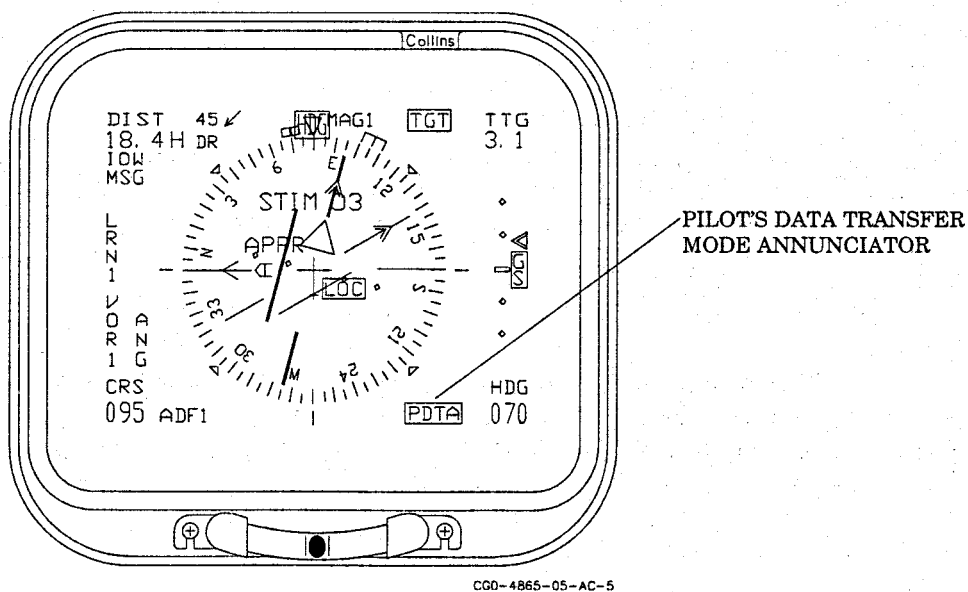
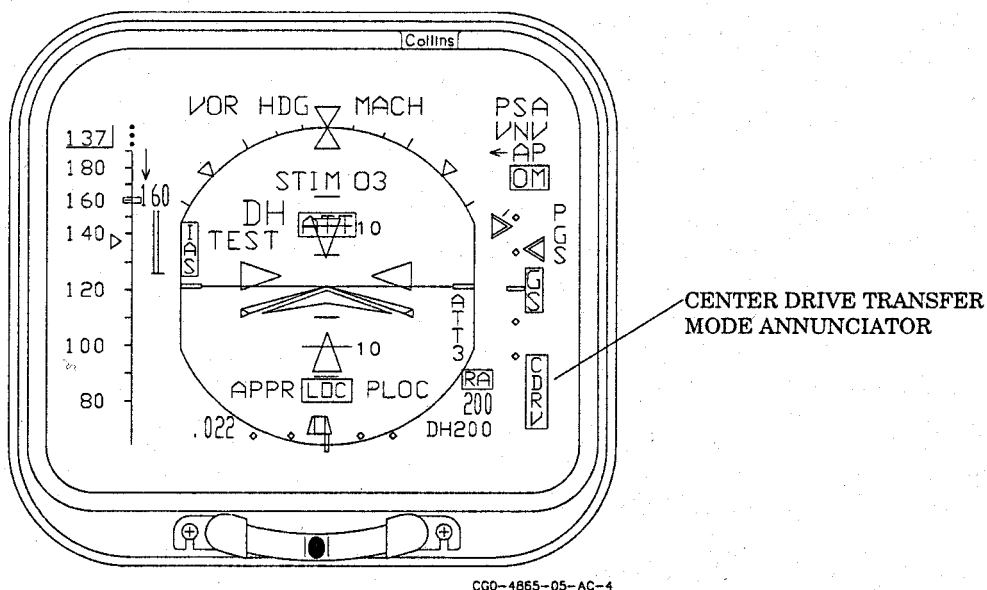


DPU Failure Example (EHSI)

In the drive transfer mode, the EFDs show a normal EADI on the top display and an EHSI on the bottom display or a composite display on either or both of the EFDs (depends on the EFD reversion switch settings as previously described for an EFD failure). In addition, the MFD changes to show the same display information as the bottom EFD and all normal MFD functions are disabled, except for the PWR button and the INT control. When both sides of the cockpit are in drive transfer mode, the MFD shows the pilot's bottom EFD display information. A yellow boxed "CDRV" annunciator shows on the EFDs on the side set to the drive transfer mode and on the MFD. If both sides are set to the drive transfer mode then a yellow boxed "PDTA" annunciator (pilot's data) shows on all four EFDs and on the MFD.

DCP-85() Display Control Panel sensor selection, EHSI display mode, and other display functions, operate in the same manner through the MPU in drive transfer mode as they do through the DPU in normal operation mode. Therefore, the EADI and EHSI displays show all the same information in the same manner as in the normal operation mode. If only one side of the cockpit is set to the drive transfer mode, the MPU is controlled by that side's DCP and uses that side's sensor data for the display information. If both sides of the cockpit are set to the drive transfer mode, then the pilot's DCP controls the MPU and the pilot's side sensors are used. Therefore, when both sides of the cockpit are in drive transfer, the EADIs and EHSIs or composite displays on both sides of the cockpit show exactly the same information from the pilot's side sensors.

Comparator monitoring is switched from the failed DPU to the MPU when the drive transfer mode is selected. If it is determined that a DPU failure, not a sensor failure, is the cause of comparator warn flags, select the drive transfer mode to disable the DPU comparators and enable the MPU comparators. In some installations, depending on the aircraft wiring, the comparator monitors in the DPU are not disabled when the drive transfer mode is selected, but the MPU monitors are turned on. In these installations, if it is determined that a DPU failure is the cause of comparator warn flags, in addition to selecting the drive transfer mode, open the failed DPU's circuit breaker to disable its comparator monitors. This allows only the MPU comparators to operate on the drive transfer side. When both sides of the cockpit are set to the drive transfer mode, comparator monitoring is disabled.



Drive Transfer Annunciation

Display Control Panel (DCP) Failure

A DCP failure shows the red boxed "CTL" annunciator in place of the decision height read-out on the EADI and the numeric course read-out on the EHSI. To regain control of the displays on the side of the cockpit with the failed DCP, set the external, cockpit mounted DCP control transfer switch for the failed side to the transfer or cross-side position. This transfers control of both the DCP and CHP controls to the working side DCP and CHP. A yellow boxed "XCTL" annunciator shows on the EFDs on the side of the cockpit that has transferred control and on the MFD.

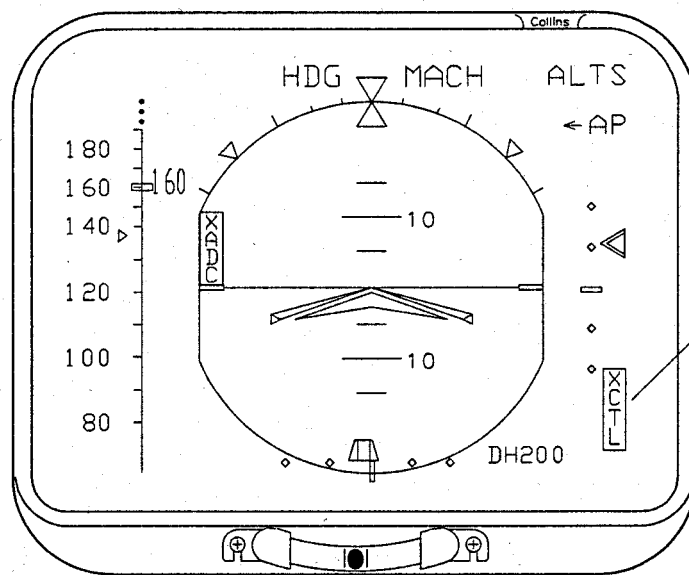
Externally, cockpit mounted DCP transfer or cross-side switches select the DCP that controls the DPU (or MPU in drive transfer mode) and therefore the EFDs. In the normal operation mode the pilot's DCP controls the pilot's side displays and the copilot's DCP controls the copilot's side displays. This is true even when one side of the cockpit is in the drive transfer mode where the MPU replaces the DPU on that side. When the DCP transfer or cross-side switch on the pilot's side is set to the transfer or cross-side mode, the copilot's DCP controls both sides of the cockpit. The opposite is true when the copilot's side is set to the transfer or cross-side position. Also, when both sides of the cockpit are set to the drive transfer mode (MPU driving all four EFDs and the MFD) then the pilot's DCP controls both sides of the cockpit unless the pilot's DCP transfer or cross-side switch is set to the transfer or cross-side position. In that case the copilot's side DCP controls both sides of the cockpit.

When the DCP transfer or cross-side mode is selected, the annunciator "XCTL" shows in yellow in the lower right corner of the cross-controlled EADI and the EHSI displays. All display parameters, except FCS mode annunciation, are controlled by the cross-side DCP and CHP. Decision height set read-out and NAV data elapsed timer functions also show in yellow on the cross-controlled EADI. However, GS and TTG NAV data and sensor annunciations show in the color appropriate for the selected sensor for that side, the same as when in normal operation. TAS from the air data systems and wind data from the LNAV systems do not go to the EFIS system through the DCP. Therefore, they are not affected by control transfers and on-side data shows on each side of the cockpit unless a cross-side system is selected for the respective sensor.

Also, the flight control system (FCS) mode annunciations which are received by the EFIS system through the DCP, are removed from the EADI display on the failed DCP side of the cockpit. However, on-side FCS mode data on the working side DCP shows on the on-side EADI display. If the on-side DCP detects a failure, the FCS mode annunciation is removed from the displays.

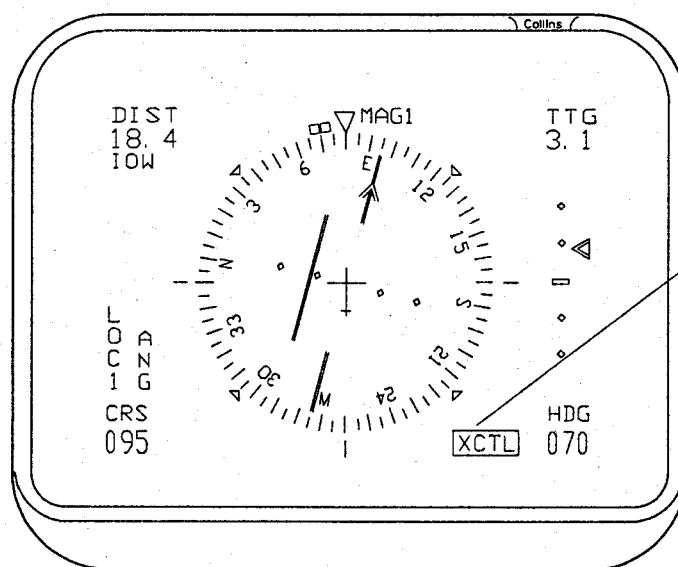
NOTE

Display brightness control functions are not transferred from one DCP to the other in the cross-side DCP modes. The DCP DIM knobs control the on-side EFD brightness in all operating modes.



CGO-4866-02-AC-2

EADI Cross-Side DCP Annunciation



CGO-4866-02-AC-1

EHSI Cross-Side DCP Annunciation

If the DCP failure is such that data from the DPU (or MPU in drive transfer mode) to the DCP is not being received, several operating functions are not available. The following table lists these functions:

FUNCTION	EFFECT / ALTERNATE ACTION
Direct to	Course deviation must be zeroed with the CRS knob.
HDG sync	Heading cursor must be zeroed with the HDG knob.
CAT II Request	CAT II request can only be reset by pushing the CAT II request button instead of being reset automatically by autopilot disengage acknowledge.
Auto BRG Menu	The bearing pointer menu is not reconfigured to exclude localizer selections. Therefore, Blank BRG pointer displays occur for these selections.

Multifunction Processor Unit (MPU) Failure

A detected MPU failure shows the annunciator "DRV" in red in the middle of the MFD and on the EFDs when the MPU is driving the displays in a drive transfer mode. When the MPU fails, the MFD and all it's functions are lost (including TCAS and weather radar) as well as the drive transfer capabilities for the EFDs. However, navigation and weather radar information are available for display on the EHSIs.

Multifunction Display (MFD) Failure

When the MFD fails, all it's functions are lost and there is no backup capability for the display. However, navigation and weather radar information are available for display on the EHSIs.

SELF-TEST

EFIS-85C(14)/86C(14) systems use internal self-test routines to verify proper operation of the EFIS systems. In addition, the displays show flags or other self-test information from the remote sensors that provide data to the system.

EFIS Self-Test

EFIS self-test is initiated by actuating an external installer supplied switch or button ("EFIS TEST" or similar). When the EFIS self-test is first initiated the following occurs:

- pilot's EADI shows a 10-degree pitch up from the current pitch attitude and a 10-degree right roll from the current roll attitude
- pilot's EHSI shows a 20-degree heading change to the right from the current heading
- copilot's EADI shows a 10-degree pitch down from the current pitch attitude and a 10-degree left roll from the current roll attitude
- copilot's EHSI shows a 20-degree heading change to the left from the current heading
- the annunciator "TEST" shows in red on the EADIs and the MFD while the test is in process.

During the EFIS self-test the various comparator monitors are tested. The 10-degree attitude and 20-degree heading changes create a compare error for the comparator monitors to detect. Properly operating attitude and heading comparator monitors show the yellow boxed "ATT" (attitude) comparator warning flag on the EADIs and the yellow boxed "HDG" (heading) comparator warning flag on the EHSIs. In addition to the EFIS comparator warn flags, if installed, remote mounted master warn and comparator warn annunciators turn on. If CAT II approach mode is selected when the EFIS self-test is initiated, the CAT II comparator monitors are also tested. Properly operating CAT II comparators show the yellow boxed "LOC", "GS", "IAS", and "RA" comparator warn flags on the EADIs and yellow boxed "LOC" and "GS" comparator warn flags on the EHSIs and, if installed, remote mounted CAT II comparator warn annunciators turn on.

Also tested when the CAT II approach mode is selected, are the CAT II lateral, vertical, (red triangle pointers) and IAS (yellow flashing arrows) excessive deviation monitors. Refer to the EADI and EHSI Category II Excessive Deviation Warnings sections of this pilot's guide for detailed explanations of the excessive deviation warning annunciations.

Push and hold the "EFIS TEST" switch/button for longer than 4 seconds to turn on the flags test portion of the self-test. During the flags test, pitch, roll, and heading information are removed from view and all active display flags show on the displays. When the self-test switch/button is released, the displays return to normal operation, except that comparator warn flags do not turn off. To turn off the comparator warn flags, push and release the remote mounted comparator warn reset switch/button.

A copyright message shows on the displays during all portions of the EFIS self-test in installations with -001 version DPU/MPU. Installations with all other versions of DPU/MPU show the copyright message only in the flags test portion of the EFIS self-test.

Sensor Self-Test

In many installations, sensors that provide data to the EFIS system have independent self-test capabilities. When EFIS detects a sensor in self-test mode, it is annunciated on the EFD and MFD displays. The following paragraphs describe the self-test annunciations for various sensors.

Attitude Heading Sensors

Collins AHS-85() Attitude Heading System—During self-test of a Collins AHS-85() Attitude Heading System, a red “STIM XX” annunciator shows in the upper middle of the EADI and EHSI displays to indicate that the AHS-85() is in the STIM mode (self-test). There are several annunciators for various self-test modes for this system. They include:

- “STIM 0A” which indicates sequence of all tests
- “STIM 01” which indicates roll tests
- “STIM 02” which indicates pitch tests
- “STIM 03” which indicates heading tests

Other Attitude Heading Systems—During self-test of non-Collins AHRS systems a red “HDG TEST” annunciator shows in the upper middle of the EHSI display and a red “ATT TEST” shows in the upper middle of the EADI display to indicate that the AHRS driving the attitude and heading displays is in the test mode.

Navigation Sensors

VOR, ILS, MLS, TCN, ADF, and LNAV—For navigation sensors in self-test mode the EADI and EHSI show the sensors fail flags non-flashing during the self-test. Vertical and lateral deviation displays are not removed from view during the test. When the sensors self-test is successfully completed, the fail flags turn off.

Radio Altitude Sensors

Radio altimeters in self-test output a fixed radio altitude value that is shown on the EADI in the radio altitude read-out annunciator in the lower right corner of the display. If the decision height is set higher than the fixed value, the “DH” annunciator shows on the display during the test. Push the RA TST button on the DCP to operate the radio altimeter self-test.

Air Data Sensors

An air data sensor in self-test mode, shows the red annunciator “ADC TEST” vertically to the right of the airspeed scale. The annunciator turns off when the self-test is successfully completed.

PROGRAMMING MFD PGE AND EMG DATA PAGES

There are two ways to program the page and emergency data pages in the MFD. One method uses a computer to create the checklist page data which is loaded into a CEU-85 or CEU-85A Checklist Entry Unit, then the data is loaded into the MFD from the CEU. The other method uses the RDP-300 Remote Data Programmer to enter the checklist page data directly into the MFD.

Programming with CEU-85 or CEU-85A Checklist Entry Units

Load the desired checklist data into the CEU-85/85A following the procedures contained in the operation section of either the CEU-85 Checklist Entry Unit Instruction Book (523-0774328) or the CEU-85A Checklist Entry Unit Instruction Book (523-0776400). Once the data is loaded into the CEU-85/85A, proceed as follows to program the MFD:

- 1.—Turn on aircraft power. Ensure that the MFD circuit breakers are in. Push the PWR button at the upper left of the MFD. Position the WXP MODE switch to OFF (the WXP is not required during programming of the MFD). Allow 15 seconds for the MFD to warm up. Adjust the INT control at the upper right of the MFD as desired.
- 2.—Push the PGE button then push the RDR button on the MFD.

NOTE

The next two steps (steps 3 and 4) must be performed within 10 seconds of each other.

- 3.—Plug the CEU-85() into the DATA jack on the MFD .
- 4.—On the MFD, simultaneously push the top line select key, the bottom line select key, and the line reverse (delta) keys. (The line select keys are located along the right side of the MFD.)
- 5.—Data transfer is complete in approximately 11 minutes maximum for CEU-85 or 7 minutes maximum for CEU-85A (depends on amount of data being loaded). The RDR mode screen reappears after data transfer is complete.
- 6.—Disconnect the CEU-85() from the MFD. Verify that data just loaded is accessible and correct by first pushing the PGE or EMG button on the MFD, then push the appropriate line select key to display the first page of the selected chapter. Use the joystick and the line advance/line reverse keys to view the various pages and/or select new chapter title pages.
- 7.—Turn off the power.

NOTE

The information loaded into the MFD will not be erased by power removal. Anything loaded during an earlier procedure will remain until reloaded.

Programming with the RDP-300 Remote Data Programmer

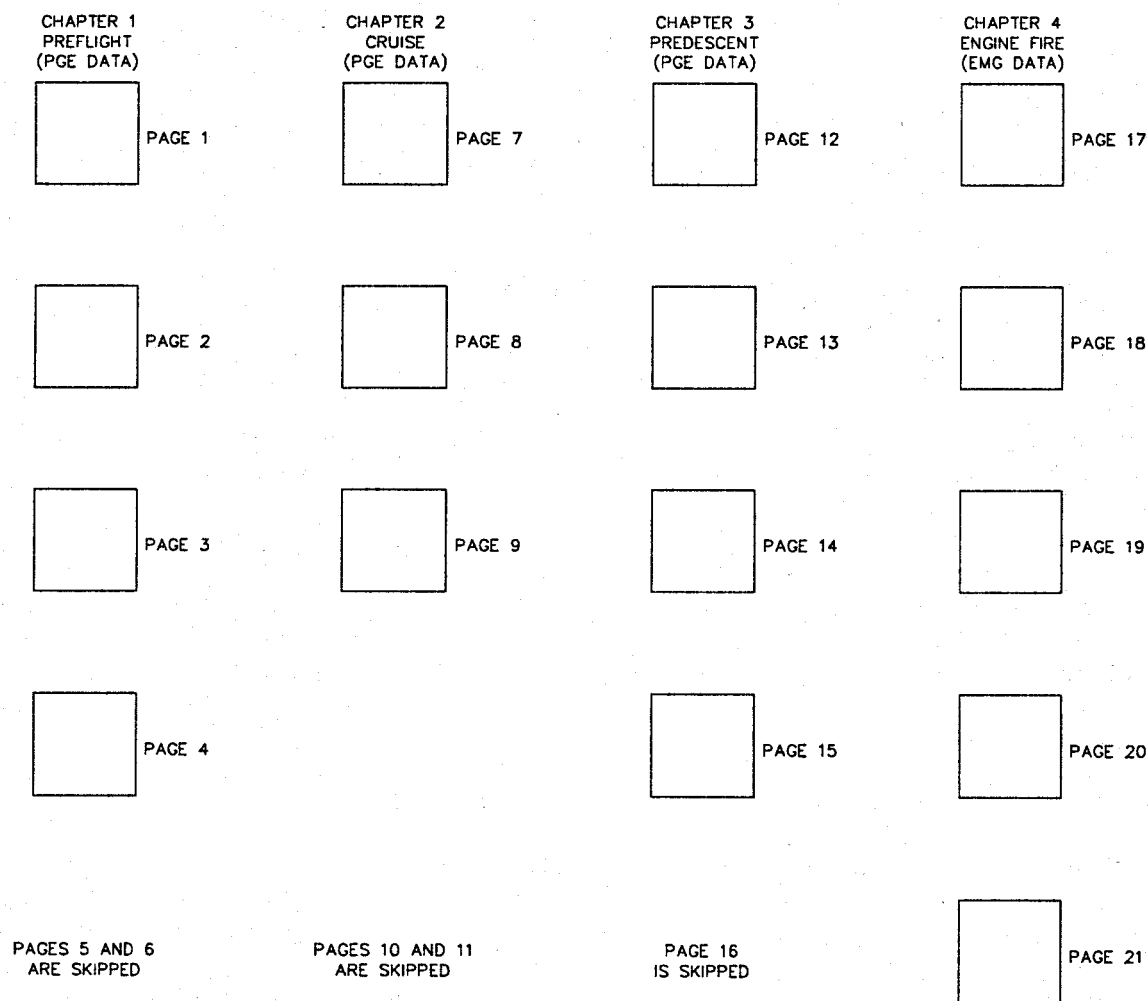
Programming Preparations

Before beginning any programming operation, prepare a draft copy listing exactly what the chapter titles are to be, and exactly what will be placed on each page within that chapter.

Make copies of the programming grid on the next page and use them to write out each page of data. Then follow the grid sheets to enter the data into the MFD. You can save the written grid sheets for future reference for reprogramming of the MFD. The grid sheets aid in keeping track of the the characters and lines used on each page and of the total pages used.

Keep in mind that there are 100 available pages and each page has 12 lines of 20 characters each. Also indicate which chapters, if any, will be emergency chapters. Chapter length depends on user requirements. You may have 1 chapter of 100 pages or 100 chapters of 1 page or any combination in between. It is recommended that pages be numbered consecutively within a chapter. It is also suggested that one or two blank pages be left between chapters to accommodate additional information in the future. See the page layout example in the figure below.

TYPICAL CHAPTER AND PAGE LAYOUT



CGO-3110-10-AC-3

PROGRAMMING GRID FOR PGE AND EMG DATA

LINES ↓	CHARACTERS →																			
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COMMENTS:

PAGE NO:

LINES ↓	CHARACTERS →																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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MFD SER NO:

A/C TAIL NO:

PAGE NO:

CGO-3110-10-AC-2

PROGRAMMING GRID FOR PGE AND EMG DATA

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COMMENTS:

PAGE NO:

LINES ↓	CHARACTERS →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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COMMENTS:

MFD SER NO:

A/C TAIL NO:

PAGE NO:

CGO-3110-10-AC-2

NOTE

When operating the MFD in the PGE and EMG modes all checklist lines are reset when another chapter is selected prior to completing the current chapters checklist. The RCL button on the MFD can return to skipped lines within the current chapter. For these reasons, some operators prefer to put an entire checklist in one chapter rather than breaking it up into different chapters.

Because there are only 100 available pages in the MFD, it is suggested that the title page be part of the checklist page (i.e. the entire page should be used for part of the checklist). Whatever is programmed into the top line of the title page shows next to the line select keys in the MFD PGE or EMG display modes. Therefore, the top line of the title page should contain enough information to describe the contents of the chapter. It is suggested that the top line be right justified so as to appear next to the appropriate line select key, or else fill in with periods "." to the right of the line. This should aid in selection of title pages.

NOTE

Page numbers for all pages (1 through 100) are visually and internally numbered at the factory for convenience. The visual page numbers show in the upper right corner of each data page. These numbers can be relocated on the page, eliminated, or changed (not recommended) to a different number as desired.

Operating the RDP-300 Remote Data Programmer

CAUTION

To prevent loss of data, it is essential that the programming procedure is followed carefully and in the order given. **Make certain that the RDP is connected to or disconnected from the MFD only while continuous power is applied to the system.** Information properly programmed into the MFD is not erased when power is removed from the MFD. Also, anything previously programmed into the MFD remains in the MFD until it is reprogrammed.

Keyboard Description

The RDP is a portable, handheld unit, similar to a calculator, that is used to input data into the memory of the MFD. The RDP should be stored in a location that is accessible to the flight crew if on-aircraft programming is desired. The following paragraphs describe the functions of the various keys on the RDP.

Letters, Numbers, and Punctuation—Letters, numbers, and punctuation may be alternately selected by shifting (SH key), much like upper and lower case on a typewriter.

PR (Page Reverse)—Used with SH (shift) key to cycle the display back to the previous page.

PF (Page Forward)—Used with SH (shift) key to advance the display to the next page.

LR (Line Reverse)—Used with SH (shift) key to go back to a line to correct or revise.

(Line Forward)—Used with SH (shift) key to move to lines down the page.

SH (Shift)—Selects “upper case” punctuation or function shown by yellow markings above the keys. Except as outlined in the following note, the SH key must be pushed once each time prior to pushing the key for the desired “upper case” punctuation or function. Unlike a typewriter, the SH key on the RDP is non locking.

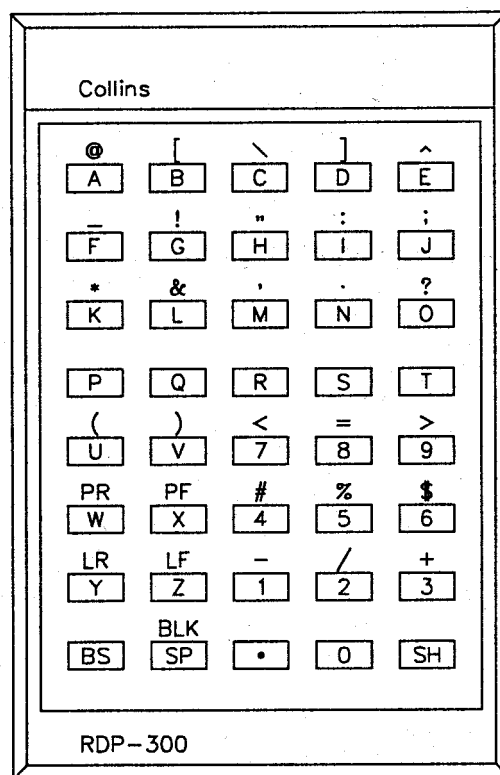
BLK (Blank)—Used with SH (shift) key to erase characters for editing or insert space between characters.

SP (Space)—Push this key to move the cursor one space to the right, without erasure.

BS (Back Space)—Push this key to move backward on a line.

NOTE

Eight keys on the RDP can be pushed and held to continuously repeat the keys function. The eight keys are “PF”, “PR”, “LF”, “LR”, “-” (dash, located above the 1), “BLK” (blank), “SP” (space), and “.” (dot). The “SH” (shift) key does not have to be held down when holding one of the “upper case” function keys down. If the SH key is pushed and then it is decided not to use the “upper case” function, push the SH key a second time to cancel the shift.



CGO-3110-10-AC-9

RDP-300 Remote Data Programmer

Operation

Before any data can be programmed into the MFD, the unit must be put into the edit mode. There are two ways to enter into the edit mode depending upon the data currently in the MFD. If the MFD is new (never programmed before), pushing the PGE button causes the display to show:

**NO PAGE
CHAPTERS FOUND. PLUG
IN RDP-300 TO EDIT.**

or pushing the EMG button will cause the display to show:

**NO EMERGENCY
CHAPTERS FOUND. PLUG
IN RDP-300 TO EDIT.**

In either case, simply plug the RDP-300 into the MFD DATA jack to enter the edit mode. If the above messages don't appear after pushing the respective PGE or EMG button, then the MFD has already been programmed (or at least a chapter has been defined) and the procedure outlined in the following paragraph should be used to enter the edit mode.

If chapter titles are displayed next to the line select keys on the MFD and a change to one of these chapters is desired, then push the line select key next to the desired chapter title, plug in the RDP-300, and make the changes. If the desired chapter title is not displayed, move the joystick up or down until the chapter title is found for the chapter that needs to be changed. Then push the line select key next to the desired chapter title, plug in the RDP-300, and make the changes. If you do not know the title of the chapter you want to change and the "NO PAGE CHAPTERS FOUND" or the "NO EMERGENCY CHAPTERS FOUND" message does not appear, then push the top line select key (upper right), plug in the RDP-300 and find the page(s) you want to change (you are now in the edit mode).

Whether you are in PGE edit mode or in EMG edit mode, all 100 pages of the checklist memory can be changed. The only time you need to know which edit mode (PGE or EMG) you are in is when you want to create a title page. If you want to make a PGE chapter title page, then the PGE edit mode must be entered. If you want to make an EMG chapter title page, then the EMG edit mode must be entered. Deleting chapter title pages from PGE or EMG can be done in either PGE or EMG edit modes.

A title page can be created at any time. The procedure is as follows:

- 1.—Enter the PGE edit mode and find the page you wish to make into a title page.
- 2.—Use the RDP keys to position the cursor over the first character on the first of the 12 lines on the page (upper left most character on the page).
- 3.—Push the top line select key at the upper right of the MFD. Observe that the cursor moves one character position to the right. This means that a PGE title page has been created. To make an EMG title page, enter the EMG edit mode and repeat the above procedure.

A title page can also be removed at any time. The procedure to delete any title page is as follows:

- 1.—Enter either PGE or EMG edit mode (either one will work for deleting a PGE or EMG title page) and find the title page by moving the joystick to the left or right.

- 2.—Use the RDP keys to position the cursor over the first character of the first line of the displayed page, and retype the character that is under the cursor using the appropriate key on the RDP. Observe that the cursor moves one character to the right. This indicates that the page is no longer a title page.

The MFD joystick is used to help find pages or chapter titles more easily. If chapters are already entered into the MFD, then pushing the PGE or EMG buttons shows up to four titles, one next to each of the line select keys. Only PGE chapters show when the MFD is in the PGE display mode and only EMG chapters show when the MFD is in the EMG display mode. Move the joystick down to show the next set of up to four chapter titles. Continue pushing the joystick down as necessary to view successive sets of titles. Move the joystick up to show the previous set of up to four titles. Continue pushing the joystick up as necessary to view previous sets of titles.

The joystick can also be used in the edit mode (RDP-300 plugged into the MFD). Move the joystick down to show the next data page, even if the next page is a title page for the PGE or EMG chapter. The cursor shows in the upper left of the display. Push the joystick up to show the previous data page. The cursor shows in the lower right of the display. Push the joystick to the right to show the next title page, regardless of whether it is a PGE title page or an EMG title page. The cursor shows in the upper left of the display. Push the joystick to the left to show the previous title page, regardless of whether it is a PGE title page or an EMG title page. The cursor shows in the upper left of the display.

NOTE

Pilot's that use the MFD for checklist purposes must consider it an aid only and realize that it contains only the data that someone has programmed into it. The Aircraft Flight Manuals contain the approved checklists.

WARNING

THE MFD EMG DISPLAY MODE SHOWS ONLY WHAT HAS BEEN PROGRAMMED INTO MEMORY AS EMERGENCY DATA. PILOTS MUST REALIZE THAT THIS DATA MAY OR MAY NOT BE APPLICABLE TO AN ACTUAL EMERGENCY. THE AIRPLANE FLIGHT MANUAL EMERGENCY CHECKLIST MUST BE REFERRED TO FOR THE PROPER CORRECTIVE ACTIONS TO TAKE IN VARIOUS EMERGENCY SITUATIONS.

Programming a New MFD Multifunction Display

The following procedures assume that the MFD is new and has not been programmed. Procedures for changing an already programmed MFD follow this section. Program a new MFD as follows:

- 1.—Turn on aircraft power. Ensure that the MFD circuit breakers are in. Push the PWR button at the upper left of the MFD. Position the WXP MODE switch to OFF (the WXP is not required during programming of the MFD). Allow 15 seconds for the MFD to warm up. Adjust the INT control at the upper right of the MFD as desired.
- 2.—Push the PGE button on the MFD. The display shows:

**NO PAGE
CHAPTERS FOUND. PLUG
IN RDP-300 TO EDIT.**

- 3.—Plug the RDP into the MFD DATA jack. Note that the first line of page 1 is displayed next to the top line select key. The characters on this first line should be: a red cursor (@ symbol) in the upper left, followed by yellow dots with the words "PAGE 1" in the upper right.

LINES ↓	CHARACTERS →																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	@	.	.	.	.	.	.	.	.	.	.		P	A	G	E				1
2																				
3																				

CGO-3110-10-AC-4

NOTE

The cursor (red @ symbol) should be in the upper left corner. If not, push the SH (shift) key then the PR (page reverse) key on the RDP until the cursor is displayed in the lower right corner of the display. Note the current page number and push SH then PF (page forward) until the cursor is at the upper left corner of page 1. After pushing the SH key, the PF or PR key can be held down for repeated function.

- 4.—Create a chapter title page as outlined below. For this example, assume the first PGE chapter title is to be called "PREFLIGHT". (You must have a chapter title at the top of the first page to be used even if the title is "PAGE 1" as shown after performing following steps a and b.) Remember that whatever is entered on the top line of the title page (in this example it is "PREFLIGHT") is shown beside the line select keys when chapter titles are called up. To enter "PREFLIGHT" as a chapter title, complete the following:
 - a. Push the BLK (blank) key once then push the BS (back space) key once. The cursor will be back in the upper left corner but the yellow dot that was in the upper left has now been erased.
 - b. Push the top line select key on the right side of the MFD. Observe that the cursor moves one character position to the right and that the space to the left of the cursor is blank. This page is now designated as a PGE chapter title page.

LINES	CHARACTERS →																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1		@	•	•	•	•	•	•	•	•	•		P	A	G	E				1
2																				
3																				

CGO-3110-10-AC-5

c. Note that PREFLIGHT has nine letters and will occupy the area now taken up by "PAGE 1" and the associated spaces. The only thing that needs to be done is to erase the yellow dots in spaces 2 through 11 of the first line and move the cursor to position 12. This is done by pushing the RDP-300 BLK key ten times and then pushing the SP (space) key once.

LINES	CHARACTERS →																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1		@	•	•	•	•	•	•	•	•	•		P	A	G	E				1
2												P	R	E	F	L	I	G	H	T
3																				

CGO-3110-10-AC-6

d. Complete the chapter title entry procedure by using the RDP keyboard to key in the word "PREFLIGHT". The letters will "write over" the data ("PAGE 1") currently occupying spaces 13 through 20 and the cursor will automatically advance to the first space in line 2.

LINES	CHARACTERS →																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1												@	P	A	G	E				1
2																				
3																				

CGO-3110-10-AC-7

e. Continue using the RDP to type in the desired PREFLIGHT checklist beginning on line 2 of the chapter title page (the chapter title page should be used as part of the checklist). Use as many pages as is needed to complete the PREFLIGHT checklist. It should be noted that all 12 lines can be used for checklist data on non-title pages.

LINES	CHARACTERS →																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1												P	R	E	F	L	I	G	H	T
2	@																			
3																				

CGO-3110-10-AC-8

NOTE

A blank (BLK) is considered as a character. A SP (space) is not considered as a character and should be used instead of blanks to separate words, symbols, numbers, etc. A BLK should only be used to "erase" a character as was done in step a.

- 5.—It is suggested that a couple of unprogrammed pages be left at the end of each chapter. This allows for expansion at a later date. To do this, push the joystick down twice or use the SH and PF keys on the RDP.
- 6.—Repeat steps 4 and 5 to create another PGE chapter title and a new checklist. Continue repeating steps 4 and 5 until all PGE chapters have been programmed.
- 7.—To program EMG chapters, push the EMG button on the MFD. The display shows the "PREFLIGHT" checklist title page that was previously programmed into the MFD as a PGE chapter title page. Use the joystick down or the RDP-300 SH and PF keys to go to a page where you want the first EMG title page to appear. Remember to skip all of the pages previously programmed during the PGE checklist programming procedure. Then repeat steps 4 through 6 above for the EMG chapter checklists.
- 8.—This completes the programming of new checklists. Disconnect the RDP from the MFD. Verify that data just programmed is accessible and correct by first pushing the PGE or EMG button on the MFD, then push the appropriate line select key to display the first page of the selected chapter. Use the joystick and the line advance/line reverse keys to view the various pages and/or select new chapter title pages.

Changing an Already Programmed MFD Multifunction Display

The following procedures assume that the MFD has already been programmed and it is desired to change some of the text. Reprogram the MFD as follows:

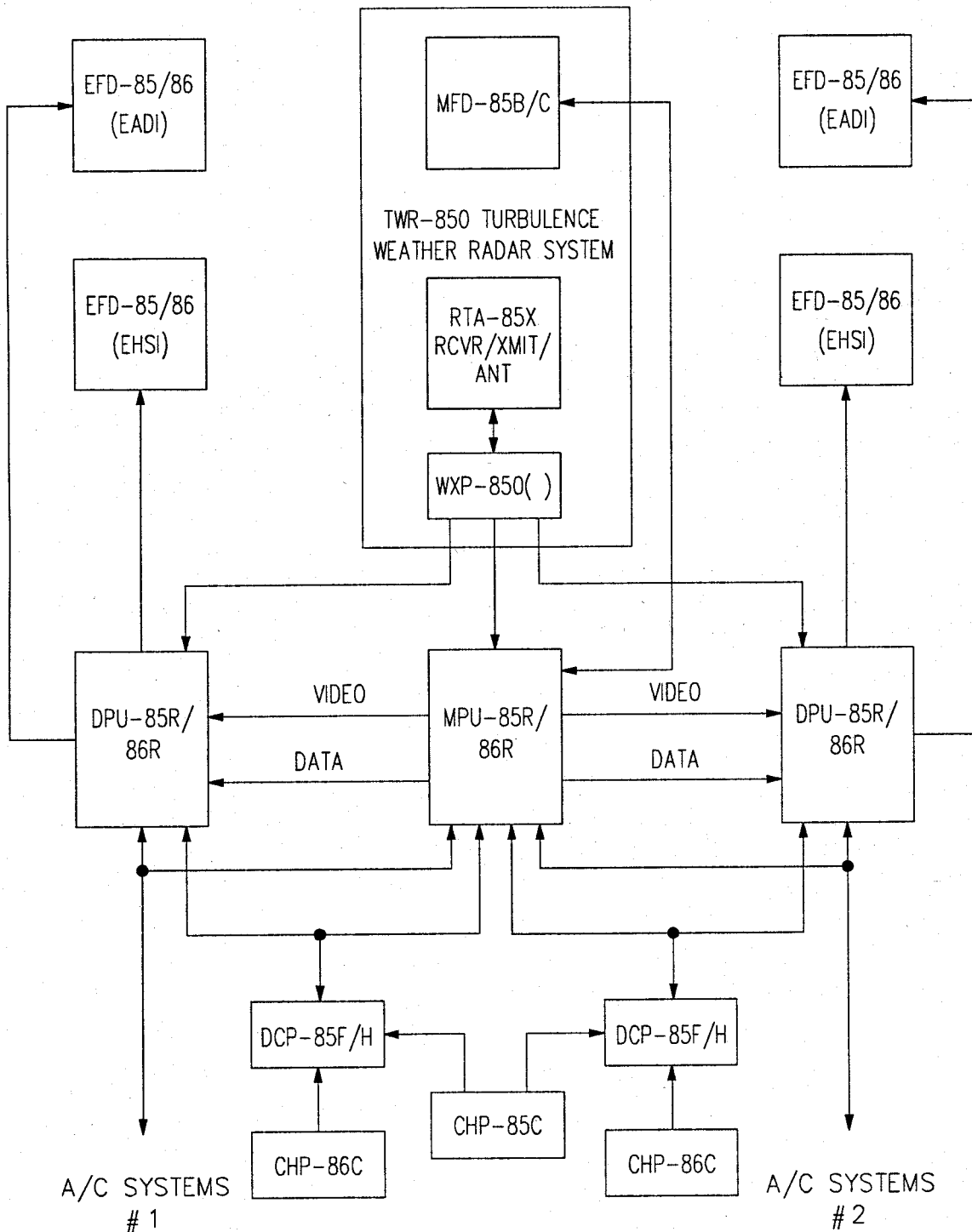
- 1.—Turn on aircraft power. Ensure that the MFD circuit breakers are in. Push the PWR button at the upper left of the MFD. Position the WXP MODE switch to OFF (the WXP is not required during programming of the MFD). Allow 15 seconds for the MFD to warm up. Adjust the INT control at the upper right of the MFD as desired.
- 2.—Push the PGE or EMG button on the MFD (depending upon what is to be changed).
- 3.—When PGE or EMG is selected, up to four chapter titles may appear next to the line select keys on the right side of the MFD. If data in one of these four chapters is to be changed, pushing the appropriate line select key causes the first page of the selected chapter to be displayed. If other chapter titles are required, movement of the joystick upward causes the next set of up to four chapter titles to be displayed. Movement of the joystick downward causes the previous four chapter titles to be displayed.
- 4.—When the desired chapter is found and the first page is called up, plug the RDP into the MFD DATA jack and make the necessary changes. If chapter title is unknown and/or nothing is displayed on the MFD, push the top line select key, plug in the RDP-300, and make the necessary changes. If all of the 100 available pages have not been used, new PGE or EMG chapters can be created by searching for a group of unprogrammed pages and then making a title page and entering checklist data as required.
- 5.—When the desired data has been entered, disconnect the RDP from the MFD. Verify that data just programmed is accessible and correct by first pushing the PGE or EMG button on the MFD, then push the appropriate line select key to display the first page of the selected chapter. Use the joystick and the line advance/line reverse keys to view the various pages and/or select new chapter title pages.

NOTE

The information programmed into the MFD will not be erased by power removal. Anything programmed during an earlier procedure will remain until reprogrammed.

EFIS SYSTEM BLOCK DIAGRAM

The following illustration is a simplified functional block diagram of the 5-tube EFIS-85C(14)/86C(14) Electronic Flight Instrument System. Supplemental information describing the system in considerable technical depth is available in the form of manuals and brochures, as well as formal training courses. Contact your local sales office for cost and availability information.



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